

claude-windows / e03c8ada-c18d-40e6-9ee1-144846fe3975

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\`e03c8ada-c18d-40e6-9ee1-144846fe3975.jsonl`  
- SHA-256: `83a89ea265f5aa4ea2da128c6eb06d03bdd18f0cea095055474094642dd3f0dd`  
- Source modified: `2026-06-22T10:15:25+00:00`  
- Imported at: `2026-07-05T16:48:19+00:00`  
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`  
- session_id: `e03c8ada-c18d-40e6-9ee1-144846fe3975`
```

Transcript

1. user (2026-06-14T10:30:31.151Z)

Confirm the just-landed Gate-A default flip (PR #146, merged to master as 570cf2b) causes no served-side regression on the real fusion path.

Context: PR #146 flipped the served default to `fusion\_hybrid` with `indexing.fusion.min\_crossings=2` (Gate-A held-shuttle guard, net_crossings>=2). The flip is validated OFFLINE (LOVO ablation, n=6 golden: ?F1 +0.074, CI [+0.042,+0.105], 6/6, p=0.031) and unit-tested, but the production `FusionSegmenter` served path (full WASB-via-WSL + Gemini handoff) has only ever been mock-tested on real video ? its byte-identity to `wasb\_hybrid` at `min\_crossings=0` was verified by #117's review but never run end-to-end with Gemini.

Task: on the GPU machine (torch+CUDA, WSL Ubuntu WASB, Gemini key in .env all present), run ONE or a few golden videos through the live `fusion\_hybrid` segmenter two ways ? `indexing.fusion.min\_crossings=0` (control = ungated, should reproduce wasb_hybrid) and `min\_crossings=2` (Gate-A default) ? and confirm: (a) the control run's served rallies match a `wasb\_hybrid` run on the same video (byte-identical seeding/handoff at mc=0); (b) the Gate-A run drops only low-crossing (held-shuttle/feed) windows, no real-rally loss vs the golden labels (or at least no F1 regression). Use the existing offline-rescore tooling where possible to avoid re-running WASB (wasb_runner has NO cache-hit early-return ? re-running overwrites the cached CSV; re-score the cached `output/wasb/\*\_wasb.csv` instead, see scratch/run_gx010125_rescore.py and backend/eval/distill_local.build_candidates). Keep artifacts isolated (per-video output dir). Record the result in docs/QUALITY_ITERATIONS.md (honest measured numbers only). If a regression appears, the rollback is a single flag: set `indexing.fusion.min\_crossings: 0` (Gate-A off) or `indexing.default\_segmenter: wasb\_hybrid` (full revert) in config.json.

This is the optional owner-gated follow-up flagged in PR #146 ? it does NOT change the merge decision; it adds served-side confidence.

2. assistant (2026-06-14T10:30:43.124Z)

I'll start by orienting myself: pulling the repo, reading the current-state/handoff, and checking what's actually available on this machine (the task assumes a GPU machine with torch+CUDA+WSL+Gemini).

3. user (2026-06-14T10:30:47.126Z)

Already up to date.

---BRANCH---

master

---HEAD---

e75ec35 feat(eval): LOS0/LOG0 ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)

570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible (#146)

7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked structural-guard decision (#145)

4. user (2026-06-14T10:30:47.146Z)

```
1      ---
2      name: session-handoff
3      description: "Living \"you are here\" handoff ? current state, next steps, open
decisions, ramp-up kit. AUTO-REFRESHED at meaningful checkpoints (PR merged, key decision, phase
shift, session wrap). \"update the handoff\" forces a full rewrite."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: 45ea3a91-654d-4972-9cd3-65f35db54caf
8      ---
9
10     # Session handoff ? sports highlight indexer (repo still named badminton-highlight-
indexer; RENAME PENDING) + siblings: sports-data-collector, rally-annotator, sports-obsreport
11
12     **Last updated:** 2026-06-14 (session wrap; master at #135; self-merged on green CI
throughout). ***? IMMEDIATE PICKUP =
13     EXECUTION-POLICY P3 (then P4)** ? a per-job, self-scheduling resilience framework
(owner's design, validated). **DESIGN
14     ([#132](https://github.com/avidullu/badminton-highlight-indexer/pull/132)
`docs/EXECUTION_POLICY.md`) + P1 the actual
15     Gemini-hang fix ([#133](https://github.com/avidullu/badminton-highlight-
indexer/pull/133)
16     `backend/infrastructure/execution_policy.py` ? `run_bounded`/`poll_until`, 100% module
coverage) + P2 the self-scheduling
17     DB substrate ([#135](https://github.com/avidullu/badminton-highlight-indexer/pull/135) ?
jobs migration v3 `policy`/`next_poll_at`,
18     `claim_due_job`/`set_job_waiting`) are MERGED.** **NOT DONE (next session, own distinct
PRs, HIGH coverage):** **P3** =
19     wire the `main.py` executor to a `claim_due_job` self-scheduling loop honoring
WAIT_AND_RESUME (release+reschedule) +
20     **resumable AI handoff** (persist per-window confirmation; resume only unconfirmed ? the
GX010125 reel becomes exhaustive
21     across resumes, no wasted Gemini spend); **P4** = apply the policy to the WSL/WASB GPU
pass + ffmpeg. **Read [[current-state]]
22     FIRST** (the top checkpoint has the full P1/P2 detail + the rationale for stopping at
P2).
23
24     **Earlier this same session (rally-quality / paper track, [[paper-coauthor-aim]]):**
analyzers/reconcilers **framework**
25     doc + a verified **multi-agent literature review** ($13 ? skeleton is standard
TAL/TAS+late-fusion, *cite don't claim*;
26     only the report-first reconciler is a novel combination) ? [#119]; **eval measurement-
honesty helpers C1?C4** (`backend/eval/
27     eval_stats.py`+`score_calibration.py`+`segmentation_metrics.py`) ? [#124] + wired into
`experiment.compare` ? [#129];
28     **FIRST golden A/B experiment** (n=6, offline) ? archived [#128]: **Gate-A (net-
crossings?2) is a measured win (+19% F1,
29     6/6, p=0.031), PROMOTED** [#131]; the window-level **learned variant FIXED** (train-fold
threshold tuning + calibration ?
30     +38% F1, un-zeroed) ? [#130]; **`docs/PER_VIDEO_OUTPUT_LAYOUT.md`** plan [#131];
**GX010125_1080p reel re-generated** into
31     isolated `output/GX010125_run` (reused cached 1080p trajectory). Prior tracks still
valid: rally-quality fusion P1?P4
32     (#114?127), UI-alpha, productization, first 4K-GPU E2E. This is the ~1-page orientation.
? **THIS IS THE OWNER'S REAL GPU
33     MACHINE** (torch+CUDA/cv2/WSL/Gemini live).
34
35     ## You are here (2026-06-13) ? ? HARDENED + M0 SCAFFOLDED + MOAT QUANTIFIED + QUALITY-
SWEPT + UI-ALPHA LANDING + 4K-E2E PROVEN
36     -1. **Recent landings since the sweep (mostly OTHER slates ? facts from git/gh, not this
session's work):**
37     - **UI-alpha:** PR-1 async jobs (#87) + **PR-2 chunked upload + highlight download
(#99, live-E2E-verified)** +
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38         docs sync (#100). Next UI slice = **PR-3 identity** (`owner_email` scoping,
JWT/header w/ `DEV_USER_EMAIL`
39         fallback, CORS narrow, rate limit). ? PR-3/deploy: `security.allowed_video_root`
MUST contain `upload_dir`.
40         - **Branding/stitcher:** #101 configurable watermark · #102 config paddings ? live
VideoCompiler · #103 bundled
41         Apache-2.0 font + fallback log · #107 `api/compile` enforces
`stitching.min_shot_count` · #108 escape Windows
42         drive-letter colon in ffmpeg filtergraph (fixed #103's watermark silently failing
on Windows).
43         - **PMF/docs:** #104 C13 field kit (flyer/playbook/answer-sheet/pre-registered
signals; associate-pay REMOVED per
44         owner) · #105 i18n plan (Kannada-first) · **OPEN #106** video-locality model
(timestamps are the product, video
45         is dead weight) ? awaiting owner.
46         - **First 4K GPU E2E (2026-06-13):** real 11.5GB GoPro match ?
`output\GX010125_highlights.mp4`, 11 rallies
47         (>7 shots) of 52 detected; 1080p-proxy-index / 4K-stitch recipe in [[gotcha-long-
runs-gpu-wsl]]. Suite 409 green.
48         - ? **IN-FLIGHT, uncommitted (sibling/owner ? do NOT clobber, not yet PR'd):** WSL
**cache-hygiene** slice on
49         `wslCommandMixin` ? `_wsl_free_gb`/`_wsl_rm_rf` + `keep_frames`/`min_free_gb` knobs
+ disk guard + post-success
50         cache delete (base/wasb_runner/tracknet_runner/models.py); `config.json` GPU knobs;
`arial.ttf`;
51         `docs/STORAGE_SHARING_MODEL.md`.
52         0. **8?9 quality sweep DONE (2026-06-11, this session's prior work):**
lint/type/coverage gates live in CI (ruff.toml,
53         mypy.ini w/ quarantine ratchet, --cov-fail-under=70); trajectory hybrids share one
engine (`trajectory_hybrid.py`,
54         equivalence-tested); **wasb_hybrid's FIRST verified live E2E run** found+fixed two
real live-path bugs (#97 WSL
55         tilde+abspath); LOVO 0.626 re-verified EXACT post-sweep. Merged #90?#98. Deferred:
ruff I/UP, yolo_hybrid fold-in,
56         pyproject (owner deselected), extra='forbid'.
57         1. **Audit + remediation DONE:** 8-auditor deep audit (all findings adversarially
verified) ? 12-PR hardening sweep
58         (both repos). Master was literally broken (compiler SyntaxError) and is now CI-
guarded (both repos have CI; merges
59         gated on verified-green). Corpus licensing corrected (23 ShuttleSet broadcast rows
demoted; commercial export =
60         owned data only). Suite: indexer 336 green, collector 136 green.
61         2. **ADR-008 ratified (owner):** training in-repo (`training/`) behind a CI-enforced
import wall; featurizer
62         single-sourced in `backend/features/rally_seq.py` (train==serve); serving consumes
versioned ARTIFACT BUNDLES only;
63         ship-gate product-side; **pre-committed repo split** (`sports-temporal-models`) at
Gen-2/second-consumer.
64         "Split driven by productionization, not isolation" (owner's framing).
65         3. **M0 executed:** canonical GT loader (`backend/eval/gt_loader.py` ? closed the two-
loader format fork, round-trip
66         tested vs collector export) · **n=6 owned corpus IN the collector SoR** (262 rallies,
Proprietary) · **Gen-0
67         harness** `python -m training.gen0.harness` (train?LOVO?gate?artifact; v0.1.0 built:
**LOVO 0.626**, floor 0.480
68         passed, sign test 5/6 p=0.109 ? ship=False with honest blockers).
69         4. **Gemini battery DONE (the moat is quantified):** clean-ish Mahadevpura ? heuristic
0.657 / Gen-0 0.744 /
70         two-stage refine 0.83 / **wrapper 0.93** ; multi-court Kushagra ? heuristic 0.157 /
Gen-0 0.561 / **wrapper 0.66**
71         (FPs = background-game pickups + dead-time merges, the contrast-metric confound).
**Ship target = beat 0.66 on
72         multi-court** ; clean footage is commoditized (serve via Gemini). Baselines committed
in `eval_baselines/`.
73         Demo reel exists: `output/MahadevpuraSingles_highlights.mp4`.

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74      5. **Gemini ops (hard-won):** prepaid/cold projects burst-throttle with a MISLEADING
"credits depleted" 429 ? ramp
75      gradually; serial works (`ai_max_workers=1` shipped); chunks re-encode to 1280
(`ai_chunk_scale_width`, PR #80);
76      generate retries = 6 attempts exponential+jitter (PR #83); model = `gemini-flash-
latest` alias (2.5-flash is
77      legacy). 503 capacity waves are real ? provider-fallback registry is load-bearing.
78
79      ## Owned-model architecture (for clarity ? owner asked)
80      **Gen-0/Gen-1 = WASB-substrate + OWNED temporal head.** WASB (frozen, MIT code, C3
checkpoint provenance = ship
81      blocker) produces trajectories; the owned IP = featurizer + head + corpus +
harness/gate. **Gen-2 (Qwen3-VL via
82      ms-swift) = pixels, drops WASB** ? that's the heavyweight step, only entered if the
lightweight head plateaus below
83      target. Gen-0 trains in MINUTES on the 2070S ($0). Multisport: WASB transfers to TT
(0.909) but not pickleball
84      (0.308) ? per-sport detectors or Gen-2.
85
86      ## ?? Next steps / open owner decisions (mirror of docs/NEXT_STEPS.md ? read that for
detail)
87      0. **? RESUME HERE ? EXECUTION-POLICY P3 then P4** (`docs/EXECUTION_POLICY.md`
$Phasing). **P3** (one PR): in
88      `backend/main.py`, replace the fire-once `_job_executor.submit(_run_job)` with a loop
that calls
89      `db.claim_due_job()` (already built, #135) and, when a long op hits its policy
budget, calls
90      `db.set_job_waiting(job_id, delay_sec)` to release+reschedule (WAIT_AND_RESUME)
instead of blocking; **+ resumable AI
91      handoff** ? persist per-window confirmation (candidate `stats` or a small table) so a
resumed handoff only does the
92      UNCONFIRMED windows (this is what makes the GX010125 reel exhaustive across resumes,
no re-spend). Tests: a fake
93      handoff that hangs ? job goes `waiting` ? re-claimed ? resumes only the unconfirmed
window. **P4** (one PR): wrap the
94      WSL/WASB `_wsl_bash` GPU pass + the ffmpeg calls in `run_bounded`/`poll_until`.
**Each = distinct self-merged PR, HIGH
95      coverage** (owner's explicit bar). Substrate (P1 `execution_policy.py` + P2
`claim_due_job`/`set_job_waiting`) is done,
96      so P3 is wiring not invention.
97      1. **Owner mission in progress: grow the golden corpus** (multi-court badminton first;
?3 matches/sport, TT next;
98      adults-only for training; per video: label ? `import-local --normalize` ? re-run
harness). Sign test reaches
99      significance around n=10 with 9 wins.
100     2. **Set the ship-gate target** (suggested: LOVO F1@tIoU0.5 ? 0.70 on multi-court folds
+ paired-sign p<0.05).
101     3. **M0.4 increment:** ActionFormer/ASFormer into the harness (still lightweight/local)
+ per-video threshold calibration.
102     4. C13 interviews + camera pilot (PMF) ? **field kit SHIPPED (#104):** flyer/visit-
playbook/answer-sheet/pre-registered
103     signals (associate-pay removed per owner; he handles engagement terms); repo rename;
rotate the chat-pasted Gemini
104     key; collector=SoR confirm.
105     5. **UI-alpha track (owner-gated):** PR-1 async jobs (#87) + PR-2 upload/download (#99)
DONE ? **next = PR-3 identity**
106     (`owner_email` scoping, JWT/header auth w/ `DEV_USER_EMAIL` dev fallback, CORS
narrow, rate limit, quota). The
107     PR2_HANDOFF "tomorrow-morning CUJ" runbook is unblocked for the owner.
108     6. **Open owner decisions:** ratify/merge **#106** video-locality model (it reframes
hosting ? timestamps are the
109     product); whether the in-flight WSL cache-hygiene slice gets PR'd by its owner or
folded into the next run.
110
111     ## Ramp-up kit

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112 - [[current-state]] FIRST (dense checkpoints) · `docs/NEXT_STEPS.md` (live plan) ·
`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`
113 (roadmap) · ADR-008 in `docs/archives/decisions/DECISIONS.md` (topology) ·
`docs/CODE_MAP.md` (note its 2026-06-10
114 sweep banner ? several documented behaviors changed).
115 - Code anchors: `training/gen0/harness.py` · `backend/features/rally_seq.py` ·
`backend/eval/gt_loader.py` ·
116 `eval_baselines/*.json` · collector `src/cli/curate.py` (reclassify-licenses, import-
local --normalize/--force).
117 - **This session's anchors:** `docs/EXECUTION_POLICY.md` +
`backend/infrastructure/execution_policy.py` (P1) +
118 `database.py::claim_due_job`/`set_job_waiting` (P2) +
`backend/main.py::run_job`/`_job_executor` (the P3 target) ·
119 `docs/ANALYZERS_AND_RECONCILERS.md` (§13 = the verified prior-art/course-corrections
review) · `backend/eval`/
120 {`experiment.py` (now emits a `honest` block: CIs + sign test + F1@k by default;
`Variant.tune_threshold`/`calibrate`),
121 `eval_stats.py`, `score_calibration.py`, `segmentation_metrics.py`} ·
`docs/archives/quality-iterations/2026-06-first-ab-experiment`/
122 (the first A/B: Gate-A +19% F1) ·
`scratch/{run_gx010125_full,run_gx010125_rescore,ab_n6_lovo}.py` (machine-side repros).
123 - Data: labels `Annotation Setup\Collect\Golden Labelled` · trajectories+Roora
`...\Collect\Trajectories` ·
124 eval manifest `output/human_lovo_manifest.json` (gitignored) · corpus DB = collector
`data/sports_metadata.db`
125 (backup: `data/sports_metadata.backup-2026-06-10.db`, gitignored).
126
127 ## Gotchas (this session's additions)
128 - Merge discipline: gate on `gh pr view --json statusCheckRollup` == SUCCESS (a piped
`gh pr checks --watch` exit code
129 once let a red PR merge ? #75/#76 incident). Collector DB: stash-and-pop, never
`checkout --` (recovered once from
130 a dangling stash). Other sessions may write [[current-state]] concurrently ? insert
checkpoints, never overwrite.
131 - PowerShell `ConvertTo-Json` reformats config.json wholesale ? edit surgically.
`Select-Object -Last N` buffers
132 background output until exit (no interim tail) ? and loses EVERYTHING if the process
is killed (idle/suspend kills
133 background runs silently); pipe long background runs through `Out-File` instead.
134 - **Sibling sessions share this checkout and flip branches mid-session** (the node24
chip did) ? re-verify
135 `git branch --show-current` before EVERY commit; the post-merge "checkout master &&
pull" scripting once left the
136 tree on master and a commit went straight to the default branch (the AI-7 process slip
? don't repeat it).
137 - Windows ffmpeg here lacks NVENC ("No capable devices found") ? use libx264 for host-
side re-encodes.
138 - The dev container claim in CLAUDE.md is stale for THIS machine: cv2/torch/numpy all
present; GPU via WSL.
139

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5. user (2026-06-14T10:30:47.528Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-451 of 1851 total (87024 tokens, cap 25000). Call Read with offset=452 limit=451 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1 ---
2 name: current-state
3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5   node_type: memory
6   type: project

```

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7      originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
**Owner adopted the HYBRID** ? personalization as a
13     flag-gated FEATURE on a generalization-first baseline + a **contribution flywheel** ("a
tool that gets better the more you help it";
14     free-tier videos pool into the owner-trained baseline). Pivot analysis
(`scratch/PERSONALIZATION_PIVOT_ANALYSIS.md`) recommended
15     hybrid-not-personalization-first; the owner's contribution-flywheel closes the flywheel-
loss objection. **MERGED this session:**
16     telemetry wire-in **#140** ; per-user promotion gate **#141** (min-N floor + structural-
guard exemption); held-out-USER learning
17     curve **#142** ("<K videos to superior quality" = falsifiable); per-user no-regression
guard **#143** ; **`docs/PERSONALIZATION_PLAN.md`
18     #144** (the checked-in project doc + locked decisions). Gemini-training-source concern =
**CLEAN** (eval-only, gitignored, M0.3 owner-gated).
19     **LOCKED owner decisions:** identity = north-star-design/current-impl (nullable user_id,
device UUID); inline classifier_json behind a
20     cloud-load interface; **`min_n` is a PROMOTION-confidence floor, NOT a service gate** (a
1-video user is always fully segmented by the
21     baseline); free-tier video pooling (consent + minors-exclusion); **Gate-A-only
structural + `min_n=5`** . ?? **IN FLIGHT (3 bg agents):**
22     quality-iterations doc PR; **Gate-A/B golden-set numbers + flip-implications**
(`scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md` ? leave-one-
23     signal-out ablation; owner: NUMBERS INFORM, FOUNDATIONS DECIDE); **ablation-layer
framework plan** (`scratch/ABLATION_FRAMEWORK_PLAN.md` ?
24     leave-one-signal-out + signal-relevance ledger over time, for the paper). **GPU + Gemini
API now AUTHORIZED for evals** (cooler boost on).
25     **Gate-A default flip = NOT done** ? gather numbers + implications first, then owner
decides (data-influenced, foundations-led). NEXT
26     personalization = Phase 1 (serving bridge: wire `LogisticRegression.load` into
`_select_for_handoff`) ? owner-gated, see [[weak-signals-retained]] +
`docs/PERSONALIZATION_PLAN.md`.
27
28     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASURED + extraction COMPLETE + telemetry #134
merged + Cooler Boost). **THE NUMBERS
29     (honest LOVO n=6, IoU?0.5, C1 bootstrap-CI + sign-test):** shuttle-only F1**0.307** ;
**+person (P3) ?F1=?0.021** (?6.7%, CI
30     [?0.165,+0.162], sign 2W/4L p=0.688, **CI-clears-0: NO**) ? NO lift, stays default-OFF
(no court polygons ? `players_in_court`
31     counts spectators; per-video wash: testlarge +0.40 / gbaaddy ?0.22). **+person+audio (P4
incremental) ?F1=?0.079** (+27.7%, F1
32     0.286?0.366** , CI [?0.054,+0.203], sign 4W/2L p=0.688, **CI-clears-0: NO**) ?
PROMISING (wins 4/6, largetest/mahadevpura
33     +0.27; endpoint 0.366 > shuttle-only 0.307) but n=6 can't confirm. **VERDICT: neither
promotable (both CIs span 0); audio is
34     the lead ? need more matches + measure audio on shuttle-only directly (drop the person
handicap).** Recorded in
35     `eval_baselines/experiment_leaderboard.json`; honest negative/uncertain results KEPT
([[paper-coauthor-aim]]). **LANDED THIS
36     SESSION:** result logged to `docs/QUALITY_ITERATIONS.md` (#137);
`experiment_leaderboard.json` tracked (#139); **weak-signals
37     audit ?** (person/audio/court-bounds all present, default-OFF, no-special-casing,
retained ? refuted=false ? [[weak-signals-retained]]);
38     7-page paper-style **quality report PDF** at
`OneDrive\Documents\Rally_Detection_Quality_Report.pdf` (gen `scratch/quality_report.py`;
39     conservative academic restyle, strict-reviewer-approved); `FusionGoldenExtract`
scheduled task **unregistered** (cleanup); tree
40     clean on `master`. ?? **NEXT (fusion track):** (1) **wire `run_telemetry` (#134) into
fusion_golden + the psi wrapper** (deferred,
41     ready, mergeable); (2) **measure audio on shuttle-only directly** (quick eval, reuses
the 6 `output/*_golden_features.json`, NO new

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42     data ? audio's standalone lift without the person handicap); (3) **grow the golden
corpus via Roora** (owner's job ? gives power
43     to confirm the +0.079 audio lead) + **activate court polygons** (re-measure person
properly ? the ?0.021 is partly a
44     `court_polygon=None` artifact); (4) future **shuttle-dropped** cue ([[weak-signals-
retained]]). **OTHER track (per
45     [[session-handoff]] ? the stated IMMEDIATE PICKUP): EXECUTION-POLICY P3 then P4**
(owner-gated; #132/#133/#135 done). **Extraction infra** (all working): detached Scheduled Task
`FusionGoldenExtract`
46     runs `scratch/extract_until_done.ps1` (self-restart loop + in-process keep-awake + GPU-
busy gate: waits if free VRAM<3500MB) ?
47     deaths were `0xC000013A` console-control kills (NOT crash/sleep/TDR/OOM ? verified via
event log), auto-recover via 3-min
48     repetition trigger + restart-on-failure. **Cooler Boost** (MSI Creator Center) dropped
85?67°C, SM ~1100?~1300MHz, velocity
49     ~26?~50 fps (the rest is Max-Q power cap, not thermal).
50
51     **Checkpoint:** 2026-06-14 (EXECUTION-POLICY framework: design + P1 + P2 SHIPPED ? **PR
#132 (design) + #133 (P1) + #135 (P2)
52     MERGED**, self-merged on green CI). Owner asked to harden the GX010125 Gemini hang as a
**policy-driven, per-job,
53     self-scheduling** system (default WAIT_AND_RESUME, ABORT, POLL_EVERY_N_SEC?, **no master
node**) with **very high coverage**
54     (failures cause delays) and **clean poll-logging** (visible but not polluting). **#132**
= design doc `docs/EXECUTION_POLICY.md`
55     (validated the owner's architecture: per-job policy = data on the job; self-schedule via
re-enqueue; no master ? rides the
56     existing single-worker executor + `jobs` table + crash sweep; DB is the coordinator).
**#133 (P1, the actual fix)** =
57     `backend/infrastructure/execution_policy.py`: `ExecutionPolicy` + `run_bounded` (HARD
`op_timeout_sec` per-attempt ? the
58     hang-killer; retries failures AND hangs; hung attempt's daemon thread abandoned) +
`poll_until` (bounded poll loop). Clocks
59     injected ? **12 tests, 100% module coverage, instant**. Refactored `gemini.py`:
`generate_content`?`run_bounded` (a
60     never-returning generate previously blocked FOREVER = the GX010125 hang; now bounded),
processing-wait?`poll_until` (**per-poll
61     logs at DEBUG, not an INFO line every 10s** ? the no-pollution ask). 23 gemini tests
stay green. **#135 (P2, substrate)** =
62     DB migration **v3** (`jobs.policy` JSON + `next_poll_at`; existing rows survive via
ALTER) + `set_job_waiting(delay_sec)`
63     (self-schedule: park 'waiting' until due) + **`claim_due_job()`** (atomic compare-and-
set claim of queued/due-waiting ?
64     running; concurrent-safe, no master) + `create_job(policy=)`/`get_job` policy.
**Additive, no executor change**; 10 tests;
65     version asserts bumped 2?3. ? **THIS IS THE OWNER'S REAL GPU MACHINE**
(torch+CUDA/cv2/WSL/Gemini live). ?? **REMAINING
66     (owner-gated, NOT done ? deliberately not rushed at session tail given the high-coverage
bar): P3** = wire the executor
67     (main.py) to a `claim_due_job` self-scheduling loop honoring WAIT_AND_RESUME
(release+reschedule) + **resumable AI handoff**
68     (persist per-window confirmation; resume only unconfirmed ? the GX010125 reel becomes
exhaustive across resumes, no wasted
69     Gemini spend); **P4** = apply the policy to the WSL/WASB GPU pass + ffmpeg. Each = its
own distinct PR. Ties [[paper-coauthor-aim]] indirectly (infra).
70
71     **Checkpoint:** 2026-06-14 (ACTED ON FIRST-A/B FINDINGS: Gate-A PROMOTED + learned FIXED
+ GX010125 run ? **PR #130 + #131
72     MERGED** to master `2e5892e`; self-merged on green CI). ? **THIS IS THE OWNER'S REAL
GPU MACHINE** (torch+CUDA, cv2, WSL
73     Ubuntu, Gemini key in .env all present) ? not the dev container the older CLAUDE.md note
assumes. **#130 = learned-variant
74     FIX**: additive/default-OFF `Variant.tune_threshold` (operating point on the TRAIN fold)
+ `calibrate` (Platt/isotonic, C2);
75     validated on n=6 golden ? **mean held-out F1 0.307?0.423 (+0.116), testlarge 0.000?0.300

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(un-zeroed), now BEATS CONTROL
76     0.388**; threshold-in-LOVO is the big lever. **131 = Gate-A PROMOTED** (first cue
promoted via the harness; doc trail in
77     QUALITY_ITERATIONS ? measured win dF1 +0.074, 6/6, p=0.031; recommended
`indexing.fusion.min_crossings=2`; **live served-default
78     flip stays OWNER-GATED** until a real fusion-path run confirms ? fusion P2 is unverified
on real video) **+ new
79     `docs/PER_VIDEO_OUTPUT_LAYOUT.md`** (one dir per video; fixes the hardcoded `"output"`
joins in `trajectory_hybrid.py:~176`
80     so `--output-dir` truly isolates). **GX010125_1080p RUN (offline, no GPU):** ? key
finding ? `wasb_runner.run_predict` has
81     **NO cache-hit early-return**, so the full pipeline would RE-RUN WASB (~tens of min on
1080p) + overwrite the cached CSV; the
82     cached trajectory is reusable ONLY via offline re-score. So ran
`scratch/run_gx010125_rescore.py` = parse cached
83     `output/wasb/GX010125_1080p__89605e80ed01_wasb.csv` ? 45 candidates ? CONTROL 45 rallies
(11.0min) vs **Gate-A 27** (9.3min),
84     Gate-A **dropped 18 low-crossing windows (~1.7min)** of likely held-shuttle/feed FPs ?
isolated `output/GX010125_run/` (no
85     pollution; no golden labels for this footage ? divergence not F1). A FULL Gemini-
confirmed highlight reel = a separate
86     ~tens-of-min GPU+API run (owner kicks off). Repro:
`scratch/{run_gx010125_rescore,ab_n6_lovo,ab_eval_demo}.py`. All in
87     isolated worktrees off master (siblings concurrent). ?? NEXT (owner-gated): real fusion-
path run to confirm Gate-A served-side
88     + flip the live default; impl `PER_VIDEO_OUTPUT_LAYOUT` L1; Tier-2/3 §13 corrections;
analyzers/reconcilers A1?A5/R1?R3. Ties [[paper-coauthor-aim]].
89
90     **Checkpoint:** 2026-06-14 (FIRST GOLDEN A/B EXPERIMENT + HARNESS HONEST-REPORTING WIRED
? **PR #128 + #129 MERGED**
91     to master `f680b8d`, self-merged on green CI per [[working-agreement-merging]]; owner-
directed momentum). Ran the
92     **first real end-to-end A/B on the n=6 golden corpus, 100% OFFLINE** (re-scored
persisted WASB trajectories via
93     `distill_local.build_candidates`, no GPU ? the "persist once re-score" thesis proven).
One candidate set re-scored
94     three ways: **Gate-A (net_crossings?2) is a MEASURED WIN ? ?F1 +0.074, 6/6 matches,
sign-test p=0.031, CI
95     [+0.042,+0.104], seg-count-ratio 1.68?1.01** (over-seg nearly eliminated); the **window-
level 5-feature logistic did
96     NOT clear the bar** (?0.081, 2W/4L; testlarge zeroed = threshold/calibration artifact ?
C2 motivation; **distinct from
97     the stronger per-frame `rally_seq_proto` 0.583** ? don't conflate). Headline: **the
framework held up ? C1 promoted one
98     variant and refused another (right call both ways); C4 surfaced over-seg tIoU hid.**
**#128** = archived posterity doc
99     `docs/archives/quality-iterations/2026-06-first-ab-experiment/` (full tables) +
archives-index + live QUALITY_ITERATIONS
100     pointer. **129 = (b) wired C1+C4 into the CORE `experiment.compare()`**
(additive/default-ON `honest_stats=True`:
101     `honest` block = per-metric bootstrap CIs + paired ?F1 CI + exact sign test + F1@k +
segment_count_ratio; `record_experiment`
102     carries honest ?F1; existing fields byte-unchanged, `honest_stats=False` = legacy
shape). Reuses #124 helpers; complements
103     sibling **127** which wired the same C1 into `fusion_compare` (no dup). 19 existing
experiment tests pass + 3 new;
104     ruff+mypy+CI all green. Repro `scratch/ab_n6_lovo.py` + `scratch/ab_eval_demo.py`
(machine-side; read gitignored
105     manifest/external golden). All done in **isolated worktrees off master** (siblings
merged #121-#127 concurrently). ?? NEXT
106     (owner-gated): promote Gate-A via the no-regression gate (it's a real measured win); fix
the logistic (C2 calibration +
107     threshold-inside-LOVO); Tier-2/3 §13 corrections; analyzers/reconcilers impl A1?A5/R1?R3
(none started). Ties [[paper-coauthor-aim]].
108

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109     **Checkpoint:** 2026-06-14 (FUSION P3/P4 MEASUREMENT BUILT + GPU EXTRACTION RUNNING ?
master `d7b1f51`; suite 514 green).
110     The "modeling" track for the person-cue (P3) + audio (P4) fusion lift ? all default-OFF,
no-special-casing, harness-measured.
111     **Merged this run (mine): #121** fusion P3 (`backend/eval/fusion_golden.py` offline
person-feature EXTRACTOR + `fusion_compare.py`
112     LOVO driver ? reuses build_candidates/PersonDetector/fusion_features/experiment.compare,
NO new scoring); **#122** P4
113     (`backend/pipeline/detectors/audio_features.py` + `backend/eval/fusion_audio.py`
incremental audio cue); **#123** progress-logging
114     + resumability (`--fresh`); **#125** single-pass
`PersonDetector.detections_over_windows` (fixed the per-window-peek 0(N)
115     pathology ? ~28s/window ? one sequential decode); **#126** `--smallest-first`; **#127**
honest ?F1 = bootstrap CI + paired
116     sign test on per-video held-out F1 (wires #124's `eval_stats.paired_delta_ci`; "never a
bare mean"). Concurrent sessions
117     merged #119 (analyzers framework), #120 (run.log), #124 (eval-honesty utils:
eval_stats/score_calibration/segmentation_metrics).
118     **? GPU EXTRACTION via DETACHED SCHEDULED TASK `FusionGoldenExtract`** (runs
`scratch/extract_until_done.ps1` ? Task Scheduler
119     owns it, OUTSIDE Claude's job object ? survives turn/context/session boundaries; in-
process keep-awake + self-restart loop until
120     6/6; clean UTF-8 logs to `output/fusion_golden.log` + `scratch/extract_wrapper.log`).
Loops `python -u -m backend.eval.fusion_golden
121     --manifest output/human_lovo_manifest.json --out-dir output --smallest-first` ? 6
`output/<name>_golden_features.json` (GITIGNORED
122     + RESUMABLE: skips any video whose JSON exists). **3/6 DONE** (testlarge, mahadevpura,
largetest); remaining
123     kushagra?gbaaddy?adarsh(last, 84k frames). **MONITOR:** JSON count + `tail
output/fusion_golden.log` + `nvidia-smi`;
124     `Get-ScheduledTask FusionGoldenExtract|Select State`. **RESTART (resumes, skips done):**
`Start-ScheduledTask -TaskName
125     FusionGoldenExtract`. **STOP:** `Stop-ScheduledTask -TaskName FusionGoldenExtract` +
`Get-Process python|Stop-Process -Force`.
126     ?? **WHY DETACHED (debugged 2026-06-14):** Claude-bg-task runs DIED TWICE silently
(largestest 61%, kushagra 6% @137s), NO Python
127     traceback AND ? verified via Windows event log ? NO sleep (Kernel-Power 42), NO TDR
(Display 4101), NO WER crash, NO RADAR-OOM
128     (no event 1003; 18GB free). Traceless vanish = external `TerminateProcess`; common
factor = both were Claude-managed bg tasks
129     (job-object kill-on-close at a context boundary fits death #1 @ the prior context
summary). Scheduled Task = immune; GPU access
130     CONFIRMED in the interactive task session (53?64% util, 3.1GB). GPU util while running =
20?64% bursty ? batch=1 Faster-RCNN
131     latency-bound (small kernels + CPU NMS + per-frame ByteTrack), NORMAL, model IS on CUDA,
NOT decode-bound; batching the forward
132     pass is the next-extraction speed lever (changes features, so NOT mid-run). ~1.5-2h
total. Stale Claude bg keep-awakes stopped. **DEATH #3 + GPU-job check (2026-06-14):** died a
3RD time (gbaaddy 69%)
133     ? `LastTaskResult=0xC000013A` = STATUS_CONTROL_C_EXIT = terminated by a **console-
control event / CTRL+C** (NOT crash/sleep/TDR/OOM
134     ? explains the no-trace vanish; killed the detached task too, so NOT Claude's job
object). GPU-contention angle CHECKED: **NO
135     competing GPU compute job** (only desktop apps + our 1 python; 5GB free; HW is a
**laptop** RTX 2070S **Max-Q**, MSI). Mitigations:
136     (1) task re-registered with **3-min repetition trigger + restart-on-failure (count 10,
1-min)** ? auto-recovers within minutes of
137     ANY death; (2) **GPU-busy gate in the ps1** (owner policy = *wait until free, auto-
resume*): before each python launch, if free
138     VRAM <3500MB (another job has the GPU) it waits 30s + re-checks, then launches. **??
WHEN IT COMPLETES ? the remaining steps for
139     the NUMBERS:** (1) `python -m backend.eval.fusion_compare --features-dir output
--record` ? honest P3 LOVO ?F1 (shuttle-only
140     vs +person) w/ CI+sign-test; (2) `python -m backend.eval.fusion_audio --augment
--compare --record` ? P4 incremental audio

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141     ?F1; (3) log both to `docs/QUALITY_ITERATIONS.md` I3 (HONEST measured numbers only ?
[[paper-coauthor-aim]]). Honest caveat:
142     NO court polygons ? `players_in_court` counts spectators (3-7 on "single-court"
Mahadevpura); robust signals = motion +
143     `shuttle_player_colocation`. GPU = RTX 2070S, person detection on Windows torch cu124
(NO WSL). Branch protocol held through
144     heavy concurrent merges (#119-127) ? [[gotcha-shared-checkout-branch-collisions]]. On
master, clean.
145     **STORAGE (this session, owner-approved "delete cache + compact WSL"):** deleted the
stale **93GB GX010125 frame cache**
146     (WSL `~/clips` 95?2.3GB; WSL fs used 152?60GB, 896GB free; C: 53?72GB free) ? kept the
2.2GB GX010125 proxy. The WSL
147     `ext4.vhdx` was `--set-sparse true` + `fstrim`'d (946GB trimmed) but this WSL build
(2.3.24) did NOT auto-deallocate on
148     shutdown ? the file is still **~221GB allocated on C:**. **DEFERRED to post-extraction:
the physical ~150GB C: reclaim via
149     `wsl --shutdown` then elevated `diskpart` ? `select vdisk
file="...\CanonicalGroupLimited.Ubuntu_79rhkp1fndgsc\LocalState\ext4.vhdx"`
150     ? `compact vdisk`** (heavy C: I/O would fight the GPU run's video reads). [[gotcha-long-
runs-gpu-wsl]] · [[cache-hygiene-policy]].
151
152     **Checkpoint:** 2026-06-14 (RALLY-DETECTION FRAMEWORK doc + LITERATURE REVIEW + eval-
honesty code ? **PR #119 + #124 MERGED**
153     to master via self-merge on green CI per [[working-agreement-merging]]; owner-directed).
**PR #119** (`docs/ANALYZERS_AND_RECONCILERS.md`):
154     reframed the analyzers/reconcilers design into a **multi-scope signal-fusion framework**
(frame detectors / window analyzers /
155     **session analyzers**); warm-up `SessionAnalyzer` + shuttle-state frame-detector folded
in) + **§13 = a verified multi-agent
156     literature-review** (8 patterns surveyed, ~40 citations adversarially checked, **ZERO
fabrications**; author-line fixes applied).
157     **Verdict (honest):** the skeleton is STANDARD prior art ? candidate-windows+scorer =
Temporal Action Localization;
158     reconciler-as-refinement = Temporal Action Segmentation post-proc; "signal not verdict"
= late/score-level fusion + stacked
159     generalization (~20 yrs old, already in racket sports). **Only the report-first
reconciler-as-linter bundle rated
160     `novel-combination`**; warm-up-as-soft-feature = thinnest gap. **Cite, don't claim**;
terminology realigned (producer?labeling
161     function, scorer?stacked meta-learner, LOVO?leave-one-group-out, tIoU?F1@k/mAP). Ties
[[paper-coauthor-aim]] (§13 is the honest
162     related-work backbone). **PR #124** implements the **Tier-1 measurement-honesty course
corrections C1?C4** as PURE ADDITIVE /
163     default-OFF helpers (no existing compare/LOVO/metrics behaviour changed):
`backend/eval/eval_stats.py` (C1 `mean_with_ci`/
164     `bootstrap_ci`/`paired_sign_test`/`paired_delta_ci` ? never a bare LOVO mean at n?6; C3
`out_of_fold_predictions`/`grouped_folds`
165     ? no stacking leakage), `score_calibration.py` (C2 Platt+isotonic+brier/ECE ? comparable
cue confidences), `segmentation_metrics.py`
166     (C4 F1@{10,25,50} + mAP@tIoU + segment_count_ratio ? make over-segmentation tIoU is
blind to visible). 24 tests; ruff+mypy clean;
167     all 4 CI lanes green (full-suite coverage floor held). Both PRs done in **isolated git
worktrees** (off master) to dodge the
168     recurring shared-checkout branch-flips (siblings merged #117 fusion-P2, #118, #120, #123
concurrently; **#116 closed ? its content
169     had landed via #117**). ?? NEXT (owner-gated): **wire C1?C4 into the DEFAULT harness
reporting** (`compare`/`run_regression.py`
170     emit CIs + F1@k/edit-score instead of a bare mean ? changes reported numbers, so its own
PR) + the Tier-2/3 corrections in §13
171     (global reconciler decode, cue-correlation modeling, temporal smoothing, external
ShuttleSet eval). Analyzers/reconcilers IMPL
172     (A1?A5/R1?R3) still owner-gated, none started.
173
174     **Checkpoint:** 2026-06-14 (RUN.LOG READABILITY for manual QA ? **PR #120 MERGED ?
master `6076eef`**; owner asked ? self-merged

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175 on verified-green CI per [[working-agreement-merging]] (squash; remote+local branch
 deleted); was branch `feat/runlog-readability`
 176 off `35a50f3`; CODE, minimal/config-gated). Owner's manual-QA loop = run the tool on a
 video ?
 177 eyeball the rally CSV against `run.log`. Two readability changes: (1) ****suppress HTTP
 chatter**** ? new central helper
 178 `backend/utils/logging\_setup.py::setup\_logging(verbose\_http=False)` raises
 `httpx/httpcore/urllib3/google_genai/google_genai/
 179 google.auth/grpc` to WARNING (NOTSET to restore), gated by NEW knob
 `logging.verbose\_http: false` (`LoggingConfig` on
 180 `AppConfig`); our `backend.\*` loggers untouched; wired into `run\_process\_and\_stitch.py`
 (at import + re-applied
 181 post-config-load). (2) ****end-of-run rally summary**** ? new pure helper
 `backend/tools/export.py::format\_rally\_summary` (next to
 182 `segments\_to\_csv`/`basic\_stats`): header (total rallies + total play HH:MM:SS) + table
 `# start end dur(s) shot\_count` from the
 183 produced play_segments so log?CSV line up row-for-row; handles metadata dict-or-JSON-
 string, missing shot_count?`-`; HH:MM:SS
 184 via local `\_fmt\_hms` matching the segmenters' `\_fmt\_ts`; logged as ONE block at end of
 `main()` (also on the all-filtered
 185 early-return). Tests: `test\_logging\_setup.py`, `test\_rally\_summary.py`, + `verbose\_http`
 default/override in `test\_config.py`.
 186 Suite ****477 green****, ruff (`.`) + mypy (`backend training`) clean. ? Commit-msg first
 attempt mangled (PowerShell `@'?'@`
 187 here-string in the *Bash* tool ? literal `@`; amended via `-F file`) ? Bash tool ?
 PowerShell. ****BRANCH-COLLISION CLEANUP**
 188 (owner asked): ****after merge, LOCAL checkout left clean ? on**
 `master`==`origin/master`==`6076eef`, 0 ahead/behind; pruned 6 stale
 189 LOCAL branches mapped to MERGED/CLOSED PRs (#92/#104/#111/#106/#102 + #116-closed-
 redundant) + the stale remote-tracking refs;
 190 ****kept**** `docs/analyzers-framework` (#119 OPEN). Working tree = only the intentional
 untracked carryovers (`scratch/`,
 191 `artifacts/`). REMOTE still has 15 prunable branches (14 merged-PR + #116) ? ****offered**
 to prune, owner did NOT answer the
 192 question ? left untouched ****** (must KEEP `feat/khelacharya-shot-intelligence` #11 =
 deliberately-retained sibling layer per
 193 [[sibling-internship-repo]], + #119). ?? NEXT: owner does the machine-side check (run on
 a real video ? eyeball the quieter log +
 194 summary block; dev container can't run the GPU pipeline); remote-branch prune available
 on request.
 195
 196 ****Checkpoint:**** 2026-06-14 (ANALYZERS DOC ? FRAMEWORK + paper framing ? ****PR #119 OPEN,**
 owner-gated, do NOT merge ******; branch
 197 `docs/analyzers-framework` off master `35a50f3`; docs-only). Owner fed two more ideas
 this session and asked to fold them in
 198 and treat the doc as a ****reusable framework / research paper****, not a project patch
 (ties [[paper-coauthor-aim]]). Extended
 199 the on-master `docs/ANALYZERS\_AND\_RECONCILERS.md` (which had landed via #117) from "two
 seams" into a ****multi-scope**
 200 signal-fusion framework ******: producers emit SIGNALS at ****3 temporal scopes**** ? frame
 detectors (`RallyCue`) / window analyzers
 201 / ****session analyzers**** ? all consumed as feature columns by the learned scorer (spine
 unchanged: ****no special-casing****, signal
 202 ?scorer, never `if/else`). Folded in: (1) ****warm-up/practice**** as a new
****SessionAnalyzer**** tier (whole-timeline span ? an
 203 `in\_warmup` membership feature the scorer DOWN-WEIGHTS, NOT a hard `if t<warmup\_end`:
 drop; measurable on golden TODAY since
 204 warm-up is already labeled negative); (2) ****shuttle-state****
 (`in\_air`/`racket\_contact`/`inactive`) placed correctly as a
 205 ****frame-scope detector**** (a MULTI_SIGNAL_FUSION_PLAN `RallyCue`, NOT a window analyzer)
 ? derive-from-existing-signals first
 206 (velocity+visibility+direction-reversals+audio thwack), dedicated per-frame model ONLY
 where WASB trajectory gaps bite, with
 207 the per-frame-label blocker flagged. Added ****\$10 generality & research positioning****
 (weakly-supervised temporal segmentation;

208 standard-vs-contributed) with an explicit ****attribution honesty caveat****: NO fabricated
 citations/results, related-work +
 209 ablations + >1-sport still owed, n=6 = direction-check only ? the honest backbone the
 paper aim needs. Diff = `2-seam?framework`
 210 (310+/137?). ? Shared checkout had flipped to master mid-task (the sibling's collision
 lesson recurred) ? re-verified branch,
 211 branched fresh, normal commit (no in-flight WIP left to isolate; #117/#118 already
 merged so all cross-links resolve). PR #116
 212 stays CLOSED (content on master via #117). ?? NEXT (owner-gated): review #119; impl
 tracks unchanged (fusion P3 LOVO, then
 213 analyzers/reconcilers A1?A5 / R1?R3 per the doc's phasing ? each ONE PR, default OFF,
 promote on measured LOVO lift).
 214
 215 ****Checkpoint:**** 2026-06-14 (FUSION P2 SHIPPED + collision cleanup + paper aim ? master
 `35a50f3`; suite 467 green; 0 open PRs).
 216 ****PR #117 MERGED = `FusionSegmenter`**** (fusion P2, registry `fusion\_hybrid`, ****default-
 OFF****): shuttle-primary (windows seed
 217 from WASB/TrackNet) + adjudicating cues ? `net\_crossings` (Gate A, always persisted,
 GPU-free via rally_gate w/ a reused
 218 axis estimate) + optional person cue (Gate B) persisting the ~5 `fusion\_features.py`
 features (pure-tested) via
 219 `PersonDetector.detections\_per\_frame` over each window's ORIGINAL-video frame range.
 Plugs in via ****two IDENTITY hooks**** on
 220 `trajectory\_hybrid` (`\_enrich\_candidate\_stats`/`\_select\_for\_handoff`) ? wasb/tracknet
 byte-identical (adversarially verified).
 221 Served gating optional+default-OFF (`fusion.min\_crossings`/`min\_players`) ? gates the
 handoff while persisting the full
 222 SUPERSET (offline substrate intact). Court mask = optional `fusion.court\_polygon`.
****Design-first**** (`fusion-p2-design`
 223 workflow) ? ****31-agent adversarial review**** (25 findings?19 confirmed; headline = NO-
 REGRESSION verified) ? fixes folded
 224 (frame-dim guards, round() frame idx, rally_gate `estimate=` reuse, honest torch
 docstrings, +6 tests). ****PR #118 MERGED**** =
 225 README index fix (QUALITY_ITERATIONS + EXPERIMENT_HARNESS P1-SHIPPED). ****? SHARED-
 CHECKOUT COLLISION (lesson):**** the
 226 spawn_task DESIGN session ran in THIS checkout, committed `0f673ff` (the analyzers
 design doc) + flipped the branch BETWEEN
 227 my branch-point check and my `checkout -b` ? my fusion branch rode on top ? ****#117's
 squash absorbed the design doc****, so
 228 ****`docs/ANALYZERS\_AND\_RECONCILERS.md` is on master via #117**** and ****PR #116 was CLOSED
 as redundant**** (content intact on
 229 master; #116's stale "OPEN do-not-merge" checkpoint below is superseded). Re-verify `git
 branch` AND that no sibling commit
 230 landed since, before `checkout -b`. ****Owner side-aim ratified: co-author a paper****
 "local, cheap & efficient rally detection
 231 via multiple clever ideas" ? [[paper-coauthor-aim]]; `docs/QUALITY\_ITERATIONS.md` is the
 ablation-ready empirical backbone
 232 (honest measured results only). ****Analyzers+reconcilers DESIGN on master (owner-gated,
 NOT started):**** injectable per-window
 233 analyzers (e.g. net-hit service-fault) + a report-first reconciliation pass; spine =
 signals?learned-scorer, ****no**
 234 special-casing** (owner directive). ?? NEXT (owner-gated): ****fusion P3**** (person
 features ? `experiment.compare` LOVO lift on
 235 the 6 golden videos ? the eager-person-compute cleanup lands here) + measure the Gate-A
 held-shuttle drop on GPU; then
 236 analyzers/reconcilers impl; P4 audio. Earlier this session: PR #114 harness, #115
 person-cue extraction (see checkpoints below).
 237
 238 ****Checkpoint:**** 2026-06-14 (DESIGN DOC: two extensibility seams ? ****PR #116 OPEN, owner-
 gated, do NOT merge****; docs-only,
 239 branch `docs/analyzers-and-reconcilers` off master `df7d5df`). Produced
 `docs/ANALYZERS\_AND\_RECONCILERS.md` (367 lines, a
 240 peer to EXPERIMENT_HARNESS + MULTI_SIGNAL_FUSION_PLAN) + a 1-line `docs/README.md`
 cross-link. ****Spine (owner directive):**
 241 NO special-casing in the decision path** ? every analyzer/reconciler emits a SIGNAL

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(value+confidence); the SCORING MODEL
242     learns to weight it (already true in miniature: `experiment.predict`/_gate_mask` treat
`net_crossings`/_players_in_court`
243     as feature columns). **Seam 1 = injectable `WindowAnalyzer`s (forward)**: a registry
(mirrors `segmenter_registry`) of pure
244     per-window cues over the already-produced per-frame-aligned signals (WASB trajectory +
`PersonDetector.detections_per_frame`
245     + `rally_gate` net); each emits an `AnalyzerResult` = confidence-scored features ?
`candidate_segments.stats`, events ?
246     `events` table (buffer-on-candidate, flush-on-confirm), optional boundary hints; worked
example = a **service-fault**
247     analyzer (shuttle halts in net region ? `service_fault` signal + `net_hit` event + END
hint); injects at the FusionSegmenter
248     `_enrich_candidate_stats` hook, flag-gated default OFF. **Seam 2 = reconciliation pass
(backward)**: a report-first "linter
249     for rally decisions" ? `ConsistencyRule`s (trailing-dead-time?truncate *(owner's misfire
fix)*, no-net-crossing?drop,
250     over-merged?split, boundary-slack?tighten) flag decision-vs-evidence conflicts;
**report-first/apply-optional, NEVER silent**;
251     expressible as a harness `Variant` transform so LOVO-measurable on the 6 golden videos
BEFORE auto-apply; idempotent, fixed
252     precedence. ? **DESIGNED AGAINST the IN-FLIGHT fusion-P2 WIP** (uncommitted in the
shared tree: `fusion_hybrid.py`,
253     `fusion_features.py`, `FusionConfig.cue_flags`, `person_detector.detections_per_frame`,
`docs/QUALITY_ITERATIONS.md`) ? that
254     WIP was kept **byte-intact** (I isolated my single README line via git plumbing; tree is
CRLF / autocrlf=true, restored
255     owner README to exact session-start bytes 3841B). **Merge-order:** the doc cross-links
QUALITY_ITERATIONS.md + the fusion
256     hook, neither on master yet ? **merge #116 after (or with) fusion-P2** (noted in the PR
body). ?? NEXT: owner reviews #116;
257     implementation is the phased A1?A3 / R1?R3 plan in the doc (each ONE PR off master,
default OFF, promoted only on measured
258     LOVO lift) ? all owner-gated, none started. Ties [[candidate-segments-finetuning]] + the
harness ([[working-agreement-merging]]).
259
260     **Checkpoint:** 2026-06-13 (RALLY-QUALITY TRACK: HARNESS + PERSON-CUE EXTRACTION SHIPPED
? **PR #114 + #115 MERGED**,
261     master `df7d5df`; suite 421?441 green; 0 open PRs). **PR #115 (fusion P1):** extracted
the torchvision person detector
262     + ByteTrack + velocity windowing out of `yolo_hybrid` into
`backend/pipeline/detectors/person_detector.py`
263     (`PersonDetector`, a reusable cue producer); `yolo_hybrid` now ORCHESTRATES only
(chunk/prefetch/AI-handoff/DB) and
264     delegates Stage-1 to `self.person`. **No behavior change** ? proven by the unchanged
`process_video` output tests (mock
265     seam moved to `self.person`); pure detection unit tests ? new
`tests/test_person_detector.py`. person_detector is
266     mypy-CLEAN (`self.model: Any`) so it needed NO quarantine; the torchvision code that put
yolo_hybrid in quarantine left
267     it, but yolo_hybrid STAYS quarantined for PRE-EXISTING orchestration debt (implicit-
Optional process_video defaults,
268     Success|Failure ABC-override mismatch, a Future[...] annotation ? flagged in mypy.ini to
ratchet separately). ruff+mypy
269     clean, CI 4/4. Docs synced (fusion-plan P1?DONE, CODE_MAP person_detector entry +
yolo_hybrid-as-orchestration, NEXT_STEPS
270     item 2 ticked). ?? **NEXT (owner-steered):** **PR 3 = fusion P2.** The cheap, highest-
confidence, PURE-LOGIC piece =
271     **wire Gate A (`min_crossings`) into the live trajectory path**
(`trajectory_hybrid`/_wasb_hybrid`) + add the knob to
272     `IndexingConfig` ? *fixes the owner's held-shuttle false positive*, default OFF,
immediately A/B-able via the harness on
273     the 6 golden trajectories. Gate B (court-mask + player-count) is **owner-gated** on the
court-mask-source decision
274     (MULTI_SIGNAL_FUSION_PLAN §9.1: manual polygon vs auto homography ? owner). **P3 learned

```

```

fusion + the SxS-on-new-videos
275     RUN need GPU/owner-side** (persist person features on real video ? train on the 6 golden
? `compare_unlabeled(CONTROL,
276     +person)` on new footage). Two PRs this session within [[working-agreement-merging]]
self-merge authority.
277
278     **Checkpoint:** 2026-06-13 (EXPERIMENT HARNESS P1 SHIPPED ? **PR #114 MERGED**, master
`ecee786`; suite 421?440 green).
279     Picked up the rally-quality track (the desktop-agent pick-up `NEXT_STEPS` flagged; both
design docs `EXPERIMENT_HARNESS.md`
280     + `MULTI_SIGNAL_FUSION_PLAN.md` arrived via the session-start `git pull`
15e4e0e?9231af2, PR #112). Built
281     `backend/eval/experiment.py` = the thin variant/experiment layer from
`EXPERIMENT_HARNESS.md` P1 (the audit's "-80%
282     already exists" HELD ? verified file:line; pure glue over
metrics/calibration/classifier/training/regression.gt_hash/
283     gemini_refine.merge_overlaps/query_candidate_segments, **NO new scoring math**):
`Variant` (JSON recipe:
284     segmenter+config_overrides+cue_flags+Gate-A `min_crossings`+Gate-B `min_players`+learned
classifier; `spec_hash`; pinned
285     `CONTROL`), `compare` (labeled A/B, **LOVO-honest** train-on-others for learned
variants, B?A deltas + `label_set_hash`),
286     **`compare_unlabeled`** (the owner's **SxS-on-NEW-videos** ask ? agreed/a_only/b_only
via match_intervals, no GT),
287     `record_experiment` (? `eval_baselines/experiment_leaderboard.json`),
`load_video_candidates` (DB bridge). 19 tests incl.
288     the load-bearing **no-regression invariant** (all cues OFF ? byte-identical to CONTROL);
ruff+mypy clean; CI 4/4. Docs
289     synced (EXPERIMENT_HARNESS P1?DONE, CODE_MAP entry + stale regression-baseline-path fix,
NEXT_STEPS item 3 ticked). Owner
290     answered ramp-up Qs: **corpus = all 6** golden videos (he said "5";
LargeTestVideo/TestLargeVideo may be cuts of one match
291     ? kept all 6 per the manifest); **merge authority = self-merge on green then continue to
PR 2** (within [[working-agreement-merging]]).
292     ?? NEXT (this session): **PR 2 = fusion P1** person/vision-cue extraction (lift
torchvision+ByteTrack+velocity out of
293     `yolo_hybrid` ? `backend/pipeline/detectors/person_detector.py` producer; yolo_hybrid
consumes it, no behavior change ?
294     regression-safe). Then P3 learned fusion trains on the 6 golden videos +
`compare_unlabeled(CONTROL,+person)` SxS on new
295     footage (GPU/owner-side). Tracks [[candidate-segments-finetuning]] + [[collector-
indexer-bridge]].
296
297     **Checkpoint:** 2026-06-13 (WORKING-TREE CLEANUP ? master `15e4e0e`, tree now CLEAN).
Cleaned up the leftover
298     uncommitted bits (the cache-hygiene CODE was already MERGED as #109/#110 per the
checkpoint below ? nothing to rescue
299     there): (1) `config.json` GPU-run knobs **reverted to committed defaults** (was
wasb_hybrid/7200/8 ? gemini/1800/1;
300     values stay recorded in the cache-hygiene checkpoint below ? re-apply
`default_segmenter: wasb_hybrid` in one line if
301     that's the intended new default; user didn't answer keep-vs-revert, took the safe
recoverable path). (2) stray root
302     `arial.ttf` (unused ? shipped watermark font is committed
`backend/assets/fonts/RobotoSlab-Regular.ttf` from #103) ?
303     moved to `scratch/`. (3) genuine untracked draft `docs/STORAGE_SHARING_MODEL.md`
(190-line storage-anchored
304     monetization model, owner co-design pending) ? branched + **PR #111 OPEN (owner-gated,
docs-only ? do NOT auto-merge**);
305     supersedes COMMERCIALIZATION.md framing if ratified ? ADR then). Master tree now shows
ONLY intentional carryovers
306     (untracked `scratch/` handoffs+repro scripts, `artifacts/` Gen-0 bundle) ? could be
gitignored for a pristine
307     `git status` if desired (offered, not done). Reminder: `python -m tools.cleanup_caches
--apply` reclaims the 96.5 GB

```

```

308     the SessionStart hook flagged.
309
310     **Checkpoint:** 2026-06-13 (CACHE HYGIENE shipped ? **PR #109 + docs #110 MERGED**,
master `15e4e0e`; + the ?5 reel +
311     manual cleanup). **#110** = diligence docs follow-up (README §Cache Hygiene + knobs +
tools/ line; CODE_MAP detector
312     self-clean, tools/cleanup_caches.py entry, common-task rows; min_shot_count API-parity
note) ? docs-only, CI 4/4 green.
313     Both repos now have **0 open PRs**. After the GX010125 E2E run left ~163 GB in WSL
`~/clips`, built a full cache-hygiene system (all on master):
314     (1) **self-cleaning runners** ? `indexing.{wasb,tracknet}.keep_frames` (default false)
deletes the frame cache +
315     staged video after a SUCCESSFUL run; on failure keeps them for resume; (2)
**tools/cleanup_caches.py** dry-run-first
316     ops tool (`--apply`/`--keep-video`/`--older-than`/`--include-wasbcache`/`--summary`);
(3) **disk guard** `min_free_gb`;
317     (4) **SessionStart nudge hook** in `.claude/settings.local.json` (local, gitignored).
Suite 409?**421 green**, ruff+mypy
318     clean. Policy saved ? [[cache-hygiene-policy]]. ? WSL-scan footgun: the owner's login
shell mangles `for e in ` loops
319     (loop var empty) ? use du/find arg-globbing. Also this session (earlier): merged
**#107** (API min_shot_count) + **#108**
320     (watermark Windows colon fix); ran GX010125 E2E on GPU ? **GX010125_highlights.mp4** (11
rallies >7 shots) +
321     **GX010125_highlights_min5.mp4** (24 rallies ?5, watermark now works ? validates #108)
via a stitch-only reuse script
322     (no re-inference); confirmed **Gemini got 1080p** (4K only at final stitch); delivered
an execution flowchart. Manual
323     WSL cleanup freed ~68 GB (kept GX010125's 93 GB frames + proxy per owner choice). ?
LOCAL uncommitted `config.json`:
324     default_segmenter=wasb_hybrid / min_shot_count=8 / wasb.timeout_sec=7200 from the run.
See [[gotcha-long-runs-gpu-wsl]].
325
326     **Checkpoint:** 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-
up session; no new code from
327     me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**.
Reconciled remote vs my last
328     session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh):
**#101** configurable branding
329     watermark . **#102** thread config stitching paddings into live VideoCompiler sites .
**#103** bundle Apache-2.0
330     default watermark font + fallback log . **#104** C13 field kit (flyer/playbook/answer-
sheet/signals) . **#105** i18n
331     design plan (Kannada-first) . **#107** enforce `stitching.min_shot_count` on
/api/compile . **#108** escape Windows
332     drive-letter colon in ffmpeg filtergraph (fixes #103 watermark on Windows). **OPEN:
#106** docs/video-locality-model
333     (awaiting owner). ? **UNCOMMITTED WIP IN TREE (sibling/owner ? DO NOT CLOBBER):** a WSL
**cache-hygiene** slice in
334     progress, built on my `wslCommandMixin` ? `base.py` (`_wsl_free_gb`/`_wsl_rm_rf`),
`wasb_runner.py`+`tracknet_runner.py`
335     (`keep_frames`/`min_free_gb` config + pre-run disk guard + post-success cache delete),
`models.py` (new fields),
336     `config.json` (the GPU-run knobs: default_segmenter=wasb_hybrid, min_shot_count=8,
wasb.timeout_sec=7200), untracked
337     `arial.ttf` (watermark font) + `docs/STORAGE_SHARING_MODEL.md`. Not yet PR'd by whoever
owns it. [[session-handoff]]
338     rewritten to this state.
339
340     **Checkpoint:** 2026-06-13 (FIRST FULL E2E GPU RUN on a real 4K GoPro match + 2 merged
fixes).
341     Owner asked to run the indexer E2E on the GPU over `F:\GoPro Hero Black
12\KhelSutra\Badminton 12
342     June 2026\GX010125.MP4` (12.8-min 4K60, 11.5GB), highlights for rallies with >7 shots,
config-driven.

```

343 - **DELIVERED:** `output\GX010125\_highlights.mp4` ? **11 rallies (shot_count 8?16) of 52 detected**;

344 true 4K 3840x2160 h264 59.94fps, 2:23, 1.15GB; stitched from the 4K ORIGINAL (proxy only for indexing).

345 wasb_hybrid: 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed rallies. min_shot_count=8

346 correctly excluded a 7-shot rally (">7" semantics). Gemini ran clean (no 503s); ~47-min wall clock.

347 - **2 PRs merged on green CI** (owner's [[working-agreement-merging]] standing authorization):

348 **#107** `/api/compile` now enforces `stitching.min\_shot\_count` (parity w/ run_process_and_stitch);

349 **#108** sticher `\_ff\_quote` escapes the Windows drive-letter colon ? fixes the on-by-default

350 bundled-font watermark (PR #103) silently failing on Windows (`fontfile='C:/...'` broke ffmpeg

351 drawtext parsing; also fixed absolute logo `movie=` paths). Both with regression tests; suite 409 green.

352 - **The hard-won run recipe is in [[gotcha-long-runs-gpu-wsl]]:** Modern Standby killed the first 2

353 overnight attempts (keep-awake fix); cv2 PNG extraction (~38min) is the real bottleneck not the GPU

354 (ffmpeg pre-extract, count-exact for CFR ? skips it); 1080p-proxy-index / 4K-stitch; per-phase

355 observability (cache manifest for WASB, handoff dir for Gemini). Reusable scripts in `scratch/`

356 (run_gx010125.py, keep_awake.ps1, probe*.py).

357 - ? LOCAL UNCOMMITTED `config.json`: default_segmenter=wasb_hybrid, min_shot_count=8, wasb.timeout_sec=7200

358 (set for this run; revert to gemini/1/1800 if not wanted). Watermark fix NOT applied to the delivered

359 reel (it ran before #108); a re-stitch would hit the DB segment cache + now-working watermark (~5min).

360

361 **Checkpoint:** 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; **PR #104 updated `9c84fea` +

362 PR #106 OPEN `85abe26`, both CI green, both AWAITING OWNER**). Owner spec: (1) remove ALL associate-pay from the

363 field kit ("something else in mind") ? flyer/playbook now "Engagement terms: discussed individually"; honesty

364 mechanics reworded ("judged on honest capture, not positive answers"); KEPT: WTP research questions (study subject)

365 + the ACADEMY's Phase-2 session fee (different party ? flagged, owner may strip too). (2) Storage/upload questions

366 answered from code: uploads ARE stored in full (`security.upload\_dir`, NO retention policy ? BACKLOG); 10GB CANNOT

367 upload (max_upload_mb=4096 rejects at init; ?90MB sequential parts; hours on Indian uplinks). (3) NEW

368 `docs/VIDEO\_LOCALITY\_MODEL.md` (#106): central insight = timestamps are the product, video is dead weight (only the

369 compiler touches original bytes, cuttable where the original lives); locality ladder L0?L4 (L2 proxy-upload reuses

370 the collector ADR-002 frame-identical contract; L1-direct = client?Gemini, zero bytes through us; L0 = owned-model

371 north star; pilot = L4 sneakernet); **frugality frame (owner): bills affordable iff GPU doesn't burn ? L2/L1 serving

372 is GPU-less by construction, Gemini €/hr, GPU stays on the owned 2070S, per-video cost ceiling required**;

373 **academy-visit dividend: Phase-2 consented recordings ARE the multi-court target corpus, show-back = live L2 dress

374 rehearsal, demo reactions = build backlog**; 10 open questions OQ1?OQ10 (retention, MD5 identity, Gemini consent,

375 key custody, packaging, review source, reel delivery, uplink data, volume, L3 survival); rec lean = L2 default

376 (NOT a greenlight). NB: #106's base commit was pushed by owner/sibling mid-session

```

(shared checkout). ?? owner:
377     review/merge #104 + #106; decide academy-fee keep/strip; the "something else" comp
structure is his to define.
378
379     **Checkpoint:** 2026-06-12 (latest ? C13 KIT REFRAMED PRODUCT-FEEDBACK-FIRST per owner
spec + **RECRUITING FLYER
380     added; PR #104 UPDATED `985a329`, CI 4/4 green, AWAITING OWNER REVIEW** ? do NOT merge
without him). Owner spec:
381     hire someone to visit academies, talk to coaches AND students (how does it help +
improvement feedback); stretch =
382     record ? pipeline ? show the academy ITS OWN sample result; UI/hosting "ready by then".
Kit now = 4 docs by audience:
383     **`PRODUCT_FIELD_ASSOCIATE_FLYER.md`** (NEW ? candidate-facing recruiting flyer, owner
fills ?/phone + recruits) .
384     **`FIELD_VISIT_PLAYBOOK.md`** (RENAMED from FIELD_RESEARCH_ASSOCIATE_FLYER ? day-one
playbook: demo-after-Q6 rule,
385     adult-18+-students-only w/ coach-points-out gating, Phase-2 show-back + scripted "can we
keep using it?" no-promise
386     reply) . answer sheet (+demo-reaction/student blocks) . PMF_FIELD_SIGNALS (verdict
machinery UNCHANGED; demo/student
387     data = product feedback only). 2nd adversarial critique (`w5aexakz2`, 30 agents): 12
confirmed+fixed (key honesty
388     fixes: "web app IS opening"? "we're building?will be able to try"; "match report"? "simple
summary"; "finds EVERY
389     rally" dropped; minors loophole in "senior/adult students" closed), 15 refuted. ??
owner: review #104 (thresholds
390     $3 + flyer copy) ? merge ? fill ?/phone ? trim 60s demo reel ? recruit.
391     Owner ratified in-chat: **PMF validation = physically visiting Bangalore academies and
reading signals.** Built the
392     decision layer on top of the existing `docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md` (which IS
the "recruit hand-out" the
393     owner remembered; the khelacharya PDFs in OneDrive are the separate internship flyer ?
see
394     [[sibling-internship-repo]]): **`docs/PMF_FIELD_SIGNALS.md`** (H1-H7 ? script-question
map, G/A/R anchors, H5
395     counting rule = pre-ladder-or-budget-converted ??300 only ? anchored ?300 NEVER counts,
pinned stopping rule n=15
396     target/logistics-only/verdict-once, numeric verdicts PROCEED / PIVOT A record+index /
PIVOT B partner-API
397     (PB VisionxPodPlay footnote now in-repo) / PIVOT C parent-payer / FALSIFIED) +
**`docs/FIELD_VISIT_ANSWER_SHEET.md`**
398     (printable per-visit form: verbatim answers, required most-negative quote, before/after-
ladder tick, OFFICE-USE
399     score grid) + flyer Q5 rewritten open-ended-first (old wording read the ?100/300/500
ladder ALOUD ? center-bias
400     manufactures the exact ?300 PROCEED threshold; caught by the 38-agent adversarial
critique `ww0rvaqf3`: 1 high/2 med/
401     2 low confirmed+fixed, 9 refuted) + Stupa added to listen-for + NEXT_STEPS item 5 = kit
COMPLETE. ?? owner: review
402     #104 thresholds ? merge ? fill flyer ?/phone placeholders ? recruit the associate ?
first 2 visits paired. PDFs of
403     flyer+answer-sheet: generate AFTER placeholders are filled (offer stands).
404
405     **Checkpoint:** 2026-06-12 (latest ? R00RA VENDOR LOOP COMPLETE: **collector #23+#24+#25
MERGED** sequentially on
406     green CI, owner pre-authorized; master `ca61405`, 0 open PRs, only owner's `.gitignore`
edit uncommitted).
407     **#23 boundary_grid_snap BLOCKER** in detect_issues (gcd-of-boundary-frame-diffs > ±0.3s
bar ? gate BLOCKED; skips
408     CSV-path/no-fps/<3 rallies) ? ? **replayed against the REAL 6-June PB+TT pilot zips:
both auto-block with "All 12
409     rally boundaries sit on a 30-frame grid (~0.50s @ 59.94fps)"** (the manual forensics,
now automated) + R00RA_BRIEF
410     Phase-B revision ($7 REQUIRED method: CVAT step 1, first/last box on exact contact/dead
frame, sparse interior OK;

```

411 \$4 grid-snap auto-detected warning; \$2 skip-rallies-in-progress-at-file-edges; \$1
annotation-optimized-copy note).

412 ****#24 prep-annotation-batch****: mirrors `//` ? proxies-
only tree w/ same layout

413 (thin orchestrator over the hardened single flow; per-video isolation; resumable "have"
skips; live 2-court E2E

414 smoke green). ****#25 ROORA_RUNBOOK.md**** (docs/annotation/): operator E2E ? send checklist
(layout?batch?manifest

415 commit?intake+email w/ the 2 non-negotiables), receive (gate decision table incl. grid-
snap, review sheet,

416 import/export on ORIGINALS), troubleshooting table for every guard. Suite ****180 green****.
?? Vendor loop is now fully

417 tooled+documented. ?? AGREED NEXT (owner, session wrap 2026-06-12): (1) owner fills the
canonical doc's TODO(avidullu)

418 markers + screenshots, labels it ****v8****; (2) ****joint TEST RUN** of the full vendor loop on
real footage BEFORE

419 anything goes to Roora** (runbook \$1 send-side end-to-end: batch layout ? prep-
annotation-batch ? manifest ?

420 mock-receive via validate-delivery on an old delivery); (3) only then send. ****v7 ANOMALY
RESOLVED (2026-06-12):****

421 the canonical vendor-facing Brief+Guide is the Google Doc at
docs.google.com/document/d/1NvAXE0q7t4Z0QddLkEfnta0DU982yDZaqkGNPMe8EY0

422 (provenance: generated from master `ca61405`; supersedes Drive v3/v5/v6/v7; owner
confirmed superset-of-v7) ? see

423 [[roora-guidelines-drive-doc]]; before sending: pick ONE version label + fill its
TODO(avidullu) markers/screenshots.

424

425 ****Checkpoint:**** 2026-06-12 (earlier ? ADR-002 IMPLEMENTED: ****collector #22 MERGED****
`a068fd1`, owner authorized

426 merge-on-green; #21 ADR + #20 + indexer #102/#103 all merged earlier today, slate was
clean). Shipped: `curate

427 prep-annotation-proxy` + `src/core/proxy.py` ? ?1080p/-g 15`/CRF 19/audio-kept re-
encode w/ `fps\_mode passthrough`

428 (NEVER `-r`), post-encode parity verify (exact frame count; fps $\pm 0.1\%$; duration like-
with-like stream/container),

429 delete-on-ANY-failure (incl. Ctrl-C), VFR refusal w/o `--allow-vfr`, provenance ?

430 `data/annotation\_proxy\_manifest.csv` (source+proxy MD5s; video_id collision guard BEFORE
encode; Excel-lock prints

431 row + keeps proxy), same-file guard normcase/realpath/samefile (NTFS case-variant
`--out`+`--force` would have let

432 `ffmpeg -y` TRUNCATE the source ? review verifier reproduced it live pre-fix). Suite
****169 green**** (+32), CI green.

433 Verified live on ffmpeg 8.1.1 (smoke clip 59.94fps: parity 179 frames both files; smoke
caught `vsync` deprecation ?

434 `fps\_mode`). Process: ultracode adversarial review workflow (`wyv8524zf`, 18 agents)
confirmed 1 high/2 med/7 low ?

435 all fixed pre-merge. Docs: INGESTION_PIPELINE step 0 + diagram leg + onboarding FAQ +
sampling-FAQ split

436 (highlights-vs-golden), ADR-002 provenance bullet as-implemented amendment,
PIPELINE_ROLE ADR refs qualified.

437 ?? REMAINING ADR-002 follow-ups (separate items, NOT started): Phase-B brief revision
(frame step 1 / exact boundary

438 frames / "we supply proxies") ? owner words vendor comms; grid-snap QA check in
validate-delivery. First real use:

439 run prep-annotation-proxy on the next Roora batch before handoff.

440

441 ****Checkpoint:**** 2026-06-12 (earlier ? ANNOTATION-PROXY DECISION RATIFIED ? ****collector
PR #21 OPEN****, docs-only

442 ****ADR-002**** in collector `docs/DECISIONS.md`). Owner agreed in-chat w/ my rec from the
assessment checkpoint below:

443 proxies generated in-repo via a future `curate prep-annotation-proxy`. ADR records the
proxy contract (SAME fps ?

444 never reduce; no trim; frame-count parity check; 1080p/CRF18-20/-g15/audio kept; intake-
manifest row w/ original+proxy

445 MD5s), the why-collector rationale (owns the vendor flow = command #0 of convert-

```

cvat?validate-delivery?import-local;
446     verification+provenance is the durable 80%, one-off scripts skip it), and the **Phase-B
necessity argument the owner
447     wanted captured: ~12-15 videos/sport × ~30 min 4K/59.94 = >100k frames/file ? vendor
frame-step-1 annotation
448     infeasible without proxies ? near-precondition, not optimization**. Follow-ups recorded
IN the ADR: Phase-B brief
449     convention line (step=1, first/last box on exact contact/dead frame) + re-add grid-snap
QA to validate-delivery.
450     Collector checkout back on master (owner's stray `.gitignore` edit untouched). ? sibling
PR #20 (below) edits
451     rally_cvat.py docstring rationale 0.5/1.0?0.5/0.5 ? no file overlap w/ #21, both docs-
true. ?? owner reviews/merges

```

6. assistant (2026-06-14T10:31:16.622Z)

This is the owner's real GPU machine, and the task is well-scoped. Let me orient on the tooling and environment before doing anything. Let me read the key files and check what's cached.

7. user (2026-06-14T10:31:17.593Z)

```

1     """Run the framework on GX010125_1080p by REUSING its cached WASB trajectory ? offline,
2     no GPU, no 4K, isolated outputs (output/GX010125_run/), zero pollution of past work.
3
4     This is the "persist detection once, re-score forever" path on a REAL, unlabeled match:
5     parse the cached 1080p trajectory -> candidate rally windows -> CONTROL vs Gate-A
6     (net-crossings >= 2, the measured precision win) side-by-side (no golden labels exist
for
7     this footage, so we report the divergence, not F1).
8     """
9     import json
10    import os
11
12    from backend.eval.distill_local import build_candidates, FEATURE_NAMES
13    from backend.eval.experiment import CONTROL, Variant, VideoCandidates,
compare_unlabeled, predict
14
15    TRAJ = "output/wasb/GX010125_1080p__89605e80ed01_wasb.csv"
16    FPS, FRAME_W = 59.94, 1920.0
17    OUT_DIR = "output/GX010125_run"
18
19
20    def _dur(ivs):
21        return sum(e - s for s, e in ivs)
22
23
24    def _write_csv(path, ivs):
25        with open(path, "w", encoding="utf-8") as f:
26            f.write("start_sec,end_sec,duration_sec\n")
27            for s, e in ivs:
28                f.write(f"{s:.3f},{e:.3f},{e - s:.3f}\n")
29
30
31    def main():
32        os.makedirs(OUT_DIR, exist_ok=True)
33        intervals, X = build_candidates(TRAJ, FPS, FRAME_W)
34        feats = {FEATURE_NAMES[k]: [r[k] for r in X] for k in range(len(FEATURE_NAMES))}
35        vc = VideoCandidates(name="GX010125_1080p", intervals=intervals, features=feats,
gts=None)
36
37        gateA = Variant(name="gateA", min_crossings=2)
38        ctrl_preds = predict(CONTROL, vc)
39        gate_preds = predict(gateA, vc)
40        sxs = compare_unlabeled(vc, CONTROL, gateA)    # agreed / a_only (dropped by Gate-A)
/ b_only

```

```

41
42     print("=" * 70)
43     print(f"GX010125_1080p ? offline re-score of the CACHED trajectory (no GPU)")
44     print("=" * 70)
45     print(f"   candidate windows from trajectory : {len(intervals)}")
46     print(f"   CONTROL rallies (merge all)           : {len(ctrl_preds)}  ({_dur(ctrl_preds)
/ 60:.1f} min)")
47     print(f"   Gate-A rallies (net_crossings>=2) : {len(gate_preds)}  ({_dur(gate_preds)
/ 60:.1f} min)")
48     print(f"   Gate-A dropped (a_only)                : {len(sxs['a_only'])} windows "
49           f"({_dur(sxs['a_only']) / 60:.1f} min of likely held-shuttle/feed FPS)")
50     print(f"   both agree (kept)                        : {sxs['num_agreed']}")
51
52     _write_csv(os.path.join(OUT_DIR, "GX010125_control_rallies.csv"), ctrl_preds)
53     _write_csv(os.path.join(OUT_DIR, "GX010125_gateA_rallies.csv"), gate_preds)
54     summary = {
55         "video": "GX010125_1080p", "source": "cached WASB trajectory (offline re-score,
no GPU)",
56         "fps": FPS, "num_candidates": len(intervals),
57         "control": {"num_rallies": len(ctrl_preds), "total_min": round(_dur(ctrl_preds)
/ 60, 2)},
58         "gateA": {"num_rallies": len(gate_preds), "total_min": round(_dur(gate_preds) /
60, 2)},
59         "dropped_windows": len(sxs["a_only"])},
60     }
61     json.dump(summary, open(os.path.join(OUT_DIR, "GX010125_rescore_summary.json"),
"w"), indent=2)
62     print(f"\n wrote isolated outputs -> {OUT_DIR}/ (control + gateA rally CSVs +
summary)")
63     print(" NOTE: no golden labels for this footage -> divergence reported, not F1. A
full")
64     print(" Gemini-confirmed highlight reel is a separate ~tens-of-min GPU+API run.")
65
66
67     if __name__ == "__main__":
68         main()
69

```

8. user (2026-06-14T10:31:18.042Z)

```

1     """Distill Gemini (teacher) into a LEARNED local rally classifier (student).
2
3     Threshold-tuning the local knobs hit a low ceiling (see calibrate_local). This goes
4     further: learn "is this WASB candidate a real rally?" from Gemini silver labels,
5     using a few resolution/fps-INVARIANT local features, so the model transfers across
6     videos and runs 100% local at inference (no Gemini, privacy lever intact).
7
8     Flow (all local at runtime):
9     WASB trajectory -> recall-oriented candidate windows (propose many; classifier
10    filters) -> per-candidate features [duration, peak_velocity, mean_velocity,
11    active_ratio, net_crossings] -> LogisticRegression(prob >= threshold) -> merge
12    -> rally windows.
13
14    Training labels come from Gemini full-context labels via the SAME IoU matching the
15    evaluator uses (training.label_candidates). Honest leave-one-video-out: train on the
16    other videos' candidates, predict the held-out one, score vs its Gemini labels, and
17    compare against the threshold baseline. Reuses classifier.py, training.label_candidates,
18    rally_gate, metrics, calibrate_wasb.load_golden_gts, gemini_refine.merge_overlaps.
19
20    Run:
21    python -m backend.eval.distill_local --manifest videos.json --model-out
output/local_rally_model.json
22    """
23    from __future__ import annotations
24

```

```

25     import json
26     import argparse
27     from typing import Dict, List, Optional, Tuple
28
29
30     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
31     from backend.pipeline.detectors import rally_gate
32     from backend.eval.calibrate_wasb import load_golden_gts
33     from backend.eval.training import label_candidates
34     from backend.eval.classifier import LogisticRegression
35     from backend.eval.metrics import score
36     from backend.eval.gemini_refine import merge_overlaps
37
38     Interval = Tuple[float, float]
39
40     FEATURE_NAMES = ["duration", "peak_velocity", "mean_velocity", "active_ratio",
41 "net_crossings"]
42     # Recall-oriented proposal: deliberately permissive so real rallies are among the
43     # candidates (the classifier removes the junk). (velocity_thresh, merge_gap, min_dur)
44     PROPOSAL = (0.005, 1.0, 0.5)
45     BASELINE_LOCAL = (0.02, 2.0, 2.0) # today's production-ish windowing, no learning
46
47     def build_candidates(trajecory_csv: str, fps: float, frame_width: float,
48         proposal: Tuple[float, float, float] = PROPOSAL) ->
49 Tuple[List[Interval], List[List[float]]]:
50     """WASB trajectory -> (candidate intervals, fps/resolution-invariant feature
51 rows)."""
52     points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
53     vt, mg, mwd = proposal
54     wins = TrackNetRunner.trajectory_to_action_windows(
55         points, fps=fps, frame_width=frame_width,
56         velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
57     intervals = [(w["start"], w["end"]) for w in wins]
58     gated = rally_gate.gate_windows(points, intervals, fps, min_crossings=0) # for net-
59 crossings
60     X: List[List[float]] = []
61     for w, g in zip(wins, gated):
62         dur = max(1e-6, w["end"] - w["start"])
63         win_frames = max(1.0, dur * fps)
64         X.append([
65             dur, # rally length (sec) ?
66             float(w["peak_velocity"]), # already normalized by
67             float(w["mean_velocity"]),
68             float(w["active_frame_count"]) / win_frames, # active-shuttle ratio ?
69             float(g.crossings), # net crossings (count) ?
70         ])
71     return intervals, X
72
73     def baseline_windows(trajecory_csv: str, fps: float, frame_width: float) ->
74 List[Interval]:
75     points = TrackNetRunner.parse_trajectory_csv(trajecory_csv)
76     vt, mg, mwd = BASELINE_LOCAL
77     wins = TrackNetRunner.trajectory_to_action_windows(
78         points, fps=fps, frame_width=frame_width,
79         velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd)
80     return [(w["start"], w["end"]) for w in wins]
81
82     def load_video(spec: Dict, iou: float = 0.5) -> Dict:

```

```

81     fps, fw = float(spec["fps"]), float(spec["frame_width"])
82     intervals, X = build_candidates(spec["trajectory"], fps, fw)
83     gts = load_golden_gts(spec["labels"])
84     y = label_candidates(intervals, gts, iou)
85     return {"name": spec.get("name", spec["trajectory"]), "intervals": intervals, "X":
86     X,
87           "y": y, "gts": gts, "baseline": baseline_windows(spec["trajectory"], fps,
88     fw)}
89
90     def predict_rallies(model: LogisticRegression, X: List[List[float]], intervals:
91     List[Interval],
92           threshold: float = 0.5) -> List[Interval]:
93         if not intervals:
94             return []
95         proba = model.predict_proba(X)
96         pos = [iv for iv, p in zip(intervals, proba) if p >= threshold]
97         return merge_overlaps(pos, gap_tol=1.0)
98
99     def run(specs: List[Dict], iou: float = 0.5, threshold: float = 0.5,
100           model_out: Optional[str] = None) -> dict:
101         videos = [load_video(s, iou) for s in specs]
102         print(f"=== Distilled local rally classifier vs Gemini silver-GT "
103               f"({len(videos)} videos, IoU={iou}, prob>={threshold}) ===")
104         for v in videos:
105             ceil = score(v["intervals"], v["gts"], iou).recall
106             print(f"[{v['name']}] {len(v['intervals'])} candidates ({sum(v['y'])} pos) | "
107                   f"proposal recall ceiling {ceil:.3f} | {len(v['gts'])} Gemini rallies")
108
109         result: dict = {"videos": [v["name"] for v in videos]}
110         if len(videos) >= 2:
111             print("\n--- HONEST leave-one-video-out: learned vs threshold baseline (held-
112             out) ---")
113             clf, base = [], []
114             for i, held in enumerate(videos):
115                 others = [v for j, v in enumerate(videos) if j != i]
116                 X = [row for v in others for row in v["X"]]
117                 y = [t for v in others for t in v["y"]]
118                 model = LogisticRegression().fit(X, y, FEATURE_NAMES)
119                 preds = predict_rallies(model, held["X"], held["intervals"], threshold)
120                 sc = score(preds, held["gts"], iou)
121                 bsc = score(held["baseline"], held["gts"], iou)
122                 clf.append(sc)
123                 base.append(bsc)
124                 print(f"  held-out {held['name']}: LEARNED P={sc.precision:.3f}
125                 R={sc.recall:.3f} F1={sc.f1:.3f}"
126                       f"    |  baseline P={bsc.precision:.3f} R={bsc.recall:.3f}
127                 F1={bsc.f1:.3f}")
128             def mean(xs, a):
129                 return sum(getattr(s, a) for s in xs) / len(xs)
130             print(f"  MEAN held-out: LEARNED P={mean(clf, 'precision'):.3f}
131                   R={mean(clf, 'recall'):.3f} "
132                   f"    F1={mean(clf, 'f1'):.3f} | baseline P={mean(base, 'precision'):.3f} "
133                   f"    R={mean(base, 'recall'):.3f} F1={mean(base, 'f1'):.3f}")
134             result["mean_learned_f1"] = mean(clf, "f1")
135             result["mean_baseline_f1"] = mean(base, "f1")
136
137         # Final model on ALL videos (for production / running on a fresh held-out video).
138         Xall = [row for v in videos for row in v["X"]]
139         yall = [t for v in videos for t in v["y"]]
140         final = LogisticRegression().fit(Xall, yall, FEATURE_NAMES)
141         assert final.w is not None # set by fit()
142         weights = dict(zip(FEATURE_NAMES, [round(float(w), 3) for w in final.w]))
143         print(f"\nfinal model standardized weights (sign/size = influence): {weights}")

```

```

139         if model_out:
140             final.save(model_out)
141             print(f"final model (all videos) -> {model_out}")
142             result["model_out"] = model_out
143         return result
144
145
146     def main() -> None:
147         ap = argparse.ArgumentParser(description="Distill Gemini labels into a local rally
classifier.")
148         ap.add_argument("--manifest", required=True,
149                         help="JSON: [{name, trajectory, fps, frame_width, labels}, ...]
(labels = Gemini CSV)")
150         ap.add_argument("--iou", type=float, default=0.5)
151         ap.add_argument("--threshold", type=float, default=0.5, help="classifier probability
cutoff")
152         ap.add_argument("--model-out", default="output/local_rally_model.json")
153         args = ap.parse_args()
154         with open(args.manifest, encoding="utf-8") as f:
155             specs = json.load(f)
156         run(specs, iou=args.iou, threshold=args.threshold, model_out=args.model_out)
157
158
159     if __name__ == "__main__":
160         main()
161

```

9. user (2026-06-14T10:31:19.461Z)

=== output/wasb CSVs ===

total 1804

```

drwxr-xr-x 1 avidu 197609      0 Jun 13 14:43 .
drwxr-xr-x 1 avidu 197609      0 Jun 14 11:51 ..
-rw-r--r-- 1 avidu 197609 1635249 Jun 13 14:43 GX010125_1080p__89605e80ed01_wasb.csv
drwxr-xr-x 1 avidu 197609      0 Jun 13 14:42 GX010125_1080p__89605e80ed01_wasb_wasbcache
-rw-r--r-- 1 avidu 197609 172447 Jun 11 18:28 mahadevpura_smoke_180s__05e0e577029d_wasb.csv
drwxr-xr-x 1 avidu 197609      0 Jun 11 18:28
mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache

```

=== output dirs ===

```

GX010125_1080p.mp4
GX010125_highlights.mp4
GX010125_highlights_min5.mp4
GX010125_run
KushagraAviHighlight-Trailor.mp4
MahadevpuraSingles_highlights.mp4
ab_experiment_n6.json
adarsh_avi_full_vs_human.csv
adarsh_avi_singles_golden_features.json
adarsh_avi_vs_human.csv
adarsh_baseline_gated.csv
adarsh_baseline_segments.csv
adarsh_gemini_fullvideo.csv
adarsh_gemini_refined.csv
adarsh_gemini_smoke.csv
adarsh_min5s_segments.csv
adarsh_tuned_segments.csv
badminton_indexer.db
e2e_reel.mp4
e2e_server_log.txt
fusion_golden.log
gbaaddy_golden_features.json
gbaaddy_vs_human.csv
gx010125_min5_stitch.log
gx010125_run.log
gx010125_transcode.log

```

human_lovo_manifest.json
kushagra_1280.mp4
kushagra_1280.report.json
kushagra_1280.report.md
kushagra_1280.summary.md
kushagra_singles_golden_features.json
kushagra_vs_human.csv
largetest_doubles_golden_features.json
largetestvideo_vs_human.csv
local_calib_manifest.json
local_rally_model.json
mahadevpura_gemini_refined.csv
mahadevpura_singles_golden_features.json
mahadevpura_smoke_180s.mp4
mahadevpura_smoke_180s.report.json
mahadevpura_smoke_180s.report.md
mahadevpura_smoke_180s.summary.md
mahadevpura_vs_human.csv
phase3_probe.json
phase3_yolo.db
ramp_probe.mp4
smoke4_log.txt
sports_indexer.db
test_indexer.db

10. assistant (2026-06-14T10:31:38.729Z)

I have cached WASB CSVs for GX010125 (unlabeled) and a mahadevpura smoke clip. The golden corpus (with labels) is referenced by the manifest. Let me read the manifest, the FusionSegmenter served path, the experiment harness, and the existing n6 LOVO repro ? plus check the environment.

11. user (2026-06-14T10:31:40.403Z)

=== human_lovo_manifest.json ===

```
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/adarsh_avi_full_traj.csv",
    "fps": 59.94, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden Labelled/Adarsh and
Avi.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/kushagra_traj.csv",
    "fps": 59.94, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/KushagraYashAviFirstVideo.rallies.csv"
  },
  {
    "name": "largetest_doubles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/largetestvideo_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/LargeTestVideo.rallies.csv"
  },
  {
    "name": "gbaaddy",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/gbaaddy_traj.csv",
    "fps": 59.94, "frame_width": 1920,
```

```

    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/gBaaddy20May2026.rallies.csv"
  },
  {
    "name": "testlarge_short",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/testlarge_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/TestLargeVideo.rallies.csv"
  },
  {
    "name": "mahadevpura_singles",
    "trajectory": "C:/Users/avidu/Projects/Annotation
Setup/Collect/Trajectories/mahadevpura_traj.csv",
    "fps": 29.97, "frame_width": 1920,
    "labels": "C:/Users/avidu/Projects/Annotation Setup/Collect/Golden
Labelled/MahadevpuraSingles.rallies.csv"
  }
]

```

=== config.json (indexing section) ===

```

{
  "default_segmenter": "fusion_hybrid",
  "use_gpu": true,
  "motion_threshold": 0.08,
  "min_segment_duration": 3.0,
  "dead_time_merge_gap": 2.0,
  "chunk_duration": 90,
  "chunk_overlap": 10,
  "detector_model": "fasterrcnn_resnet50_fpn",
  "detector_score_threshold": 0.5,
  "yolo_velocity_threshold": 0.15,
  "yolo_frame_skip": 3,
  "yolo_tracker": "bytetrack.yaml",
  "yolo_prefetch_workers": 2,
  "min_action_window_duration": 2.0,
  "log_yolo_candidates": true,
  "store_yolo_candidates": true,
  "ai_handoff_padding": 2.0,
  "ai_max_workers": 1,
  "tracknet": {
    "wsl_distro": "Ubuntu",
    "repo_dir": "~/models/TrackNetV4",
    "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
    "conda_env": "TrackNetV4",
    "weights_path": "",
    "queue_length": 5,
    "stage_in_wsl": true,
    "wsl_stage_dir": "~/clips",
    "timeout_sec": 1800,
    "velocity_threshold": 0.02
  },
  "wasb": {
    "wsl_distro": "Ubuntu",
    "repo_dir": "~/models/WASB-SBDT",
    "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
    "conda_env": "wasb",
    "weights_path": "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
    "sport": "badminton",
    "wsl_stage_dir": "~/clips",
    "timeout_sec": 1800,
    "velocity_threshold": 0.02
  },
  "fusion": {

```

```

        "substrate": "wasb",
        "min_crossings": 2,
        "min_players": 0,
        "cue_flags": {}
    }
}

```

12. user (2026-06-14T10:31:41.091Z)

```

1      """Multi-signal fusion segmenter (MULTI_SIGNAL_FUSION_PLAN P2).
2
3      Shuttle stays PRIMARY: this reuses an existing trajectory substrate (WASB by default,
4      TrackNet optional) to seed candidate windows exactly as `wasb_hybrid`/`tracknet_hybrid`
5      do ? it subclasses the same `TrajectoryHybridSegmenter` engine. It then ENRICHES each
6      shuttle-seeded window via two overridable seams the engine exposes:
7
8      - ``_enrich_candidate_stats`` ? always adds the **net-crossing count** (Gate A signal,
9      GPU-free, computed from the trajectory we already have); and, only when the person cue
10     is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
11     (``fusion_features``), persisting them into ``candidate_segments.stats`` for the learned
12     fusion (P3). The person path lazily builds ONE `PersonDetector` and only then loads
the
13     torchvision/supervision model ? so the default (person-OFF) path never loads a
detector
14     model nor touches the GPU. (torch/cv2 themselves are imported by the registry
regardless,
15     via the pre-existing yolo_hybrid ? person_detector chain; the lazy part is the model.)
16     - ``_select_for_handoff`` ? optional served Gate A/B: when
``indexing.fusion.min_crossings``
17     (and, with the person cue, ``min_players``) are raised above 0, sub-threshold windows
are
18     dropped from the AI handoff (the held-shuttle / one-sided-feed false-positive fix).
19     **Persisted candidates remain the full superset** regardless, so the offline
experiment
20     harness can re-score any variant.
21
22     Defaults reproduce the substrate exactly: net_crossings/person features are persisted
but
23     nothing is gated (thresholds 0, person OFF), so `FusionSegmenter` with default config
seeds
24     and hands off identically to `wasb_hybrid`. Registered as ``fusion_hybrid``, NOT the
default
25     segmenter ? it is opt-in. Promotion to default happens only on measured LOVO lift
through the
26     experiment harness (EXPERIMENT_HARNESS.md), never silently.
27     """
28     from typing import Any, Dict, List, Optional, Tuple
29
30     from backend.pipeline.segmenters.base import segmenter_registry
31     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
32     from backend.pipeline.detectors.base import DetectorRunner
33     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
34     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, TrackNetConfig
35     from backend.pipeline.detectors import rally_gate
36     from backend.pipeline.detectors import fusion_features as ff
37
38
39     class FusionSegmenter(TrajectoryHybridSegmenter):
40         """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate
B)."""
41
42         family = "fusion"
43         display_label = "Fusion"
44
45         def __init__(self, db: Any, config: Any):

```

```

46         super().__init__(db, config)
47         # Built lazily only if the person cue is enabled (no detector model loaded
otherwise).
48         self._person: Optional[Any] = None
49         # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-
estimate it
50         # over the whole trajectory once per candidate window (it depends only on
`points`).
51         self._axis_cache_key: Optional[int] = None
52         self._axis_cache: Optional[tuple] = None
53
54         @property
55         def name(self) -> str:
56             return "fusion_hybrid"
57
58         @property
59         def description(self) -> str:
60             return ("Multi-signal fusion: shuttle trajectory (WASB/TrackNet) seeds candidate
rallies, "
61                     "net-crossing Gate A + optional person-occupancy Gate B adjudicate them,
and the "
62                     "AI provider sets semantic boundaries.")
63
64         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
65             substrate = idx_cfg.get("fusion", {}).get("substrate", "wasb")
66             if substrate == "tracknet":
67                 cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
68                 return TrackNetRunner(cfg), {
69                     "weights_path": cfg.weights_path,
70                     "queue_length": cfg.queue_length,
71                     "fusion_substrate": "tracknet",
72                 }
73             wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
74             return WasbRunner(wcfg), {"weights_path": wcfg.weights_path, "fusion_substrate":
"wasb"}
75
76         def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
77                                     frame_width: float, video_path: str,
78                                     idx_cfg: Dict[str, Any]) -> Dict[str, Any]:
79             stats = super()._enrich_candidate_stats(cw, points, fps, frame_width,
video_path, idx_cfg)
80             # Gate-A signal ? net-crossing count for this window. Pure, GPU-free, always
persisted
81             # (single service feed crosses ~once; a real rally crosses >=2x). axis/net auto-
estimated
82             # over the whole trajectory by rally_gate (camera-agnostic); reused across
windows.
83             gated = rally_gate.gate_windows(points, [(cw["start"], cw["end"])], fps,
84                                                  min_crossings=0,
estimate=self._axis_estimate(points))
85             stats["net_crossings"] = float(gated[0].crossings) if gated else 0.0
86
87             fcfg = idx_cfg.get("fusion", {})
88             if fcfg.get("cue_flags", {}).get("person", False):
89                 stats.update(self._person_features(cw, points, fps, frame_width, video_path,
fcfg))
90             return stats
91
92         def _axis_estimate(self, points: List[Any]) -> tuple:
93             """Cache rally_gate's (axis, net, span) play-axis estimate per trajectory so
it's
94             computed once per video, not once per candidate window (it depends only on
points)."""
95             key = id(points)

```

```

96         if self._axis_cache_key != key or self._axis_cache is None:
97             self._axis_cache_key = key
98             self._axis_cache = rally_gate.estimate_play_axis(points)
99         est = self._axis_cache
100         assert est is not None
101         return est
102
103     def _person_features(self, cw: Dict[str, Any], points: List[Any], fps: float,
104                         frame_width: float, video_path: str,
105                         fcfg: Dict[str, Any]) -> Dict[str, float]:
106         """GPU path (person cue ON): run the person detector over this window's
ORIGINAL-video
107         frame range and compute the scale/translation-invariant person features. Lazily
builds
108         ONE PersonDetector. ? real-video verification is owner-side; this wiring is
mock-tested."""
109         # Lazy import so the default (person-OFF) path never pulls torch/cv2 through
this module.
110         from backend.pipeline.detectors.person_detector import PersonDetector
111         if self._person is None:
112             self._person = PersonDetector(self.config)
113
114         frame_skip = self.config.get("indexing", {}).get("yolo_frame_skip", 3)
115         # round (not truncate): int() would map e.g. 1.999s*30 to frame 59 not 60 ? a
116         # window-edge off-by-one that drops the boundary frame from the shuttle
alignment.
117         start_f, end_f = round(cw["start"] * fps), round(cw["end"] * fps)
118         det = self._person.detections_per_frame(video_path, start_f, end_f, frame_skip)
119         fw = det["frame_width"] or frame_width
120         fh = det["frame_height"]
121         # If we can't trust the frame dims, every normalised feature would silently
collapse
122         # to -0 ? return an explicit all-zero person-feature row instead of a half-baked
one.
123         if fw <= 0 or fh <= 0:
124             return {k: 0.0 for k in ff.PERSON_FEATURE_NAMES}
125
126         poly = fcfg.get("court_polygon")
127         polygon = [(float(p[0]), float(p[1])) for p in poly] if poly else None
128         net = (fcfg.get("net_axis", "y"), float(fcfg.get("net_position", 0.5)))
129         shuttle_in = [p for p in points if start_f <= p.frame <= end_f]
130         return ff.compute_person_features(det["frames"], shuttle_in, fw, fh,
131                                         court_polygon=polygon, net=net)
132
133     def _select_for_handoff(self, windows: List[Dict[str, Any]],
134                           window_stats: List[Dict[str, Any]],
135                           idx_cfg: Dict[str, Any]) -> List[int]:
136         """Served Gate A/B over the (already-enriched) per-window stats. Default
thresholds
137         (0) keep every window ? identical to the substrate. Persisted candidates are
138         untouched (full superset)."""
139         fcfg = idx_cfg.get("fusion", {})
140         min_crossings = int(fcfg.get("min_crossings", 0))
141         min_players = int(fcfg.get("min_players", 0))
142         person_on = bool(fcfg.get("cue_flags", {}).get("person", False))
143
144         keep: List[int] = []
145         for i, st in enumerate(window_stats):
146             if min_crossings > 0 and st.get("net_crossings", 0.0) < min_crossings:
147                 continue
148             if person_on and min_players > 0 and st.get("players_in_court", 0.0) <
min_players:
149                 continue
150             keep.append(i)
151         return keep

```

```
152
153
154     segmenter_registry.register(FusionSegmenter)
155
```

13. user (2026-06-14T10:31:41.485Z)

```
1     """First full A/B experiment on the n=6 golden corpus ? runs entirely offline
2     (persisted WASB trajectories, no GPU). One consistent candidate set per video
3     (recall-oriented proposal windowing, the persisted pipeline's own), re-scored three
4     ways, then graded with the C1 honest aggregation + C4 segmentation metrics + a C3
5     out-of-fold learned mAP. Prints a report and writes output/ab_experiment_n6.json.
6
7     Honest: prints whatever the real golden data produces. No claims before measurement.
8     """
9     import json
10
11     from backend.eval.distill_local import FEATURE_NAMES, build_candidates
12     from backend.eval.calibrate_wasb import load_golden_gts
13     from backend.eval.training import label_candidates
14     from backend.eval.classifier import LogisticRegression
15     from backend.eval.experiment import CONTROL, Variant, VideoCandidates, evaluate_variant
16     from backend.eval.metrics import score
17     from backend.eval import segmentation_metrics as sm
18     from backend.eval import eval_stats as es
19
20     IOU = 0.5
21     MANIFEST = "output/human_lovo_manifest.json"
22     OUT = "output/ab_experiment_n6.json"
23
24
25     def load_corpus():
26         specs = json.load(open(MANIFEST))
27         videos, pooled = [], {"X": [], "y": [], "group": [], "interval": []}
28         for s in specs:
29             fps, fw = float(s["fps"]), float(s["frame_width"])
30             intervals, X = build_candidates(s["trajectory"], fps, fw)
31             gts = load_golden_gts(s["labels"])
32             y = label_candidates(intervals, gts, IOU)
33             feats = {FEATURE_NAMES[k]: [row[k] for row in X] for k in
range(len(FEATURE_NAMES))}
34             videos.append(VideoCandidates(name=s["name"], intervals=intervals,
features=feats, gts=gts))
35             for iv, row, yi in zip(intervals, X, y):
36                 pooled["X"].append(row); pooled["y"].append(yi)
37                 pooled["group"].append(s["name"]); pooled["interval"].append(iv)
38         return videos, pooled
39
40
41     def c4_aggregate(ev, videos):
42         """Mean F1@k + mean segment-count ratio across videos for one variant's
predictions."""
43         by_name = {v.name: v for v in videos}
44         f1k = {0.1: [], 0.25: [], 0.5: []}
45         ratios = []
46         for name, preds in ev["predictions"].items():
47             gts = by_name[name].gts
48             got = sm.f1_at_overlaps(preds, gts)
49             for k in f1k:
50                 f1k[k].append(got[k])
51             ratios.append(sm.segment_count_ratio(preds, gts))
52         mean = lambda xs: sum(xs) / len(xs) if xs else 0.0
53         return {"f"f1@{int(k*100)}": round(mean(v), 3) for k, v in f1k.items()} | \
54             {"seg_count_ratio": round(mean(ratios), 2)}
55
```

```

56
57 def oof_learned_map(pooled, videos):
58     """C3 + C4: out-of-fold learned candidate proba ? mAP@tIoU per video, averaged."""
59     def fit_predict(train_idx, test_idx):
60         m = LogisticRegression().fit([pooled["X"][i] for i in train_idx],
61                                     [pooled["y"][i] for i in train_idx], FEATURE_NAMES)
62         return list(m.predict_proba([pooled["X"][i] for i in test_idx]))
63     oof = es.out_of_fold_predictions(pooled["group"], fit_predict)
64     by_name = {v.name: v for v in videos}
65     maps = []
66     for name in by_name:
67         scored = [(pooled["interval"][i], float(oof[i]))
68                  for i in range(len(oof)) if pooled["group"][i] == name and oof[i] is
not None]
69         maps.append(sm.mean_average_precision(scored, by_name[name].gts))
70     return sum(maps) / len(maps) if maps else 0.0
71
72
73 def c1_block(a_f1, b_f1):
74     d = es.paired_delta_ci(a_f1, b_f1, seed=0)
75     return {
76         "mean_a": round(sum(a_f1) / len(a_f1), 3),
77         "mean_b": round(sum(b_f1) / len(b_f1), 3),
78         "mean_delta": round(d["mean_delta"], 3),
79         "ci": [round(d["ci_low"], 3), round(d["ci_high"], 3)],
80         "ci_excludes_zero": d["ci_excludes_zero"],
81         "sign": {k: int(v) if k != "p_value" else round(v, 3) for k, v in
d["sign_test"].items()},
82     }
83
84
85 def main():
86     videos, pooled = load_corpus()
87     print("=" * 80)
88     print(f"FIRST A/B EXPERIMENT ? n={len(videos)} golden corpus (offline, persisted
trajectories)")
89     print("=" * 80)
90     for v in videos:
91         ceil = score(v.intervals, v.gts, IOU).recall
92         print(f"    {v.name:<22} {len(v.intervals):>4} candidates {len(v.gts):>3} rallies
"
93             f"proposal-recall-ceiling={ceil:.3f}")
94
95     control = evaluate_variant(videos, CONTROL, IOU)
96     gateA = evaluate_variant(videos, Variant(name="gateA", min_crossings=2), IOU)
97     learned = evaluate_variant(videos, Variant(name="learned",
feature_names=FEATURE_NAMES), IOU, honest=True)
98
99     order = [p["video"] for p in control["per_video"]]
100    f1 = {"CONTROL": {p["video"]: p["f1"] for p in control["per_video"]},
101         "gateA": {p["video"]: p["f1"] for p in gateA["per_video"]},
102         "learned": {p["video"]: p["f1"] for p in learned["per_video"]}}
103
104    def col(name):
105        return [f1[name][v] for v in order]
106
107    print("\n--- per-video held-out F1 ---")
108    print(f"    {'video':<22}{'CONTROL':>9}{'GateA':>9}{'learned':>9}")
109    for v in order:
110        print(f"    {v:<22}{f1['CONTROL'][v]:>9.3f}{f1['gateA'][v]:>9.3f}{f1['learned'][v]:>9.3f}")
111
112    ab_gate = c1_block(col("CONTROL"), col("gateA"))
113    ab_learned = c1_block(col("CONTROL"), col("learned"))
114    print("\n--- A/B 1: CONTROL vs Gate-A (static) ? C1 honest aggregation ---")

```

```

115         print(f" meanF1 {ab_gate['mean_a']} -> {ab_gate['mean_b']}
?F1={ab_gate['mean_delta']:+} "
116         f"CI{ab_gate['ci']} excl0={ab_gate['ci_excludes_zero']}
sign={ab_gate['sign']}")
117         print("--- A/B 2: CONTROL vs learned fusion (LOVO) ? C1 honest aggregation ---")
118         print(f" meanF1 {ab_learned['mean_a']} -> {ab_learned['mean_b']}
?F1={ab_learned['mean_delta']:+} "
119         f"CI{ab_learned['ci']} excl0={ab_learned['ci_excludes_zero']}
sign={ab_learned['sign']}")
120
121         c4 = {name: c4_aggregate(ev, videos)
122               for name, ev in (("CONTROL", control), ("gateA", gateA), ("learned",
learned))}
123         print("\n--- C4 over-segmentation-aware metrics (mean across videos) ---")
124         for name in ("CONTROL", "gateA", "learned"):
125             print(f" {name:<9} {c4[name]}")
126         oof_map = oof_learned_map(pooled, videos)
127         print(f" learned out-of-fold (C3) mAP@tIoU[.3-.7] = {oof_map:.3f}")
128
129         result = {
130             "corpus": order, "n": len(videos), "iou": IOU,
131             "aggregate_f1": {k: round(sum(col(k)) / len(order), 3) for k in ("CONTROL",
"gateA", "learned")},
132             "per_video_f1": f1,
133             "ab_control_vs_gateA": ab_gate,
134             "ab_control_vs_learned": ab_learned,
135             "c4_metrics": c4,
136             "learned_oof_map": round(oof_map, 3),
137         }
138         json.dump(result, open(OUT, "w"), indent=2)
139         print(f"\nwrote {OUT}")
140
141
142     if __name__ == "__main__":
143         main()
144

```

14. assistant (2026-06-14T10:32:19.249Z)

Now I understand the architecture. The key realization: the offline harness (``build_candidates`` + ``experiment.predict``) reimplements gating but does NOT exercise the real ``FusionSegmenter._enrich_candidate_stats`/`_select_for_handoff`` served code. To close the served-side gap properly, I should drive the **actual** ``FusionSegmenter`` methods over the cached trajectory. Let me read the base engine to understand the production windowing path and whether I can drive it offline.

15. user (2026-06-14T10:32:19.919Z)

```

1     """Shared engine for trajectory-substrate hybrid segmenters.
2
3     tracknet_hybrid and wasb_hybrid were copy-paste twins ? CODE_MAP carried a
4     standing footgun: "any P3/P4 fix must be applied to BOTH". This base class is
5     now the single home for the whole pipeline shape:
6
7     healthcheck gate -> P2 trajectory -> candidate windows (+ persistence)
8     -> P3 AI-provider handoff over padded clips -> P4 DB population/linking
9
10    A concrete subclass supplies only its identity and runner wiring:
11    - ``family`` ? config section under ``indexing.*``, output subdir, and
12      the ``<family>-hybrid-<model>`` detector tag in metadata.
13    - ``display_label`` ? human label for log lines ("WASB", "TrackNet").
14    - ``name`` / ``description`` properties (the registry contract).
15    - ``_build_runner(idx_cfg)`` ? returns (DetectorRunner, detector-specific
16      detection_params extras such as weights_path).
17

```

```

18     This base is deliberately NOT registered; only concrete subclasses register.
19     """
20     import os
21     import time
22     import logging
23     import concurrent.futures
24     from typing import Any, Dict, List, Optional, Tuple
25
26     import cv2
27
28     from backend.pipeline.segmenters.base import BasePlaySegmenter
29     from backend.utils.video import get_video_duration, extract_video_chunk
30     from backend.sports import sport_registry
31     from backend.providers import provider_registry
32     from backend.pipeline.detectors.base import DetectorRunner
33
34     logger = logging.getLogger(__name__)
35
36
37     def _fmt_ts(seconds: float) -> str:
38         seconds = max(0, int(seconds))
39         h, rem = divmod(seconds, 3600)
40         m, s = divmod(rem, 60)
41         return f"{h:02d}:{m:02d}:{s:02d}"
42
43
44     class TrajectoryHybridSegmenter(BasePlaySegmenter):
45         """Trajectory-detector hybrid: precise candidate rallies + AI semantic
46         boundaries."""
47
48         #: config section under `indexing` (velocity_threshold lives there), the
49         #: output subdir for trajectories, and the metadata detector-tag prefix.
50         family: str = ""
51         #: human-readable label used in log lines.
52         display_label: str = ""
53
54         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
55         Any]]:
56             """Return (runner, detector-specific detection_params extras)."""
57             raise NotImplementedError
58
59         @staticmethod
60         def _probe_fps_width(video_path: str) -> tuple:
61             """Return (fps, frame_width) for a video, with sensible fallbacks."""
62             cap = cv2.VideoCapture(video_path)
63             fps = cap.get(cv2.CAP_PROP_FPS) if cap.isOpened() else 0.0
64             width = cap.get(cv2.CAP_PROP_FRAME_WIDTH) if cap.isOpened() else 0.0
65             cap.release()
66             return (fps if fps > 0 else 30.0, width if width > 0 else 1280.0)
67
68         @staticmethod
69         def _shot_count_from(segment: Dict[str, Any], duration: float) -> int:
70             """Provider-reported exchange count, else a duration-based estimate.
71
72             One canonical implementation ? the twins had drifted (wasb relied on
73             ``int(None)`` raising into the fallback; same result, by accident).
74             """
75             shot_count = segment.get("shuttle_exchange_count")
76             if shot_count is None:
77                 return max(3, int(duration / 0.8))
78             try:
79                 return int(shot_count)
80             except (ValueError, TypeError):
81                 return max(3, int(duration / 0.8))

```

```

81         # --- Fusion seams (identity by default ? wasb_hybrid/tracknet_hybrid unchanged)
82         -----
83         def _enrich_candidate_stats(self, cw: Dict[str, Any], points: List[Any], fps: float,
84                                     frame_width: float, video_path: str,
85                                     idx_cfg: Dict[str, Any]) -> Dict[str, Any]:
86             """Stats dict persisted into ``candidate_segments.stats`` for one window. The
87             BASE
88             returns exactly the 4 motion keys, so the trajectory twins persist byte-
89             identical
90             rows; ``FusionSegmenter`` overrides this to add ``net_crossings`` (+ person-cue
91             features). ``points`` are the whole-video trajectory points
92             (TrajectoryPoint)."""
93             return {
94                 "peak_velocity": cw["peak_velocity"],
95                 "mean_velocity": cw["mean_velocity"],
96                 "active_frame_count": cw["active_frame_count"],
97                 "track_id_count": cw["track_id_count"],
98             }
99
100         def _select_for_handoff(self, windows: List[Dict[str, Any]],
101                                window_stats: List[Dict[str, Any]],
102                                idx_cfg: Dict[str, Any]) -> List[int]:
103             """Indices of the candidate windows that proceed to the AI handoff (and thus may
104             become play_segments). The BASE sends ALL windows (no gating ? twins unchanged);
105             ``FusionSegmenter`` overrides this to apply Gate A/B when thresholds are raised.
106             Persisted candidates are NOT affected ? they remain the full superset for
107             offline
108             re-scoring."""
109             return list(range(len(windows)))
110
111         def process_video(self, video_id: str, video_path: str, # type: ignore[override]
112                           player_pool: Optional[List[str]] = None,
113                           max_frames: Optional[int] = None) -> List[Dict[str, Any]]:
114             # override-ignore: the ABC declares the Result contract (Success|Failure)
115             # but every live segmenter still returns a bare list ? the documented
116             # deliberate-incremental gap (CODE_MAP gotchas; main.py handles both).
117             label = self.display_label
118             # --- Sport profile + AI provider wiring (identical across segmenters) ---
119             sport_name = self.config.get("sport", "badminton")
120             try:
121                 sport_profile = sport_registry.get(sport_name)
122             except KeyError as e:
123                 logger.error(str(e))
124                 return []
125
126             idx_cfg = self.config.get("indexing", {})
127             provider_name = self.config.get("ai_provider", idx_cfg.get("ai_provider",
128 "gemini"))
129             model_name = self.config.get("ai_model", idx_cfg.get("ai_model",
130 "gemini-2.5-flash"))
131
132             api_key_env_var = f"{{provider_name.upper()}}_API_KEY"
133             api_key = os.environ.get(api_key_env_var)
134             if not api_key:
135                 logger.error(
136                     f"{{api_key_env_var}} environment variable is not set! "
137                     f"To run the {{provider_name}} handoff, set your API key. "
138                     f"In PowerShell: $env:{{api_key_env_var}}=\"your_api_key_here\"")
139             )
140             return []
141
142             try:
143                 provider = provider_registry.get(provider_name, api_key=api_key,
144 model_name=model_name)
145             except KeyError as e:
146                 logger.error(str(e))

```

```

138         return []
139
140     video_duration = get_video_duration(video_path)
141     if video_duration <= 0:
142         logger.error("Invalid video duration.")
143         return []
144     fps, frame_width = self._probe_fps_width(video_path)
145
146     # --- Runner config + windowing thresholds ---
147     runner, runner_params = self._build_runner(idx_cfg)
148
149     velocity_thresh = idx_cfg.get(self.family, {}).get("velocity_threshold", 0.02)
150     merge_gap = idx_cfg.get("dead_time_merge_gap", 2.0)
151     min_window_duration = idx_cfg.get("min_action_window_duration", 2.0)
152     handoff_padding = idx_cfg.get("ai_handoff_padding", 2.0)
153     ai_max_workers = idx_cfg.get("ai_max_workers", 5)
154     log_candidates = idx_cfg.get("log_yolo_candidates", True)
155     store_candidates = idx_cfg.get("store_yolo_candidates", True)
156
157     detection_params = {
158         "detector": self.name,
159         "velocity_thresh": velocity_thresh,
160         "merge_gap": merge_gap,
161         "min_action_window_duration": min_window_duration,
162         **runner_params,
163     }
164
165     # --- Phase 1: env healthcheck (fail fast with a clear message) ---
166     ok, msg = runner.healthcheck()
167     if not ok:
168         logger.error("%s WSL environment not ready: %s", label, msg)
169         return []
170     logger.info("%s WSL env OK: %s", label, msg)
171
172     # --- Phase 2: trajectory -> candidate rally windows ---
173     # Trajectory detectors process the whole video in one pass (they carry
174     # their own frame buffering), so unlike yolo_hybrid we don't pre-chunk.
175     t_start = time.time()
176     out_dir = os.path.join("output", self.family)
177     csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
178     if not csv_path:
179         logger.error("%s produced no trajectory; aborting.", label)
180         return []
181
182     points = runner.parse_trajectory_csv(csv_path)
183     logger.info("Parsed %d trajectory points (%d visible).",
184                 len(points), sum(1 for p in points if p.visible))
185
186     all_candidate_windows = runner.trajectory_to_action_windows(
187         points, fps=fps, frame_width=frame_width, chunk_start=0.0,
188         velocity_thresh=velocity_thresh, merge_gap=merge_gap,
189         min_window_duration=min_window_duration, chunk_index=0,
190     )
191     logger.info("Phase 2 (%s) completed in %.1fs. Candidate windows: %d",
192                 label, time.time() - t_start, len(all_candidate_windows))
193
194     if log_candidates:
195         for cw in all_candidate_windows:
196             logger.info(f"[{label} rally]
197 {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} "
198 f"({cw['end'] - cw['start']:.1f}s) |
199 frames={cw['active_frame_count']} "
200 f"peak_v={cw['peak_velocity']:.3f}
201 mean_v={cw['mean_velocity']:.3f}")

```

```

200         # Per-window stats via the enrichment hook ? base = the 4 motion keys; the
fusion
201         # segmenter adds net_crossings + person-cue features. Computed once, reused for
both
202         # persistence and the handoff gate below.
203         window_stats = [
204             self._enrich_candidate_stats(cw, points, fps, frame_width, video_path,
idx_cfg)
205             for cw in all_candidate_windows
206         ]
207
208         # Persist raw candidates for the fine-tuning dataset (same table as YOLO).
209         candidate_ids: List[Optional[int]] = [None] * len(all_candidate_windows)
210         if store_candidates:
211             for i, cw in enumerate(all_candidate_windows):
212                 candidate_ids[i] = self.db.add_candidate_segment(
213                     video_id, detector=self.name,
214                     start_time=cw["start"], end_time=cw["end"],
215                     chunk_index=cw["chunk_index"], confidence=min(1.0,
cw["peak_velocity"]),
216                     stats=window_stats[i], detection_params=detection_params,
217                 )
218
219         if not all_candidate_windows:
220             return []
221
222         # --- Phase 3: AI provider handoff over padded candidate clips ---
223         # Which windows reach the handoff (base = all; fusion may apply Gate A/B).
Persisted
224         # candidates above stay the full superset ? only the served play_segments are
gated.
225         handoff_indices = self._select_for_handoff(all_candidate_windows, window_stats,
idx_cfg)
226         handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
227         os.makedirs(handoff_dir, exist_ok=True)
228         logger.info("Phase 3: %s validation on %d candidate windows...",
229                     provider_name, len(handoff_indices))
230
231         def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
232             w_start = max(0, window["start"] - handoff_padding)
233             w_end = min(video_duration, window["end"] + handoff_padding)
234             w_dur = w_end - w_start
235             w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
236             if not extract_video_chunk(video_path, w_start, w_dur, w_path,
reencode=True):
237                 return []
238             prompt = sport_profile.get_prompt(chunk_duration=w_dur)
239             segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
240                                                  label=f"Handoff {wi+1}")
241             valid = []
242             for seg in segments:
243                 try:
244                     seg["start_time"] = float(seg["start_time"]) + w_start
245                     seg["end_time"] = float(seg["end_time"]) + w_start
246                     seg["_candidate_id"] = candidate_ids[wi]
247                     valid.append(seg)
248                 except (ValueError, TypeError, KeyError) as e:
249                     logger.warning("Dropping segment with unparseable timestamps: %s ?
%s", seg, e)
250             try:
251                 os.remove(w_path)
252             except OSError:
253                 pass
254             return valid
255

```

```

256         ai_segments: List[dict] = []
257         with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
258             futures = [executor.submit(process_candidate_window, wi,
all_candidate_windows[wi])
259                             for wi in handoff_indices]
260             for future in futures:
261                 ai_segments.extend(future.result())
262             try:
263                 os.rmdir(handoff_dir)
264             except OSError:
265                 pass
266             logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
267
268             # --- Phase 4: DB population + candidate linking ---
269             segments_data = []
270             for segment in ai_segments:
271                 try:
272                     start = float(segment["start_time"])
273                     end = float(segment["end_time"])
274                     dur = end - start
275                     metadata = {
276                         "shot_count": self._shot_count_from(segment, dur),
277                         "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
278                         "detector": f"{self.family}-hybrid-{model_name}",
279                     }
280                     segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
281                     if segment_id:
282                         candidate_id = segment.get("_candidate_id")
283                         if candidate_id is not None:
284                             self.db.link_candidate_to_segment(candidate_id, segment_id)
285                         segments_data.append({
286                             "id": segment_id, "start_time": start, "end_time": end,
287                             "duration": dur, "metadata": metadata,
288                         })
289                 except Exception as e:
290                     logger.warning("Skipping invalid segment struct: %s", e)
291
292             return segments_data
293

```

16. assistant (2026-06-14T10:33:11.577Z)

Now I fully understand the served path. Critical detail: the **served** windowing uses production params (``velocity_thresh=0.02, merge_gap=2.0, min_window_duration=2.0`` from config), which differ from the offline harness's recall-oriented proposal ``(0.005, 1.0, 0.5)``. So the faithful served-side test must window with production params and then call the **real** ``FusionSegmenter._enrich_candidate_stats`/`_select_for_handoff``. Let me check how the app loads config and instantiates segmenters so I reproduce production exactly, and confirm the environment (numpy/cv2/rally_gate).

17. user (2026-06-14T10:33:12.994Z)

```

1     from abc import ABC, abstractmethod
2     from typing import List, Dict, Any, Type, Optional
3     from backend.infrastructure.database import Database
4     from backend.results import Result # new standardized contract (Result[Value] ==
Success[Value] | Failure)
5
6     class BasePlaySegmenter(ABC):
7         def __init__(self, db: Database, config: Dict[str, Any]):
8             self.db = db

```

```

9         self.config = config
10
11     @property
12     @abstractmethod
13     def name(self) -> str:
14         """Unique identifier for this segmenter."""
15         pass
16
17     @property
18     @abstractmethod
19     def description(self) -> str:
20         """Human-readable explanation of this segmenter."""
21         pass
22
23     @abstractmethod
24     def process_video(self, video_id: str, video_path: str, player_pool: List[str] =
None, max_frames: int = None) -> "Result[List[Dict[str, Any]]]":
25         """
26         Processes a video to detect play segments, populates SQLite database,
27         and returns a Result.
28
29         Use isinstance(result, Failure) (or result.is_success()) to distinguish
30         "no segments found" from "something went wrong" (missing key, healthcheck
31         failure, etc.). This is the start of the standardized error/result contract.
32         """
33         pass
34
35     class SegmenterRegistry:
36         def __init__(self):
37             self._segmenters: Dict[str, Type[BasePlaySegmenter]] = {}
38
39         def register(self, segmenter_cls: Type[BasePlaySegmenter]):
40             """Registers a new segmenter class under its name property."""
41             try:
42                 temp_inst = segmenter_cls(None, {})
43                 name = temp_inst.name
44             except Exception as e:
45                 raise ValueError(f"Failed to resolve name property from segmenter class
{segmenter_cls.__name__}: {e}") from e
46             self._segmenters[name] = segmenter_cls
47
48         def get(self, name: str) -> Optional[Type[BasePlaySegmenter]]:
49             return self._segmenters.get(name)
50
51         def list_available(self) -> Dict[str, str]:
52             """Returns a dict of registered segmenter names and their descriptions."""
53             res = {}
54             for name, cls in self._segmenters.items():
55                 try:
56                     temp_inst = cls(None, {})
57                     desc = temp_inst.description
58                 except Exception:
59                     desc = "No description available."
60                 res[name] = desc
61             return res
62
63     segmenter_registry = SegmenterRegistry()
64

```

18. user (2026-06-14T10:33:15.064Z)

=== how config is loaded + segmenter instantiated in run_process_and_stitch ===

19. user (2026-06-14T10:33:15.174Z)

```

1      """Rally-start gate: suppress between-point false fires using shuttle net-crossings.
2
3      WASB (shuttle-only) over-detects: when a player flicks/tosses the shuttle back to
4      the opponent for service after losing a point, that single trajectory looks like a
5      rally. The discriminator is structural, not duration-based: a real rally has the
6      shuttle crossing the net MULTIPLE times (each shot lands on the other side); a
7      single service feed crosses ~once.
8
9      This module is camera-agnostic and needs NO new model ? it works on the trajectory
10     we already produce (Frame,Visibility,X,Y). It estimates the play axis (the image
11     axis with the larger shuttle spread) and the net position (median shuttle coord on
12     that axis ? the shuttle clusters at the two ends and crosses the net region often,
13     so the median sits near the net), then counts sign changes of (coord - net) across
14     the visible points inside a candidate window, with a deadband to ignore jitter.
15
16     Gate A in the over-fire fix spectrum (B = player-stance state machine, future).
17     Pure functions only ? no I/O, unit-tested; intended as a post-filter on the
18     windows from trajectory_to_action_windows, or to fold into it later.
19     """
20     from __future__ import annotations
21
22     from dataclasses import dataclass
23     from typing import List, Optional, Sequence, Tuple
24
25     Interval = Tuple[float, float]
26
27
28     @dataclass
29     class GatedWindow:
30         start: float
31         end: float
32         crossings: int
33         visible_points: int
34         kept: bool
35
36
37     def _spread(vals: Sequence[float]) -> float:
38         """Robust spread = p95 - p5 (ignores a few outlier detections)."""
39         if len(vals) < 2:
40             return 0.0
41         s = sorted(vals)
42         lo = s[max(0, int(0.05 * (len(s) - 1)))]
43         hi = s[min(len(s) - 1, int(0.95 * (len(s) - 1)))]
44         return hi - lo
45
46
47     def _median(vals: Sequence[float]) -> float:
48         if not vals:
49             return 0.0
50         s = sorted(vals)
51         n = len(s)
52         return s[n // 2] if n % 2 else 0.5 * (s[n // 2 - 1] + s[n // 2])
53
54
55     def estimate_play_axis(points) -> Tuple[str, float, float]:
56         """Return (axis 'x'/'y', net_value, span) from visible points.
57
58         Play axis = the image axis along which the shuttle travels most (larger spread).
59         For a baseline camera that's Y (players top/bottom); for a side camera, X.
60         """
61         xs = [p.x for p in points if p.visible]
62         ys = [p.y for p in points if p.visible]
63         sx, sy = _spread(xs), _spread(ys)
64         if sx >= sy:
65             return "x", _median(xs), sx

```

```

66         return "y", _median(ys), sy
67
68
69     def count_net_crossings(window_points, axis: str, net: float, deadband: float) -> int:
70         """Count sign changes of (coord - net) across visible points, ignoring the
71         deadband zone near the net (so jitter at the net doesn't inflate the count)."""
72         crossings = 0
73         last_side = None
74         for p in window_points:
75             if not p.visible:
76                 continue
77             v = (p.x if axis == "x" else p.y) - net
78             if abs(v) < deadband:
79                 continue
80             side = v > 0
81             if last_side is not None and side != last_side:
82                 crossings += 1
83             last_side = side
84         return crossings
85
86
87     def _points_in(points, fps: float, start: float, end: float):
88         return [p for p in points if start * fps <= p.frame <= end * fps]
89
90
91     def gate_windows(points, windows: List[Interval], fps: float,
92                     min_crossings: int = 2, deadband_frac: float = 0.06,
93                     estimate: Optional[Tuple[str, float, float]] = None) ->
List[GatedWindow]:
94         """Annotate each (start,end) window with its shuttle net-crossing count and a
95         keep decision (crossings >= min_crossings). Deadband is a fraction of the play
96         axis span. Net/axis are estimated once over the whole trajectory (stable).
97
98         `estimate` lets a caller scoring MANY windows off one trajectory pass a precomputed
99         `(axis, net, span)` (from `estimate_play_axis`) so the O(P log P) estimate isn't
redone
100         per window ? same result, avoids the per-window re-estimation."""
101         axis, net, span = estimate if estimate is not None else estimate_play_axis(points)
102         deadband = deadband_frac * span
103         out: List[GatedWindow] = []
104         for (s, e) in windows:
105             wp = _points_in(points, fps, s, e)
106             c = count_net_crossings(wp, axis, net, deadband)
107             vis = sum(1 for p in wp if p.visible)
108             out.append(GatedWindow(start=s, end=e, crossings=c, visible_points=vis,
109                                   kept=c >= min_crossings))
110         return out
111
112
113     def apply_gate(points, windows: List[Interval], fps: float,
114                   min_crossings: int = 2, deadband_frac: float = 0.06) -> List[Interval]:
115         """Convenience: return only the windows that pass the crossing gate."""
116         return [(g.start, g.end) for g in
117                 gate_windows(points, windows, fps, min_crossings, deadband_frac) if g.kept]
118

```

20. assistant (2026-06-14T10:33:38.009Z)

rally_gate is pure (no numpy needed). `FusionSegmenter(None, {})` is instantiable. The two seams only read the passed-in `idx\_cfg`. Now let me find the app's config loader so I reproduce production `idx\_cfg` exactly (not hand-build it), and confirm cached golden trajectories + the score/label tooling exist.

21. user (2026-06-14T10:33:41.668Z)

```

=== config loader ===
__init__.py
__pycache__
loader.py
models.py
---
backend/config/loader.py:6:This replaces scattered json.load + .get() patterns throughout the
codebase.
backend/config/loader.py:50:    return AppConfig().model_dump(mode="json")
backend/config/loader.py:53:def load_config(path: str | Path | None = None) -> AppConfig:
backend/config/loader.py:69:        file_data = json.load(f)
backend/config/loader.py:100:    cfg_path.write_text(cfg.model_dump_json(indent=4),
encoding="utf-8")
backend/config/models.py:240:class AppConfig(BaseModel):
backend/config/models.py:269:    # via model_dump, so legacy chained access keeps working ?
returning the Pydantic
backend/config/models.py:289:        return val.model_dump(mode="json")

=== top-level config.json keys ===
['sport', 'ai_provider', 'ai_model', 'validation', 'indexing', 'output', 'stitching']

=== golden trajectories present? ===
adarsh_avi_singles      traj=OK labels=OK fps=59.94
kushagra_singles        traj=OK labels=OK fps=59.94
largetest_doubles        traj=OK labels=OK fps=29.97
gbaaddy                  traj=OK labels=OK fps=59.94
testlarge_short          traj=OK labels=OK fps=29.97
mahadevpura_singles      traj=OK labels=OK fps=29.97

```

22. user (2026-06-14T10:33:44.839Z)

```

=== env: numpy/cv2/torch present? ===
numpy 2.4.6
cv2 4.13.0
=== load_golden_gts + score + label_candidates signatures ===
33:def load_golden_gts(csv_path: str) -> List[Interval]:
104:def score(preds: List[Interval], gts: List[Interval],
50:def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
39:def merge_overlaps(intervals: List[Interval], gap_tol: float = 0.0) -> List[Interval]:

```

23. assistant (2026-06-14T10:33:53.155Z)

All data present (6/6 golden trajectories + labels), numpy/cv2 available. Let me read the config loader and the AppConfig dict-like access so I instantiate the segmenter exactly as production does.

24. user (2026-06-14T10:33:54.111Z)

```

1      """Configuration loader and saver.
2
3      Provides a single place to load validated AppConfig with built-in defaults
4      merged with optional file overrides (config.json).
5
6      This replaces scattered json.load + .get() patterns throughout the codebase.
7      """
8
9      from __future__ import annotations
10
11     import json
12     import logging
13     from pathlib import Path
14     from typing import Any
15
16     from pydantic import BaseModel

```

```

17
18     from .models import AppConfig
19
20     logger = logging.getLogger(__name__)
21
22     DEFAULT_CONFIG_PATH = Path("config.json")
23
24
25     def _unknown_key_paths(data: dict[str, Any], model_cls: type[BaseModel], prefix: str =
26     "") -> list[str]:
27         """Dotted paths in `data` that the typed config will not honor.
28
29         Two silent failure modes exist without this check: a typo'd TOP-LEVEL key is
30         kept by AppConfig's ``extra="allow"`` but never read by typed code, and a
31         typo'd key INSIDE a section is silently dropped by Pydantic. Either way the
32         user's intent is lost ? collect every such key so load_config can warn.
33         """
34         unknown: list[str] = []
35         fields = model_cls.model_fields
36         for key, val in data.items():
37             if key not in fields:
38                 unknown.append(prefix + key)
39                 continue
40             ann = fields[key].annotation
41             if isinstance(val, dict) and isinstance(ann, type) and issubclass(ann,
42             BaseModel):
43                 unknown.extend(_unknown_key_paths(val, ann, f"{prefix}{key}.")
44         return unknown
45
46     def _get_builtin_defaults() -> dict[str, Any]:
47         """Return a plain dict of the current built-in defaults.
48
49         The AppConfig model is the single source of truth for defaults.
50         """
51         return AppConfig().model_dump(mode="json")
52
53     def load_config(path: str | Path | None = None) -> AppConfig:
54         """Load configuration with the following precedence:
55
56         1. Built-in defaults defined in the Pydantic models.
57         2. Values from the JSON file (if present and readable).
58         3. (Future) environment / CLI overrides.
59
60         Missing file or unreadable file ? returns a fully defaulted AppConfig.
61         Unknown keys in the file are tolerated (extra="allow" on the model).
62         """
63         cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
64
65         file_data: dict[str, Any] = {}
66         if cfg_path.exists():
67             try:
68                 with open(cfg_path, "r", encoding="utf-8") as f:
69                     file_data = json.load(f)
70             except Exception as exc:
71                 # Non-fatal: continue with defaults + whatever we could parse.
72                 # In production we might log here.
73                 logger.warning(f"Failed to read {cfg_path}: {exc}. Using defaults + partial
74         data.")
75
76         if file_data:
77             unknown = _unknown_key_paths(file_data, AppConfig)
78             if unknown:
79                 logger.warning(

```

```

79         "[CONFIG WARNING] %s contains keys the typed config does not "
80         "recognize (typo'd or removed knobs ? top-level extras are kept "
81         "but unread; unknown section keys are silently dropped): %s",
82         cfg_path, ", ".join(sorted(unknown)),
83     )
84
85     # Merge: start from defaults, overlay file data (Pydantic will validate/coerce).
86     merged = {**_get_built_in_defaults(), **file_data}
87
88     return AppConfig.model_validate(merged)
89
90
91 def save_default_config(path: str | Path | None = None) -> Path:
92     """Write a fresh default config.json.
93
94     Used by setup.py diagnostics and the --setup CLI path.
95     Overwrites if the file exists.
96     """
97     cfg_path = Path(path) if path is not None else DEFAULT_CONFIG_PATH
98     cfg = AppConfig()
99     cfg_path.parent.mkdir(parents=True, exist_ok=True)
100    cfg_path.write_text(cfg.model_dump_json(indent=4), encoding="utf-8")
101    return cfg_path
102
103
104    # Backwards-compatible alias used by some older call sites during transition.
105    def get_default_config() -> dict[str, Any]:
106        """Return defaults as a plain dict (for transitional use only)."""
107        return _get_built_in_defaults()
108

```

25. user (2026-06-14T10:33:54.793Z)

```

240    class AppConfig(BaseModel):
241        """Root configuration object.
242
243        All runtime configuration should be accessed via an instance of this class
244        (or the loader) rather than raw dict lookups.
245        """
246
247        sport: Sport = "badminton"
248        ai_provider: str = "gemini"
249        # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable
250        # the alias never deprecates underneath us. Eval reports record the run date
251        ai_model: str = "gemini-flash-latest"
252
253        validation: ValidationConfig = Field(default_factory=ValidationConfig)
254        indexing: IndexingConfig = Field(default_factory=IndexingConfig)
255        output: OutputConfig = Field(default_factory=OutputConfig)
256        stitching: StitchingConfig = Field(default_factory=StitchingConfig)
257        security: SecurityConfig = Field(default_factory=SecurityConfig)
258        logging: LoggingConfig = Field(default_factory=LoggingConfig)
259
260        # Allow extra keys during transition so existing config.json files do not break.
261        model_config = {
262            "extra": "allow",
263            "validate_assignment": True,
264        }
265
266        # Temporary migration shim for legacy dict-style access (config.get("section",
267        {}).get(...)).
268        # Will be removed once all call sites (validators, segmenters, etc.) move to typed
269        # attribute access / the typed config. A nested *section* is returned as a plain

```

```

dict
269     # via model_dump, so legacy chained access keeps working ? returning the Pydantic
270     # section model instead breaks it (a model has no .get), which bit every validator
on
271     # real runs. Scalar leaves are returned as-is.
272     def get(self, key: str, default: Any = None) -> Any:
273         _MISSING = object()
274         val: Any = _MISSING
275         if key in type(self).model_fields or hasattr(self, key):
276             val = getattr(self, key, _MISSING)
277         if val is _MISSING:
278             # Legacy convenience: find a leaf key inside any nested section. Sections
are
279             # discovered dynamically from the model fields, so a future top-level
section
280             # (e.g. `audio`, `jobs`) is covered automatically without editing a
hardcoded list.
281             for fname in type(self).model_fields:
282                 section = getattr(self, fname, None)
283                 if isinstance(section, BaseModel) and hasattr(section, key):
284                     val = getattr(section, key)
285                     break
286             if val is _MISSING:
287                 return default
288             if isinstance(val, BaseModel):
289                 return val.model_dump(mode="json")
290             return val
291
292     def __getitem__(self, key: str) -> Any:
293         val = self.get(key)
294         if val is None and key not in type(self).model_fields:
295             # Re-raise like a real dict for unknown top-level
296             raise KeyError(key)
297         return val
298
299
300     # Convenience type alias for callers that want to annotate
301     Config = AppConfig
302

```

26. assistant (2026-06-14T10:34:30.152Z)

Everything needed is present and I now know how to reproduce production exactly: `load\_config()`
? `config.get("indexing")` is a plain dict, and the segmenter seems only read the passed-in
`idx\_cfg`. Let me confirm the runner's parse/windowing signatures and the `score` function so my
harness calls them correctly.

27. user (2026-06-14T10:34:33.003Z)

```

=== parse_trajectory_csv + trajectory_to_action_windows signatures ===
233:     def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
252:     def trajectory_to_action_windows(

```

```

=== trajectory_to_action_windows full signature ===
def trajectory_to_action_windows(
    points: List[TrajectoryPoint],
    fps: float,
    frame_width: float,
    chunk_start: float = 0.0,
    velocity_thresh: float = 0.02,
    merge_gap: float = 2.0,
    min_window_duration: float = 2.0,

```

```

=== metrics.score ===
def score(preds: List[Interval], gts: List[Interval],

```

```

        iou_threshold: float = 0.5,
        match_result: Optional[MatchResult] = None) -> Score:
    """Compute precision/recall/F1, mean matched IoU, and boundary errors.

    Pass a precomputed `match_result` to avoid re-matching (e.g. when also
    extracting the failure lists); otherwise matching is done here.
    """
    mr = match_result if match_result is not None else match_intervals(preds, gts,
iou_threshold)

    tp = len(mr.matches)
    fp = len(mr.unmatched_pred_indices)
    fn = len(mr.unmatched_gt_indices)

    precision = tp / (tp + fp) if (tp + fp) > 0 else 0.0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0.0
    f1 = (2 * precision * recall / (precision + recall)) if (precision + recall) > 0 else 0.0

    if mr.matches:
        mean_iou = sum(m.iou for m in mr.matches) / tp
        mean_start_err = sum(abs(preds[m.pred_index][0] - gts[m.gt_index][0]) for m in
mr.matches) / tp
        mean_end_err = sum(abs(preds[m.pred_index][1] - gts[m.gt_index][1]) for m in mr.matches)
/ tp
    else:
        mean_iou = mean_start_err = mean_end_err = 0.0

    return Score(
        iou_threshold=iou_threshold,
        num_pred=len(preds), num_gt=len(gts),
        true_positives=tp, false_positives=fp, false_negatives=fn,
        precision=precision, recall=recall, f1=f1,
        mean_iou=mean_iou, mean_start_error=mean_start_err, mean_end_error=mean_end_err,
    )

def pr_curve(preds: List[Interval], gts: List[Interval],
            thresholds: Optional[List[float]] = None) -> List[Score]:
    """Score across a sweep of IoU thresholds (default 0.1..0.9 step 0.1).

    Lets you see strict-vs-lenient behaviour rather than a single operating point.
    """
    if thresholds is None:
        thresholds = [round(0.1 * i, 1) for i in range(1, 10)]
    return [score(preds, gts, iou_threshold=t) for t in thresholds]

=== label_candidates ===
def label_candidates(candidate_intervals: List[Interval], gt_intervals: List[Interval],
                    iou_threshold: float = 0.5) -> List[int]:
    """Label each candidate window 1 (real rally) or 0 (spurious).

    A window is positive iff it is matched one-to-one to a ground-truth rally at
    `iou_threshold` ? the *same* matching the evaluator uses for true positives,
    so the classifier is trained on exactly what we score. Substrate-agnostic:
    works for YOLO or WASB-derived windows alike.
    """
    mr = match_intervals(candidate_intervals, gt_intervals, iou_threshold)
    positive = {m.pred_index for m in mr.matches}
    return [1 if i in positive else 0 for i in range(len(candidate_intervals))]

def build_training_set(
    candidate_rows: List[Dict],
    gt_intervals: List[Interval],
    iou_threshold: float = 0.5,

```

```

    feature_fn: Callable[[Dict], List[float]] = extract_yolo_features,
) -> Tuple[List[List[float]], List[int], List[Interval]]:
    """Turn one video's candidate rows + ground truth into (X, y, intervals).

    `feature_fn` is pluggable so a WASB substrate can supply shuttle-motion
    features without changing the labeling logic. Returns the feature matrix `X`,
    binary labels `y`, and the candidate intervals (handy for downstream merging).
    """
    intervals: List[Interval] = [
        (float(r["start_time"]), float(r["end_time"])) for r in candidate_rows
    ]
    y = label_candidates(intervals, gt_intervals, iou_threshold)
    X = [feature_fn(r) for r in candidate_rows]
    return X, y, intervals

```

28. assistant (2026-06-14T10:35:12.153Z)

I have everything. Let me confirm `wasb\_hybrid` inherits the base seams unchanged (so the mc=0 identity claim is real), then write the verification harness.

29. user (2026-06-14T10:35:12.968Z)

```

1      """WASB-based hybrid play segmenter.
2
3      Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via
4      WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained
5      badminton weights and runs today, so this is the live shuttle-motion substrate.
6
7      For the per-video *calibration* goal we mainly use Phase 2 (trajectory ->
8      candidate windows); the windowing thresholds (velocity_thresh / merge_gap /
9      min_window_duration) are what get tuned against golden rally labels.
10
11     The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB)
12     lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the
13     WASB runner and identity.
14     """
15     from typing import Any, Dict, Tuple
16
17     from backend.pipeline.segmenters.base import segmenter_registry
18     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
19     from backend.pipeline.detectors.base import DetectorRunner
20     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
21
22
23     class WasbHybridSegmenter(TrajectoryHybridSegmenter):
24         family = "wasb"
25         display_label = "WASB"
26
27         @property
28         def name(self) -> str:
29             return "wasb_hybrid"
30
31         @property
32         def description(self) -> str:
33             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
34                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
35
36         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
37             cfg = WasbConfig.from_indexing_cfg(idx_cfg)
38             return WasbRunner(cfg), {"weights_path": cfg.weights_path}
39
40

```

```
41     segmenter_registry.register(WasbHybridSegmenter)
42
```

30. assistant (2026-06-14T10:35:40.460Z)

Confirmed: `WasbHybridSegmenter` inherits both base seams unchanged, and at `min\_crossings=0` the fusion windowing params equal wasb's ? so the mc=0 identity is real and I'll demonstrate it empirically rather than assert it. Now I'll write the served-side verification harness.

31. assistant (2026-06-14T10:36:23.412Z)

```
"""Served-side verification of the Gate-A default flip (PR #146).
```

Confirms the LIVE `FusionSegmenter` served path (not the offline harness's recall-oriented proposal) behaves as the flip promises, WITHOUT re-running WASB (wasb_runner has no cache-hit early-return ? re-running would overwrite the cached CSV). We reuse each golden video's CACHED WASB trajectory and drive the REAL served code: the same production windowing `process\_video` uses, then the actual `FusionSegmenter.\_enrich\_candidate\_stats` (net_crossings) and `\_select\_for\_handoff` (the served Gate-A gate) ? the exact methods Gemini's handoff set is built from.

Two questions (the owner-gated follow-up flagged in #146):

- (a) At indexing.fusion.min_crossings=0 (ungated control), does the fusion handoff set match a wasb_hybrid run on the same trajectory? (byte-identical seeding + handoff at mc=0 ? the no-regression invariant on the real served path.)
- (b) At min_crossings=2 (the Gate-A default), does it drop ONLY low-crossing (held-shuttle/feed) windows ? no real-rally loss vs golden labels ? i.e. no F1 regression?

Config is loaded EXACTLY as production (`load\_config()`); segmenters come from the registry. Honest measured numbers only. Artifacts isolated under output/gateA_served_verify/. No GPU, no Gemini, no WASB re-run.

```
"""
from __future__ import annotations
```

```
import copy
import json
import os
from typing import Any, Dict, List, Tuple
```

```
from backend.config.loader import load_config
from backend.pipeline.segmenters.base import segmenter_registry
```

Import for side-effect registration of both segmenters.

```
import backend.pipeline.segmenters.fusion_hybrid # noqa: F401
import backend.pipeline.segmenters.wasb_hybrid # noqa: F401
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
from backend.eval.calibrate_wasb import load_golden_gts
from backend.eval.training import label_candidates
from backend.eval.metrics import score
```

```
Interval = Tuple[float, float]
MANIFEST = "output/human_lovo_manifest.json"
OUT_DIR = "output/gateA_served_verify"
IOU = 0.5
```

```
def win_params(idx_cfg: Dict[str, Any], family: str) -> Tuple[float, float, float]:
    """Reproduce TrajectoryHybridSegmenter.process_video's windowing knobs exactly."""
    vt = idx_cfg.get(family, {}).get("velocity_threshold", 0.02)
    mg = idx_cfg.get("dead_time_merge_gap", 2.0)
    mwd = idx_cfg.get("min_action_window_duration", 2.0)
    return float(vt), float(mg), float(mwd)
```

```

def windows_for(points, fps: float, fw: float, vt: float, mg: float, mwd: float) ->
List[Dict[str, Any]]:
    return TrackNetRunner.trajectory_to_action_windows(
        points, fps=fps, frame_width=fw, chunk_start=0.0,
        velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd, chunk_index=0,
    )

def ivs(windows: List[Dict[str, Any]], idxs) -> List[Interval]:
    return [(windows[i]["start"], windows[i]["end"]) for i in idxs]

def run_video(spec: Dict[str, Any], fusion, wasb, base_idx_cfg) -> Dict[str, Any]:
    name = spec["name"]
    fps, fw = float(spec["fps"]), float(spec["frame_width"])
    points = TrackNetRunner.parse_trajectory_csv(spec["trajectory"])
    gts = load_golden_gts(spec["labels"])

    # --- Served windowing for each segmenter, from its own family config ---
    f_vt, f_mg, f_mwd = win_params(base_idx_cfg, fusion.family) # family="fusion"
    w_vt, w_mg, w_mwd = win_params(base_idx_cfg, wasb.family) # family="wasb"
    f_windows = windows_for(points, fps, fw, f_vt, f_mg, f_mwd)
    w_windows = windows_for(points, fps, fw, w_vt, w_mg, w_mwd)

    # (a) Same seeding: fusion and wasb produce identical candidate windows.
    f_iv = [(w["start"], w["end"]) for w in f_windows]
    w_iv = [(w["start"], w["end"]) for w in w_windows]
    seeding_identical = f_iv == w_iv

    # Real served enrichment (net_crossings) via the actual FusionSegmenter method.
    f_stats = [fusion._enrich_candidate_stats(cw, points, fps, fw, "", base_idx_cfg)
                for cw in f_windows]

    # Handoff selection through the REAL served gate at mc=0 and mc=2.
    cfg0 = copy.deepcopy(base_idx_cfg); cfg0.setdefault("fusion", {})[ "min_crossings" ] = 0
    cfg2 = copy.deepcopy(base_idx_cfg); cfg2.setdefault("fusion", {})[ "min_crossings" ] = 2
    keep0 = fusion._select_for_handoff(f_windows, f_stats, cfg0)
    keep2 = fusion._select_for_handoff(f_windows, f_stats, cfg2)
    # wasb_hybrid's handoff = the base seam (all windows, no gating).
    wasb_keep = wasb._select_for_handoff(w_windows, [{ } for _ in w_windows], base_idx_cfg)

    # (a) Handoff identity at mc=0: same windows AND same selected set as wasb_hybrid.
    handoff_identical = (seeding_identical and list(keep0) == list(range(len(f_windows)))
                        and list(wasb_keep) == list(range(len(w_windows))))

    # (b) Label every window once vs golden; classify each Gate-A drop as FP or real-rally loss.
    labels = label_candidates(f_iv, gts, IOU)
    dropped = [i for i in keep0 if i not in set(keep2)]
    drops = [{"start": round(f_windows[i]["start"], 2), "end": round(f_windows[i]["end"], 2),
        "net_crossings": f_stats[i].get("net_crossings", 0.0),
        "is_real_rally": bool(labels[i])} for i in dropped]
    real_rally_losses = [d for d in drops if d["is_real_rally"]]

    s0 = score(ivs(f_windows, keep0), gts, IOU)
    s2 = score(ivs(f_windows, keep2), gts, IOU)

    return {
        "name": name, "fps": fps,
        "num_gts": len(gts), "num_windows": len(f_windows),
        "seeding_identical_to_wasb": seeding_identical,
        "handoff_identical_at_mc0": handoff_identical,
        "win_params_fusion": [f_vt, f_mg, f_mwd], "win_params_wasb": [w_vt, w_mg, w_mwd],
        "handoff_mc0": len(keep0), "handoff_mc2": len(keep2),
        "dropped_by_gateA": len(dropped),
        "real_rally_losses": len(real_rally_losses),
    }

```

```

    "drops": drops,
    "mc0": {"P": round(s0.precision, 3), "R": round(s0.recall, 3), "F1": round(s0.f1, 3),
            "TP": s0.true_positives, "FP": s0.false_positives, "FN": s0.false_negatives},
    "mc2": {"P": round(s2.precision, 3), "R": round(s2.recall, 3), "F1": round(s2.f1, 3),
            "TP": s2.true_positives, "FP": s2.false_positives, "FN": s2.false_negatives},
    "delta_f1": round(s2.f1 - s0.f1, 3),
}

def main() -> None:
    os.makedirs(OUT_DIR, exist_ok=True)
    config = load_config() # exactly production
    idx_cfg = config.get("indexing", {}) # plain dict, identical to served path
    FusionCls = segmenter_registry.get("fusion_hybrid")
    WasbCls = segmenter_registry.get("wasb_hybrid")
    fusion = FusionCls(None, config)
    wasb = WasbCls(None, config)

    specs = json.load(open(MANIFEST, encoding="utf-8"))
    print("=" * 92)
    print("SERVED-SIDE Gate-A verification ? live FusionSegmenter over CACHED golden
trajectories")
    print(f"(config.indexing.fusion = {idx_cfg.get('fusion')})")
    print("=" * 92)

    results = [run_video(s, fusion, wasb, idx_cfg) for s in specs]

    print(f"\n{'video':<22}{ 'win':>4}{ 'mc0':>5}{ 'mc2':>5}{ 'drop':>5}{ 'lost':>5}"
          f"{'F1@mc0':>8}{ 'F1@mc2':>8}{ '?F1':>7} seed=wasb handoff==wasb@mc0")
    all_identical = True
    total_lost = 0
    for r in results:
        all_identical &= r["seeding_identical_to_wasb"] and r["handoff_identical_at_mc0"]
        total_lost += r["real_rally_losses"]
        print(f"{'name':<22}{ 'num_windows':>4}{ r['handoff_mc0']:>5}{ r['handoff_mc2']:>5}"
              f"{'dropped_by_gateA':>5}{ r['real_rally_losses']:>5}"
              f"{'mc0':['F1']:>8.3f}{ r['mc2']['F1']:>8.3f}{ r['delta_f1']:>+7.3f}"
              f" {str(r['seeding_identical_to_wasb']):>5}"
              f"{'handoff_identical_at_mc0':>5}")

    n = len(results)
    mean_f1_0 = sum(r["mc0"]["F1"] for r in results) / n
    mean_f1_2 = sum(r["mc2"]["F1"] for r in results) / n
    mean_p0 = sum(r["mc0"]["P"] for r in results) / n
    mean_p2 = sum(r["mc2"]["P"] for r in results) / n
    mean_r0 = sum(r["mc0"]["R"] for r in results) / n
    mean_r2 = sum(r["mc2"]["R"] for r in results) / n

    print("-" * 92)
    print(f"MEAN over n={n}: P {mean_p0:.3f}->{mean_p2:.3f} R {mean_r0:.3f}->{mean_r2:.3f}"
          "
          f"F1 {mean_f1_0:.3f}->{mean_f1_2:.3f} (? {mean_f1_2 - mean_f1_0:+.3f})")
    print(f"\n(a) mc=0 seeding+handoff byte-identical to wasb_hybrid on all videos : "
          f"{'YES ?' if all_identical else 'NO ?'}")
    print(f"(b) real-rally windows dropped by Gate-A (mc=2) across corpus : "
          f"{total_lost} {'(none ? drops are held-shuttle/feed FPs) ?' if total_lost == 0 else
'?'})")
    print(f" F1 regression on any video : "
          f"{'NONE ?' if all(r['delta_f1'] >= 0 for r in results) else 'YES ?'}")

    summary = {
        "purpose": "served-side Gate-A flip verification (PR #146) via live FusionSegmenter "
        "over cached golden trajectories ? no WASB re-run, no Gemini",
        "config_fusion": idx_cfg.get("fusion"),
        "iou": IOU, "n": n,
    }

```

```

    "mc0_seeding_handoff_identical_to_wasb_all_videos": all_identical,
    "total_real_rally_losses_gateA": total_lost,
    "no_f1_regression_any_video": all(r["delta_f1"] >= 0 for r in results),
    "mean_f1_mc0": round(mean_f1_0, 3), "mean_f1_mc2": round(mean_f1_2, 3),
    "mean_delta_f1": round(mean_f1_2 - mean_f1_0, 3),
    "per_video": results,
}
json.dump(summary, open(os.path.join(OUT_DIR, "served_gateA_summary.json"), "w"), indent=2)
print(f"\nwrote {OUT_DIR}/served_gateA_summary.json")

if __name__ == "__main__":
    main()

```

32. user (2026-06-14T10:36:23.442Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\served_gateA_verify.py (file state is current in your context ? no need to Read it back)

33. assistant (2026-06-14T10:36:27.158Z)

Now let me run the served-side verification harness.

34. user (2026-06-14T10:36:34.735Z)

```

=====
SERVED-SIDE Gate-A verification ? live FusionSegmenter over CACHED golden trajectories
(config.indexing.fusion = {'substrate': 'wasb', 'velocity_threshold': 0.02, 'cue_flags': {},
'min_crossings': 2, 'min_players': 0, 'court_polygon': None, 'net_axis': 'y', 'net_position':
0.5})
=====

```

video	win	mc0	mc2	drop	lost	F1@mc0	F1@mc2	?F1	seed=wasb	
handoff==wasb@mc0										
adarsh_avi_singles	97	97	76	21	7	0.632	0.628	-0.004	True	True
kushagra_singles	2	2	2	0	0	0.000	0.000	+0.000	True	True
largetest_doubles	52	52	42	10	5	0.535	0.484	-0.051	True	True
gbaaddy	22	22	20	2	0	0.086	0.088	+0.003	True	True
testlarge_short	5	5	5	0	0	0.182	0.182	+0.000	True	True
mahadevpura_singles	24	24	23	1	0	0.364	0.370	+0.007	True	True

MEAN over n=6: P 0.317->0.337 R 0.290->0.261 F1 0.300->0.292 (? -0.008)

- (a) mc=0 seeding+handoff byte-identical to wasb_hybrid on all videos : YES ?
- (b) real-rally windows dropped by Gate-A (mc=2) across corpus : 12 ?
- F1 regression on any video : YES ?

wrote output/gateA_served_verify/served_gateA_summary.json

35. assistant (2026-06-14T10:37:32.395Z)

Important ? this is a substantive finding, not a rubber-stamp. Let me read it carefully:

- ***(a) holds****: at `mc=0` the fusion seeding + handoff is byte-identical to `wasb\_hybrid` on all 6 videos. ? The no-regression invariant on the real served path is confirmed.
- ***(b) does NOT reproduce the offline win****: on the ****production**** windowing, Gate-A drops ****12 windows that actually match golden rallies**** (real-rally loss), mean F1 goes 0.300?0.292 (****?0.008****), with largetest_doubles ?0.051.

The offline `+0.074` was measured on the recall-oriented **proposal** windowing `(0.005,1.0,0.5)`; production serves a coarse windowing `(0.02,2.0,2.0)`. I must verify this is a real windowing-driven effect and not a harness artifact before recording it. Let me re-run the offline LOVO baseline and inspect the dropped real-rally windows.

36. user (2026-06-14T10:37:38.382Z)

=== Re-run offline ab_n6_lovo (proposal windowing) to confirm +0.074 baseline ===

FIRST A/B EXPERIMENT ? n=6 golden corpus (offline, persisted trajectories)

	172 candidates	96 rallies	proposal-recall-ceiling=0.833
adarsh_avi_singles	172 candidates	96 rallies	proposal-recall-ceiling=0.833
kushagra_singles	20 candidates	32 rallies	proposal-recall-ceiling=0.125
largetest_doubles	86 candidates	49 rallies	proposal-recall-ceiling=0.816
gbaaddy	82 candidates	48 rallies	proposal-recall-ceiling=0.312
testlarge_short	14 candidates	6 rallies	proposal-recall-ceiling=0.500
mahadevpura_singles	57 candidates	31 rallies	proposal-recall-ceiling=0.645

--- per-video held-out F1 ---

video	CONTROL	GateA	learned
adarsh_avi_singles	0.597	0.722	0.613
kushagra_singles	0.154	0.163	0.163
largetest_doubles	0.593	0.667	0.535
gbaaddy	0.231	0.274	0.222
testlarge_short	0.300	0.375	0.000
mahadevpura_singles	0.455	0.571	0.308

--- A/B 1: CONTROL vs Gate-A (static) ? C1 honest aggregation ---

meanF1 0.388 -> 0.462 ?F1=+0.074 CI[0.042, 0.104] excl0=True sign={'wins': 6, 'losses': 0, 'ties': 0, 'n_effective': 6, 'p_value': 0.031}

--- A/B 2: CONTROL vs learned fusion (LOVO) ? C1 honest aggregation ---

meanF1 0.388 -> 0.307 ?F1=-0.081 CI[-0.173, -0.002] excl0=True sign={'wins': 2, 'losses': 4, 'ties': 0, 'n_effective': 6, 'p_value': 0.688}

--- C4 over-segmentation-aware metrics (mean across videos) ---

CONTROL	{'f1@10': 0.645, 'f1@25': 0.603, 'f1@50': 0.388, 'seg_count_ratio': 1.68}
gateA	{'f1@10': 0.748, 'f1@25': 0.698, 'f1@50': 0.462, 'seg_count_ratio': 1.01}
learned	{'f1@10': 0.564, 'f1@25': 0.509, 'f1@50': 0.307, 'seg_count_ratio': 0.89}
learned out-of-fold (C3)	mAP@tIoU[.3-.7] = 0.245

wrote output/ab_experiment_n6.json

37. user (2026-06-14T10:37:38.590Z)

=== Dropped real-rally windows (net_crossings) on production windowing ===

adarsh_avi_singles lost= 7 net_crossings of lost real rallies: [1.0, 0.0, 1.0, 1.0, 1.0, 1.0, 1.0]

dropped-FPs=14 (correct drops) total-windows=97 gts=96
dropped-FPs=0 (correct drops) total-windows=2 gts=32

largetest_doubles lost= 5 net_crossings of lost real rallies: [1.0, 1.0, 1.0, 1.0, 1.0]

dropped-FPs=5 (correct drops) total-windows=52 gts=49
dropped-FPs=2 (correct drops) total-windows=22 gts=48
dropped-FPs=0 (correct drops) total-windows=5 gts=6
dropped-FPs=1 (correct drops) total-windows=24 gts=31

38. assistant (2026-06-14T10:38:39.393Z)

Now the picture is rigorous and clear. Let me state what the two runs establish:

- **Offline harness reproduced exactly**: proposal windowing `(0.005,1.0,0.5)`, CONTROL F1 0.388 ? Gate-A 0.462, $^{+0.074}$, 6/6, $p=0.031$, seg_count_ratio $^{1.68 \pm 0.10}$ (it was cleaning up heavy over-segmentation).

- **Served path**: production windowing `(0.02,2.0,2.0)` already merges aggressively (low over-segmentation), so Gate-A has little FP to remove and instead **drops 12 real-rally windows** whose net_crossings are 0?1 (almost all exactly 1) ? mean F1 $^{0.008}$, worst on doubles.

The lost rallies sit at `net\_crossings=1`, which suggests `min\_crossings=1` might be the right served operating point. Let me sweep `min\_crossings ? {0,1,2,3}` on the production windowing to give the owner the operating-point curve ? that's directly decision-relevant.

39. assistant (2026-06-14T10:38:57.768Z)

"""Operating-point sweep for Gate-A on the SERVED (production) windowing.

served_gateA_verify.py showed the Gate-A default (min_crossings=2) drops real-rally windows on the production windowing because their WASB trajectory registered only 0-1 net crossings. This sweeps min_crossings to find the served operating point: for each threshold, mean F1 + total real-rally windows lost vs the mc=0 control. Reuses the same live FusionSegmenter seams. No GPU / Gemini / WASB re-run.

"""

```
from __future__ import annotations
```

```
import copy
```

```
import json
```

```
from typing import Any, Dict, List
```

```
from backend.config.loader import load_config
```

```
from backend.pipeline.segmenters.base import segmenter_registry
```

```
import backend.pipeline.segmenters.fusion_hybrid # noqa: F401
```

```
from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
```

```
from backend.eval.calibrate_wasb import load_golden_gts
```

```
from backend.eval.training import label_candidates
```

```
from backend.eval.metrics import score
```

```
MANIFEST = "output/human_lovo_manifest.json"
```

```
IOU = 0.5
```

```
THRESHOLDS = [0, 1, 2, 3]
```

```
def main() -> None:
```

```
    config = load_config()
```

```
    idx_cfg = config.get("indexing", {})
```

```
    fusion = segmenter_registry.get("fusion_hybrid")(None, config)
```

```
    specs = json.load(open(MANIFEST, encoding="utf-8"))
```

```
    vt = idx_cfg.get(fusion.family, {}).get("velocity_threshold", 0.02)
```

```
    mg = idx_cfg.get("dead_time_merge_gap", 2.0)
```

```
    mwd = idx_cfg.get("min_action_window_duration", 2.0)
```

```
    per_video: List[Dict[str, Any]] = []
```

```
    for s in specs:
```

```
        fps, fw = float(s["fps"]), float(s["frame_width"])
```

```
        points = TrackNetRunner.parse_trajectory_csv(s["trajectory"])
```

```
        gts = load_golden_gts(s["labels"])
```

```
        windows = TrackNetRunner.trajectory_to_action_windows(
```

```
            points, fps=fps, frame_width=fw, chunk_start=0.0,
```

```
            velocity_thresh=vt, merge_gap=mg, min_window_duration=mwd, chunk_index=0)
```

```
        stats = [fusion.enrich_candidate_stats(cw, points, fps, fw, "", idx_cfg) for cw in
```

```
        windows]
```

```
        iv = [(w["start"], w["end"]) for w in windows]
```

```
        labels = label_candidates(iv, gts, IOU)
```

```
        row: Dict[str, Any] = {"name": s["name"], "n_win": len(windows), "n_gts": len(gts)}
```

```
        base_keep = None
```

```
        for mc in THRESHOLDS:
```

```
            cfg = copy.deepcopy(idx_cfg); cfg.setdefault("fusion", {})["min_crossings"] = mc
```

```
            keep = fusion._select_for_handoff(windows, stats, cfg)
```

```
            if mc == 0:
```

```
                base_keep = set(keep)
```

```
            sc = score([iv[i] for i in keep], gts, IOU)
```

```
            lost = sum(1 for i in base_keep if i not in set(keep) and labels[i] == 1)
```

```
            row[f"mc{mc}"] = {"kept": len(keep), "F1": round(sc.f1, 3),
```

```
                            "P": round(sc.precision, 3), "R": round(sc.recall, 3),
```

```
                            "real_lost_vs_mc0": lost}
```

```
        per_video.append(row)
```

```

n = len(per_video)
print("=" * 88)
print("Gate-A operating-point sweep on the SERVED (production) windowing "
      f"(vt={vt}, merge_gap={mg}, min_win={mwd})")
print("=" * 88)
print(f"{'metric':<20}" + "".join(f"{'mc='+str(mc):>14}" for mc in THRESHOLDS))
mean_f1 = {mc: sum(v[f'mc{mc}']['F1'] for v in per_video) / n for mc in THRESHOLDS}
mean_p = {mc: sum(v[f'mc{mc}']['P'] for v in per_video) / n for mc in THRESHOLDS}
mean_r = {mc: sum(v[f'mc{mc}']['R'] for v in per_video) / n for mc in THRESHOLDS}
tot_lost = {mc: sum(v[f'mc{mc}']['real_lost_vs_mc0'] for v in per_video) for mc in THRESHOLDS}
print(f"{'mean F1':<20}" + "".join(f"{'mean_f1[mc]:>14.3f}" for mc in THRESHOLDS))
print(f"{'mean precision':<20}" + "".join(f"{'mean_p[mc]:>14.3f}" for mc in THRESHOLDS))
print(f"{'mean recall':<20}" + "".join(f"{'mean_r[mc]:>14.3f}" for mc in THRESHOLDS))
print(f"{'real rallies lost':<20}" + "".join(f"{'tot_lost[mc]:>14d}" for mc in THRESHOLDS))
print(f"{'?F1 vs mc=0':<20}" + "".join(f"{'mean_f1[mc]-mean_f1[0]:>14.3f}" for mc in THRESHOLDS))
print("\n(mc=2 is the current served default from PR #146; mc=0 = Gate-A OFF = wasb_hybrid)")

if __name__ == "__main__":
    main()

```

40. user (2026-06-14T10:38:57.804Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\served_gateA_sweep.py (file state is current in your context ? no need to Read it back)

41. user (2026-06-14T10:39:15.591Z)

```

=====
Gate-A operating-point sweep on the SERVED (production) windowing (vt=0.02, merge_gap=2.0, min_win=2.0)
=====

```

metric	mc=0	mc=1	mc=2	mc=3
mean F1	0.300	0.303	0.292	0.269
mean precision	0.317	0.327	0.337	0.338
mean recall	0.290	0.288	0.261	0.226
real rallies lost	0	1	12	28
?F1 vs mc=0	+0.000	+0.003	-0.008	-0.031

(mc=2 is the current served default from PR #146; mc=0 = Gate-A OFF = wasb_hybrid)

42. assistant (2026-06-14T10:39:40.446Z)

The operating-point curve is decisive and decision-relevant. On the ****served**** (production) windowing:

	mc=0 (off)	mc=1	**mc=2 (shipped)**	mc=3
mean F1	0.300	**0.303**	0.292	0.269
precision	0.317	0.327	0.337	0.338
recall	0.290	0.288	0.261	0.226
real rallies lost	0	1	12	28
?F1 vs off	?	+0.003	**?0.008**	?0.031

So at the served windowing, `mc=2` is mildly net-negative and drops 12 real rallies; `mc=1` would be the better operating point if Gate-A is kept on. Let me record this honestly in `docs/QUALITY\_ITERATIONS.md`. Let me read its structure first.

43. user (2026-06-14T10:39:42.675Z)

```

=== file size + headings ===
1:# Rally-detection quality iterations ? living log
25:## Baselines that drive strategy (IoU 0.5 vs human golden labels)
41:## Current track (live) ? newest first
43:### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
feature *(2026-06-14)*
52: promote noise and can't disable a structural guard (the proven Gate-A held-shuttle guard
stays on regardless of a thin
70:**Gate-A-only structural + `min_n = 5`**.
72:### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio = promising
(not promotable) *(2026-06-14)*
91:### Gate-A flipped to the SERVED DEFAULT (owner-approved) *(2026-06-14)*
92:**Gate-A (net-crossings ? 2) is now the DEFAULT detection behaviour.** The owner approved
promoting the
95:- `backend/config/models.py` ? `FusionConfig.min_crossings` default **0 ? 2** (Gate A on
wherever
103:Gate A (ungated handoff, reproduces the substrate byte-for-byte), or
`indexing.default_segmenter: wasb_hybrid`
116:### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
117:**Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The measured
win
120:bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
125:CONTROL 45 rallies vs **Gate-A 27** ; Gate-A dropped **18 low-crossing windows (~1.7 min)**
of likely
135:**Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
138:### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
140:**Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test **p =
0.031**, CI
146:Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure is
the C2 (calibration)
149:### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF *(2026-06-14)*
173:### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
174:**Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
188:**Data support for Gate A.** The same archive shows real rallies on the multi-court matches
contain
192:**Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
210:Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics); and
`fusion.velocity_threshold`
215:### Person cue extracted to a reusable producer (fusion P1) ?
[#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13
221:### Experiment harness: A/B without rollback risk ?
[#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
231:## Archived predecessors (the historical trail)
245:## Standing lessons (carry-forward)
258: Gate A leans on this.
264:- **Gate A already existed, just wasn't wired to production** ? the cheapest precision win
is plumbing, not
---- total lines ----
265 docs/QUALITY_ITERATIONS.md

```

44. assistant (2026-06-14T10:39:47.163Z)

Let me read the existing Gate-A entries to match style and place my entry correctly.

45. user (2026-06-14T10:39:48.034Z)

```

41      ## Current track (live) ? newest first
42
43      ### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
feature *(2026-06-14)*
44      The owner adopted the **HYBRID** posture: personalization as a **flag-gated FEATURE on a

```

```

generalization-first baseline**,
45     paired with a **contribution flywheel** so per-user gains and the shared baseline
compound together (full design in
46     [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md), merged
[#144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).
47     This checkpoint landed the three **honest-eval RAILS** that keep a per-user/small-N
tuning layer from promoting noise or
48     silently regressing a user:
49
50     - **Per-user promotion gate ? [#141](https://github.com/avidullu/badminton-highlight-
indexer/pull/141)**
51         (`backend/eval/per_user_gate.py`): a **min-N floor** + **structural-guard exemption**
? per-user/small-N tuning can't
52         promote noise and can't disable a structural guard (the proven Gate-A held-shuttle
guard stays on regardless of a thin
53         user corpus).
54     - **Held-out-USER learning curve ? [#142](https://github.com/avidullu/badminton-
highlight-indexer/pull/142)**
55         (`backend/eval/per_user_eval.py`): turns the vague "<K videos to superior quality"
claim into a **falsifiable release
56         number** (the `smallest_k_with_lift` measured on held-out users).
57     - **Per-user no-regression guard ? [#143](https://github.com/avidullu/badminton-
highlight-indexer/pull/143)**: a
58         retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen
per-user CONTROL; a regressing
59         candidate is refused).
60
61     **Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
**not** new measured
62     rally-detection wins. No F1 moved this checkpoint; what landed is the discipline that
lets per-user tuning ship without
63     regressing the baseline or over-claiming on n=6-per-user data.
64
65     **Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
66     use (single-user local)*; the inline per-user classifier sits **behind a cloud-load
interface** (local store = one impl);
67     **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-
video user is ALWAYS fully segmented by
68     the baseline batteries); **free-tier videos ARE pooled** into the shared baseline
(consent baked into free terms,
69     minors-exclusion enforced at the pooling query); and the previously-pending structural-
guard sub-decision is now LOCKED ?
70     **Gate-A-only structural + `min_n = 5`**.
71
72     ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
73     The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
74     replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
75     test; never a bare mean at n=6)**:
76
77     | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
78     |---|---|---|---|---|---|
79     | shuttle-only | 0.307 | ? | ? | ? | ? |
80     | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
81     | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
82
83     **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
84     default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?

```

```

`players_in_court` counts spectators);
85     per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
86     (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
87     confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
88     *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
89     results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
90
91     ### Gate-A flipped to the SERVED DEFAULT (owner-approved) *(2026-06-14)*
92     **Gate-A (net-crossings ? 2) is now the DEFAULT detection behaviour.** The owner
approved promoting the
93     validated held-shuttle guard from "recommended config" to the live served default. The
flip is **two minimal,
94     config-reversible changes**:
95     - `backend/config/models.py` ? `FusionConfig.min_crossings` default **0 ? 2** (Gate A on
wherever
96     `fusion_hybrid` serves).
97     - `config.json` ? `indexing.default_segmenter` **`gemini` ? `fusion_hybrid`** (+ an
explicit `indexing.fusion`
98     block making the served operating point visible). With default config,
`FusionSegmenter` reproduces the WASB
99     substrate's seeding/handoff exactly *except* that sub-2-crossing windows are dropped
from the AI handoff ? the
100     held-shuttle / one-sided-feed / warm-up false-positive fix.
101
102     **Reversible ? rollback is a single flag** (no code change): set
`indexing.fusion.min_crossings: 0` to disable
103     Gate A (ungated handoff, reproduces the substrate byte-for-byte), or
`indexing.default_segmenter: wasb_hybrid`
104     to revert the served segmenter entirely. **Gate B stays OFF** (`min_players = 0`, person
cue OFF). **The
105     persisted candidate superset is unchanged** ? gating filters only the AI handoff, so the
offline harness can
106     still re-score any variant; the offline ablation harness is untouched (it sets
`Variant.min_crossings`
107     explicitly and never reads `FusionConfig`). The validation evidence is unchanged from
the promotion below
108     (?F1 +0.074, 6/6, p=0.031, CI [+0.042, +0.105], over-seg 1.68?1.01) plus the GX010125
45?27 corroboration.
109     **Recorded:** a `promotion` record appended to
`eval_baselines/experiment_leaderboard.json`. ? **Honesty
110     boundary:** the offline ?F1 is on the recall-oriented *proposal* candidate set, and the
production fusion
111     segmenter's served path is still only mock-tested on real video (P2). The gate's effect
is **byte-reversible**,
112     so this ships behind a one-flag rollback; a golden *served* regression on the fusion
path remains a worthwhile
113     optional follow-up (owner-gated GPU+Gemini run) to confirm no served-side regression,
but it does not block the
114     flip ? the offline 6/6 + p=0.031 ablation and the unit tests are the gate.
115
116     ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
117     **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
118     (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
119     [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest
120     bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
121     folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the

```

```

recall-oriented *proposal*
122     candidate set; flipping the **live served default** stayed owner-gated at the time of
this entry ? **now resolved:
123     the owner approved the flip; see the SERVED-DEFAULT entry above.** **Corroborated on
real footage:** offline
124     re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated
`output/GX010125_run/`) ? 45 candidates ?
125     CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7
min)** of likely
126     held-shuttle/feed FPS (no golden labels for this footage ? divergence, not F1).
127
128     **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
129     first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
130     additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train**
fold) + `calibrate`
131     (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423
(+0.116)** , testlarge
132     0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)** . Threshold-in-LOVO
is the big lever;
133     calibration is available too.
134
135     **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
136     ([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are
self-contained.
137
138     ### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
139     First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB
trajectories, no GPU).
140     **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test
**p = 0.031** , CI
141     [+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01** . The window-
level 5-feature logistic
142     **did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the
stronger per-frame
143     prototype). The framework held up: **C1 promoted one variant and refused another; C4
surfaced over-segmentation
144     tIoU hid.** Full record + tables ?
145     [`archives/quality-iterations/2026-06-first-ab-experiment/`](archives/quality-
iterations/2026-06-first-ab-experiment/README.md).
146     Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure
is the C2 (calibration)
147     + threshold-in-loop task list.
148
149     ### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF
*(2026-06-14)*
150     **Why.** A multi-agent literature review (folded into
151     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13) confirmed the
framework's *skeleton* is
152     standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion
? cite, don't claim),
153     and flagged that our **measurement** is walking into known small-N traps that would make
"promote only on
154     measured lift" promote noise or hide a real win. Four corrections, shipped as **pure,
additive helpers** (no
155     existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):

```

46. assistant (2026-06-14T10:40:28.876Z)

The existing flip entry (lines 110?114) explicitly anticipated this exact follow-up: "the offline ?F1 is on the recall-oriented proposal candidate set? a golden served regression on the fusion path remains a worthwhile optional follow-up." My run answers it. Let me record the honest result at the top of the live track.

47. user (2026-06-14T10:40:53.416Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

48. assistant (2026-06-14T10:41:00.272Z)

Now let me update the live memory checkpoint with this finding.

49. user (2026-06-14T10:41:15.489Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

50. user (2026-06-14T10:41:21.810Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-14 (PERSONALIZATION Phase-0 LANDED + quality checkpoint).
**Owner adopted the HYBRID** ? personalization as a
```

51. user (2026-06-14T10:41:38.079Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

52. assistant (2026-06-14T10:41:43.033Z)

Recorded. Let me verify the working-tree state (I have not touched `config.json` ? the rollback is owner-gated).

53. user (2026-06-14T10:41:44.696Z)

```
M docs/QUALITY_ITERATIONS.md
?? artifacts/
--- confirm config.json untouched (still mc=2) ---
min_crossings = 2
```

54. assistant (2026-06-14T10:42:18.692Z)

Verification complete and recorded. Only `docs/QUALITY\_ITERATIONS.md` is modified; `config.json` is untouched (still `mc=2`), scratch scripts and artifacts are in gitignored dirs. Here's what I found.

Served-side Gate-A verification ? result

I drove the **live** `FusionSegmenter` served seams* (`\_enrich\_candidate\_stats` ? `\_net\_crossings`, `\_select\_for\_handoff` ? the gate) over each golden video's **cached** WASB trajectory, at the **production windowing** `process\_video` actually uses (`vt=0.02, merge\_gap=2.0, min\_win=2.0`, loaded via `load\_config()`). No WASB re-run, no GPU, no Gemini. Repro: [served_gateA_verify.py](scratch/served_gateA_verify.py) + [served_gateA_sweep.py](scratch/served_gateA_sweep.py).

(a)** No-regression invariant ? CONFIRMED ?.** At `min\_crossings=0`, `fusion\_hybrid`'s seeding *and**** handoff set are ****byte-identical** to `wasb\_hybrid` on all 6 golden videos**. The "fusion adds nothing when the gates are off" property is now verified on the **live** served code, not just by #117's review.

(b)** The `+0.074` win does NOT transfer to the served path ??.** That win was measured on the eval harness's recall-oriented *proposal**** windowing `(0.005,1.0,0.5)`, where CONTROL over-segments (seg-ratio 1.68) and Gate-A's job is cleaning up fragmentation (?1.01). Production serves a ****coarse**** windowing that already merges aggressively ? so Gate-A has little FP to remove and instead trims ****real rallies**** whose WASB trajectory registered only 0?1 net crossings:

`min_crossings`	mean F1	precision	recall	real-rally windows lost	?F1	
0 (off = `wasb_hybrid`)	0.300	0.317	0.290	0	?	
1	**0.303**	0.327	0.288	1	**+0.003**	
2 (shipped #146)	0.292	0.337	0.261	**12**	**?0.008**	
3	0.269	0.338	0.226	28	?0.031	

The shipped `mc=2` is a small mean-F1 ****regression** (?0.008)** and drops ****12** windows that match golden rallies** (worst on doubles, `largetest\_doubles` ?0.051`). Honest caveats: deltas are **\*\*within n=6 noise\*\*** (not catastrophic), but `mc=2` is **not** a served improvement ? and because a dropped candidate ****never reaches Gemini****, each lost rally is an ****unrecoverable** served recall loss**, while the precision "gain" partly duplicates what Gemini already does at the handoff. So the served tradeoff is ****? neutral****.

****Root cause:**** eval-harness windowing ? served windowing. Gate-A's value is conditional on over-segmentation; the coarse served windowing doesn't over-segment, so the gate inverts.

This is exactly the served-side follow-up the flip entry said was owed ? it doesn't unwind the merge, but the served numbers don't support `mc=2` as the default. The decision on what to do with the just-merged default is yours:

55. user (2026-06-14T10:42:30.440Z)

The user did not answer the questions.

56. assistant (2026-06-14T10:42:54.148Z)

I'll proceed with the safe default: no unilateral change to a production quality-default, and no commit without your go-ahead. Here's where things stand.

Done

- ****Served-side verification ran**** (offline, reusing cached golden trajectories ? no WASB re-run, no GPU, no Gemini).
- ***(a) Confirmed****: at `mc=0`, `fusion\_hybrid` is byte-identical to `wasb\_hybrid` (seeding + handoff) on all 6 golden videos ? the no-regression invariant holds on the live served code.
- ***(b) Surfaced honestly****: the offline `+0.074` win does ****not**** transfer to the served (production) windowing; the shipped `mc=2` is a within-noise mean-F1 regression (?0.008) that drops 12 real-rally windows. `mc=1` is the better served operating point.
- ****Recorded**** in docs/QUALITY_ITERATIONS.md (top of live track) and in the memory checkpoint (durable regardless of commit).

Working-tree state (nothing pushed, nothing committed)

- `docs/QUALITY\_ITERATIONS.md` ? ****modified, uncommitted**** (the honest record).
- `config.json` ? ****untouched****, still `min\_crossings: 2`. I did not flip the default; that's your call.
- Repro scripts (`scratch/served\_gateA\_{verify,sweep}.py`) + artifacts (`output/gateA\_served\_verify/`) ? in gitignored dirs.

Your call (whenever ? not blocking)

The single-flag options are in the question above; my recommendation is ****`mc=1`**** (keeps the held-shuttle guard's intent ? drops 0-crossing dead feeds ? without over-dropping real

1-crossing rallies). The deeper fix worth a separate item: ****align the experiment-harness windowing with the served windowing****, so a harness promotion actually predicts served behavior ? that mismatch is the real lesson here, beyond this one threshold.

Tell me which option you want (or "commit the doc") and I'll apply it on a branch + PR.

57. user (2026-06-14T10:54:22.141Z)

definitely commit the doc!

58. assistant (2026-06-14T10:54:52.743Z)

On it. Following the shared-checkout branch protocol (branch from `origin/master` explicitly, stage only the doc, re-verify HEAD before pushing). Let me fetch and check the remote first.

59. user (2026-06-14T10:54:56.235Z)

```
--- current branch ---
master
--- local HEAD ---
e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
--- origin/master HEAD ---
e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
--- did origin/master move / touch the doc? ---
--- working tree ---
  M docs/QUALITY_ITERATIONS.md
?? artifacts/
```

60. assistant (2026-06-14T10:55:01.061Z)

Clean: `origin/master == local HEAD == e75ec35`, no sibling commits, no remote changes to the doc. Branching from `origin/master` and committing the doc only.

61. user (2026-06-14T10:55:02.493Z)

```
M      docs/QUALITY_ITERATIONS.md
branch 'docs/served-gate-a-verification' set up to track 'origin/master'.
Switched to a new branch 'docs/served-gate-a-verification'
--- staged path (explicit) ---
M  docs/QUALITY_ITERATIONS.md
?? artifacts/
```

62. user (2026-06-14T10:55:24.807Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_servedgatea.txt (file state is current in your context ? no need to Read it back)

63. assistant (2026-06-14T10:55:18.906Z)

On the branch with only the doc staged (`artifacts/` untracked, not staged). Committing via a message file to avoid shell-quoting issues.

64. assistant (2026-06-14T10:55:24.795Z)

docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to production windowing

Owner-gated follow-up flagged in PR #146. Drove the live FusionSegmenter served seams (`_enrich_candidate_stats -> net_crossings, _select_for_handoff -> the gate`) over each golden video's CACHED WASB trajectory at the production windowing process_video uses (`vt=0.02, merge_gap=2.0, min_win=2.0`). No WASB re-run / GPU / Gemini.

- (a) CONFIRMED: at min_crossings=0, fusion_hybrid seeding+handoff is byte-identical to wasb_hybrid on all 6 golden videos ? the no-regression invariant verified on live served code, not just by review.
- (b) FINDING: the +0.074 win was measured on the harness's recall-oriented proposal windowing (over-segments, seg-ratio 1.68 -> 1.01). The coarse served windowing already merges aggressively, so Gate-A trims real rallies (net_crossings 0-1) instead of FPs. Candidate-handoff F1 (IoU0.5, n=6): mc0 0.300 | mc1 0.303 (+0.003, 1 lost) | mc2 0.292 (-0.008, 12 lost) | mc3 0.269 (-0.031, 28 lost). Within n=6 noise, but mc2 is not a served improvement and drops genuine rallies.

Root cause: eval-harness windowing != served windowing. Honest record only; config.json untouched (mc=2 stays) ? the rollback/retune is owner-gated.
Repro: scratch/served_gateA_{verify,sweep}.py.

Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

65. user (2026-06-14T10:55:30.968Z)

```
[docs/served-gate-a-verification f6acde5] docs(quality): served-side Gate-A check ? flip's
offline win doesn't transfer to production windowing
  1 file changed, 47 insertions(+)
--- re-verify: HEAD is mine, only my commit ahead of origin/master ---
docs/served-gate-a-verification
f6acde5 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing
--- files in the commit (must be ONLY the doc) ---
f6acde5 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing
  docs/QUALITY_ITERATIONS.md | 47 +++++
  1 file changed, 47 insertions(+)
```

66. user (2026-06-14T10:56:04.590Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\prbody_servedgatea.md (file state is current in your context ? no need to Read it back)

67. assistant (2026-06-14T10:55:54.678Z)

Commit is clean: `f6acde5`, one file (`docs/QUALITY\_ITERATIONS.md`, +47), only my commit ahead of `origin/master`. Now pushing and opening a PR (docs-only, one work item).

68. assistant (2026-06-14T10:56:04.569Z)

Served-side Gate-A verification (follow-up to #146)

Owner-gated follow-up flagged by the Gate-A **SERVED-DEFAULT** entry in `QUALITY\_ITERATIONS.md`. Confirms whether the `min\_crossings=2` default flip causes a served-side regression on the **live** `FusionSegmenter` path ? driven over each golden video's **cached** WASB trajectory at the production windowing `process\_video` uses (`vt=0.02, merge\_gap=2.0, min\_win=2.0`, via `load\_config()`). No WASB re-run / GPU / Gemini.

(a) No-regression invariant ? CONFIRMED ?

At `min\_crossings=0`, `fusion\_hybrid` seeding **and** handoff are byte-identical to `wasb\_hybrid` on all 6 golden videos. The "fusion adds nothing when the gates are off" property is now verified on the live served code, not just by #117's review.

(b) The +0.074 win does NOT transfer to the served windowing ??

The offline win was measured on the harness's recall-oriented **proposal** windowing `(0.005,1.0,0.5)` where CONTROL over-segments (seg-ratio 1.68?1.01 = Gate-A cleaning fragmentation). Production serves a **coarse** windowing that already merges aggressively, so Gate-A has little FP to remove and instead trims **real rallies** (net_crossings 0?1).

Candidate-handoff F1 (pre-Gemini, IoU ? 0.5, n=6):

```
| `min_crossings` | mean F1 | precision | recall | real-rally windows lost | ?F1 |  
|---|---|---|---|---|---|  
| 0 (off = `wasb_hybrid`) | 0.300 | 0.317 | 0.290 | 0 | ? |  
| 1 | **0.303** | 0.327 | 0.288 | 1 | **+0.003** |  
| **2 (shipped #146)** | 0.292 | 0.337 | 0.261 | **12** | **?0.008** |  
| 3 | 0.269 | 0.338 | 0.226 | 28 | ?0.031 |
```

Within n=6 noise, but `mc=2` is **not** a served improvement and drops genuine rallies (worst: `largetest\_doubles` ?0.051). A dropped candidate never reaches Gemini ? **unrecoverable served recall loss**, while the precision "gain" partly duplicates what Gemini already does.

Root cause: eval-harness windowing ? served windowing ? the harness promotes/refuses variants on a windowing production doesn't serve.

Scope

Docs-only. `config.json` is untouched (`mc=2` stays); the rollback/retune (recommended `mc=1`) is owner-gated. Repro: `scratch/served\_gateA\_{verify,sweep}.py` (gitignored); artifacts in `output/gateA\_served\_verify/`.

? Generated with [Claude Code](https://claude.com/claude-code)

69. user (2026-06-14T10:56:13.778Z)

```
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/served-gate-a-verification  
remote:  
branch 'docs/served-gate-a-verification' set up to track 'origin/docs/served-gate-a-verification'.  
To https://github.com/avidullu/badminton-highlight-indexer.git  
  * [new branch]      docs/served-gate-a-verification -> docs/served-gate-a-verification  
--- create PR ---  
https://github.com/avidullu/badminton-highlight-indexer/pull/152
```

70. assistant (2026-06-14T10:56:25.476Z)

PR opened: [#152](https://github.com/avidullu/badminton-highlight-indexer/pull/152). Let me check CI status (gating explicitly on the status rollup, per the merge-discipline gotcha).

71. user (2026-06-14T10:56:47.787Z)

```
{"checks":[{"conclusion":"SUCCESS","name":"Ruff  
lint","status":"COMPLETED"}, {"conclusion":"","name":"Mypy (lenient  
baseline)","status":"IN_PROGRESS"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke  
(no GPU stack)","status":"COMPLETED"}, {"conclusion":"","name":"Full test suite (offline,  
mocked)","status":"IN_PROGRESS"}], "mergeStateStatus":"UNKNOWN", "mergeable":"UNKNOWN"}
```

72. assistant (2026-06-14T10:56:57.989Z)

CI running (2 of 4 green, mypy + full suite in progress). Let me wait for it to finish.

73. user (2026-06-14T10:56:58.734Z)

```
<tool_use_error>Blocked: sleep 110 followed by: gh pr view 152 --json statusCheckRollup --jq  
'[.statusCheckRollup[]][name,status,conclusion]' head -40. To wait for a condition, use Monitor  
with an until-loop (e.g. `until <check>; do sleep 2; done`). To wait for a command you started,  
use run_in_background: true. Do not chain shorter sleeps to work around this  
block.</tool_use_error>
```

74. user (2026-06-14T10:57:05.016Z)

```
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Mypy  
(lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU
```

```
stack)},{\"conclusion\":\"\",\"name\":\"Full test suite (offline,  
mocked)\"}],\"mergeStateStatus\":\"UNSTABLE\",\"mergeable\":\"MERGEABLE\"}
```

75. assistant (2026-06-14T10:57:15.996Z)

3 of 4 green; full test suite still running (`UNSTABLE` = pending check). I won't merge until all are SUCCESS. Let me wait on it with an until-loop.

76. user (2026-06-14T10:58:29.242Z)

```
waited ~60s
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Mypy (lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU stack)"}, {"conclusion":"SUCCESS","name":"Full test suite (offline, mocked)"}], "mergeStateStatus":"CLEAN", "mergeable":"MERGEABLE"}
```

77. assistant (2026-06-14T10:58:37.628Z)

All 4 CI lanes green, `mergeStateStatus: CLEAN`. This is a docs-only honest record ? within my standing authority to merge on verified-green CI, and you clearly want it landed. Merging (squash) and cleaning up.

78. user (2026-06-14T10:58:47.361Z)

```

tests/test_fusion_hybrid.py      | 86 +-----
tests/test_served_gate_a_regression.py | 160 ++++++
8 files changed, 478 insertions(+), 157 deletions(-)
create mode 100644 backend/eval/served_gate_a.py
create mode 100644 tests/test_served_gate_a_regression.py
--- back to master, ff pull ---
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
--- verify landed + clean ---
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
?? artifacts/
--- tidy temp commit/PR message files ---
removed temp msg files (repro scripts kept)

```

79. assistant (2026-06-14T10:59:21.313Z)

Important: while my docs PR was in flight, a **sibling session merged** [\[#151\]\(https://github.com/avidullu/badminton-highlight-indexer/pull/151\)](https://github.com/avidullu/badminton-highlight-indexer/pull/151) ? a revert of the Gate-A flip that independently reached the same conclusion ("confirmed served regression (eval?serve)") and even added a ``backend/eval/served_gate_a.py`` module + regression test. My docs PR #152 landed cleanly on top (``53bc743``, CI green). Let me reconcile ? my doc entry currently presents the rollback as still-pending, which may now be stale. Let me see what #151 actually did.

80. user (2026-06-14T10:59:24.083Z)

```
=== config.json now (did #151 flip mc back?) ===  
Traceback (most recent call last):  
  File "<string>", line 1, in <module>  
    import json;f=json.load(open('config.json'))['indexing'];print('default_segmenter  
=',f['default_segmenter']);print('fusion.min_crossings = ',f['fusion']['min_crossings'])  
~^^^^^^^^^^^  
KeyError: 'fusion'  
default_segmenter = gemini  
  
=== #151 commit body ===
```

revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression (eval?serve) (#151)

Restore the pre-#146 no-regression production state. The #146 flip promoted Gate A (net-crossings >= 2) from a recommended config to the live served default, on the strength of an offline +0.074 ?F1. A served-side check (the #148 litmus + an independent served run) found the lift does NOT transfer: on the production COARSE windowing Gate A is neutral-to-negative (mean served ?F1 -0.008, ~12 real rallies dropped, 2/6 losses).

Root cause (eval?serve, not a bug): the +0.074 is a property of the offline harness's permissive PROPOSAL windowing (vt=0.005, gap=1.0, min=0.5), which emits many short windows with held-shuttle junk for Gate A to cut. The served COARSE windowing (vt=0.02, gap=2.0, min=2.0) already under-segments (over-seg ratio < 1), so there is little junk left and the gate trims borderline true windows instead.

Reverted (back to the pre-#146 state):

- config.json: indexing.default_segmenter fusion_hybrid -> gemini (the prior served detector); the #146-added indexing.fusion block removed.
- backend/config/models.py: FusionConfig.min_crossings default 2 -> 0 (Gate A OFF by default). Gate A is NOT deleted ? it stays opt-in (indexing.fusion.min_crossings: 2).
- tests/test_config.py, tests/test_fusion_hybrid.py: restored to the Gate-A-OFF defaults (test_fusion_hybrid.py is byte-identical to its pre-#146 form). The persisted-superset invariant is preserved ? gating only ever filtered the AI handoff, never persistence.
- docs/QUALITY_ITERATIONS.md: the Gate-A SERVED-DEFAULT entry now reads REVERTED with the honest eval?serve reason; the PROMOTED entry's stale "owner approved the flip" pointer is corrected.
- eval_baselines/experiment_leaderboard.json: the #146 promotion record carries a "reverted (eval?serve)" note.

The #148 served litmus (backend/eval/served_gate_a.py + tests/test_served_gate_a_regression.py) is the standing guard that caught this and pins the honest measured behaviour.

ruff + mypy clean; full suite 605 passed, 2 skipped (golden-data-gated).

Co-authored-by: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

=== #151 files changed ===

2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression (eval?serve) (#151)

backend/config/models.py	33 +++-----
config.json	8 +--
docs/QUALITY_ITERATIONS.md	63 ++++++++-----
eval_baselines/experiment_leaderboard.json	1 +
tests/test_config.py	38 ++++++-----
tests/test_fusion_hybrid.py	86 ++-----

6 files changed, 72 insertions(+), 157 deletions(-)

81. assistant (2026-06-14T11:00:06.634Z)

Now I have the full picture. A sibling session did the complete rollback in parallel:

- **PR #148** productionized a served-side litmus: `backend/eval/served\_gate\_a.py` + `tests/test\_served\_gate\_a\_regression.py` (standing guard).
- **PR #151** reverted #146: `config.json` ? `default\_segmenter: gemini` (the `indexing.fusion` block removed ? that's why my probe hit `KeyError: 'fusion'`), `FusionConfig.min\_crossings` default 2?0 (Gate A now opt-in), and it already updated the SERVED-DEFAULT doc entry to "REVERTED". Its independent numbers **match mine** (?0.008, ~12 rallies dropped, 2/6 losses).

So my just-merged entry is now **stale** ? it says "config.json untouched (mc=2 stays)" and frames the rollback as pending, when it's already done. That misstates the live state. I need to reconcile the doc to keep it honest. Let me read the current top sections to fix it precisely without contradicting #151's wording.

82. user (2026-06-14T11:00:07.755Z)

43 ### Served-side Gate-A check on the LIVE FusionSegmenter ? the flip's offline win does

NOT transfer to the production windowing *(2026-06-14)*

44 The owner-gated follow-up flagged by the SERVED-DEFAULT entry below: confirm the live
`FusionSegmenter` served path has no
45 regression from the Gate-A default flip (PR #146). Driven the **real** served seams
(`enrich\_candidate\_stats` ?
46 `net\_crossings`, `\_select\_for\_handoff` ? the gate) over each golden video's **cached**
WASB trajectory, at the **production**
47 windowing **process_video** actually uses (`velocity_thresh=0.02, merge_gap=2.0,
min_window=2.0` from config) ? no WASB
48 re-run (the runner has no cache-hit early-return), no GPU, no Gemini. Config loaded
exactly as production (`load\_config`);
49 segmenters from the registry. Repro: `scratch/served\_gateA\_verify.py` +
`scratch/served\_gateA\_sweep.py`; artifacts isolated
50 in `output/gateA\_served\_verify/`.
51
52 - **(a)** No-regression invariant CONFIRMED on real served code. At `min\_crossings = 0`,
`fusion\_hybrid`'s seeding **and**
53 handoff set are **byte-identical** to `wasb\_hybrid` on all 6 golden videos (same
candidate windows; handoff = all windows =
54 the base seam). The "fusion adds nothing when the gates are off" property is now
verified on the **live** served path, not
55 only by code review. ?
56 - **(b)** The +0.074 win does NOT transfer to the served windowing. The offline
promotion (+0.074, 6/6, p=0.031) was measured
57 on the eval harness's **recall-oriented** **proposal** windowing `(0.005, 1.0, 0.5)`,
where CONTROL **over-segments**
58 (segment-count ratio 1.68) and Gate-A's job is to clean up that fragmentation (?
1.01). The **served** path uses a coarse
59 windowing `(0.02, 2.0, 2.0)` that already merges aggressively (low over-
segmentation), so Gate-A has little FP to remove
60 and instead trims **real** rallies whose WASB trajectory registered only 0?1 net
crossings. **Candidate-handoff-level F1**
61 (pre-Gemini, IoU ? 0.5) on the served windowing, n=6:
62
63 | min_crossings | mean F1 | precision | recall | real-rally windows lost vs off | ?F1
vs off |
64 |---|---|---|---|---|---|
65 | 0 (Gate-A OFF = `wasb\_hybrid`) | 0.300 | 0.317 | 0.290 | 0 | ? |
66 | 1 | **0.303** | 0.327 | 0.288 | 1 | **+0.003** |
67 | **2** (shipped default, PR #146) | 0.292 | 0.337 | 0.261 | **12** | **?0.008** |
68 | 3 | 0.269 | 0.338 | 0.226 | 28 | ?0.031 |
69
70 At the shipped `mc=2`, mean F1 goes **0.300 ? 0.292 (?0.008)** and Gate-A drops **12**
windows that match golden rallies
71 (net_crossings almost all exactly 1.0), concentrated in doubles (`largetest\_doubles`
?0.051). All deltas are **tiny** / within
72 n=6 noise, so this is not a catastrophic regression ? but `mc=2` is **not** a
served-side **improvement**, and the
73 held-shuttle "guard" is removing genuine rallies, not just feeds.
74 - **Served relevance (honest).** A dropped candidate window **never** reaches Gemini, so
each lost real rally is an
75 **unrecoverable** served recall loss; the precision "gain", by contrast, is partly
what Gemini already does at the handoff ?
76 so the served-side tradeoff is at best neutral and plausibly net-negative, i.e. the
candidate-level ?0.008 likely
77 **understates** the served downside.
78 - **Root cause = eval ? serve windowing.** The harness promotes/refuses variants on a
windowing the production path does not
79 serve. Gate-A's value is conditional on over-segmentation; the coarse served windowing
doesn't over-segment, so the gate
80 inverts. [[decision-rigor-norm]]: the foundational issue (harness windowing ? served
windowing) outweighs the single
81 proposal-windowing number.
82
83 **Recommendation (owner-gated, single-flag).** `mc=1` is the better served operating

```

point if Gate-A stays on (?flat F1, only
84     1 real rally lost, +0.010 precision); `mc=0` is the clean rollback. Either is a one-line
config change (no code):
85     `indexing.fusion.min_crossings: 1` or `: 0` (or full revert `indexing.default_segmenter:
wasb_hybrid`). The deeper fix is to
86     **align the experiment-harness windowing with the served windowing** (or re-run
promotions on the production windowing) before
87     treating a harness lift as a served default. The offline ablation and unit tests remain
valid for what they measured; this
88     adds the served-side number the flip entry said was still owed.
89
90     ### PERSONALIZATION Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
feature *(2026-06-14)*
91     The owner adopted the **HYBRID** posture: personalization as a **flag-gated FEATURE on a
generalization-first baseline**,
92     paired with a **contribution flywheel** so per-user gains and the shared baseline
compound together (full design in
93     [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md), merged
[144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).
94     This checkpoint landed the three **honest-eval RAILS** that keep a per-user/small-N
tuning layer from promoting noise or
95     silently regressing a user:
96
97     - **Per-user promotion gate ? [141](https://github.com/avidullu/badminton-highlight-
indexer/pull/141)**
98     (`backend/eval/per_user_gate.py`): a **min-N floor** + **structural-guard exemption**
? per-user/small-N tuning can't
99     promote noise and can't disable a structural guard (the proven Gate-A held-shuttle
guard stays on regardless of a thin
100     user corpus).
101     - **Held-out-USER learning curve ? [142](https://github.com/avidullu/badminton-
highlight-indexer/pull/142)**
102     (`backend/eval/per_user_eval.py`): turns the vague "<K videos to superior quality"
claim into a **falsifiable release
103     number** (the `smallest_k_with_lift` measured on held-out users).
104     - **Per-user no-regression guard ? [143](https://github.com/avidullu/badminton-
highlight-indexer/pull/143)**: a
105     retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen
per-user CONTROL; a regressing
106     candidate is refused).
107
108     **Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
**not** new measured
109     rally-detection wins. No F1 moved this checkpoint; what landed is the discipline that
lets per-user tuning ship without
110     regressing the baseline or over-claiming on n?6-per-user data.
111
112     **Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
113     use (single-user local)*; the inline per-user classifier sits **behind a cloud-load
interface** (local store = one impl);
114     **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-
video user is ALWAYS fully segmented by
115     the baseline batteries); **free-tier videos ARE pooled** into the shared baseline
(consent baked into free terms,
116     minors-exclusion enforced at the pooling query); and the previously-pending structural-
guard sub-decision is now LOCKED ?
117     **Gate-A-only structural + `min_n = 5`**.
118
119     ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
120     The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
121     replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign

```

```

122     test; never a bare mean at n=6)**:
123
124     | Variant (honest LOV0, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
125     |---|---|---|---|---|---|
126     | shuttle-only | 0.307 | ? | ? | ? | ? |
127     | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
128     | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
129
130     **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
131     default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
132     per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
133     (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
134     confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
135     *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
136     results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
137
138     ### Gate-A served-default flip ? **REVERTED (eval?serve regression)** *(2026-06-14)*
139     **The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.**
A served-side check ?
140     the repeatable litmus in [#148](https://github.com/avidullu/badminton-highlight-
indexer/pull/148)
141     (`backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`),
corroborated by an independent
142     served run ? found Gate A is **not a win on the served path**: on the production
(COARSE) windowing it is

```

83. assistant (2026-06-14T11:01:03.752Z)

Now I understand the full state: the doc now has my entry (stale ? says "mc=2 stays / decision pending") AND #151's authoritative "REVERTED" entry below it. My entry misstates the live state. I'll reconcile it to reflect reality (flip already reverted by #151, corroborated by my independent run) ? keeping its unique value (the byte-identity (a) confirmation + the mc-sweep) but fixing the stale framing.

84. user (2026-06-14T11:01:11.605Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

85. user (2026-06-14T11:01:22.296Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

86. assistant (2026-06-14T11:01:40.142Z)

Now let me fix the memory checkpoint ? it also has the stale "decision pending / not committed yet" framing.

87. user (2026-06-14T11:01:49.684Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

88. assistant (2026-06-14T11:01:54.567Z)

Memory updated. Now commit the doc reconciliation ? fetching first since sibling sessions are clearly active.

89. user (2026-06-14T11:01:57.656Z)

```
--- local vs origin/master ---
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
--- branch from origin/master ---
branch 'docs/served-gate-a-reconcile' set up to track 'origin/master'.
Switched to a new branch 'docs/served-gate-a-reconcile'
M docs/QUALITY_ITERATIONS.md
?? artifacts/
```

90. assistant (2026-06-14T11:02:08.289Z)

docs(quality): reconcile served-side Gate-A entry with the #151 revert

My #152 entry was authored while the rollback decision looked pending; a sibling session had already (a) productionized the served litmus in #148 (backend/eval/served_gate_a.py + tests/test_served_gate_a_regression.py) and (b) REVERTED the #146 flip in #151 (config default_segmenter -> gemini, the indexing.fusion block removed, FusionConfig.min_crossings default 2 -> 0 so Gate A is OFF-by-default-but-opt-in).

This corrects the now-stale framing so the log stays honest:

- header/intro reframed as the INDEPENDENT measurement that corroborates #148/#151 (our numbers match: served ?F1 -0.008, ~12 rallies dropped, 2/6 losses);
- the "Recommendation (decision pending)" paragraph -> "Outcome: REVERTED in #151" with the actual config state and a cross-link to the REVERTED entry;
- points at the productionized guard backend/eval/served_gate_a.py (#148), not only the scratch repros.

Measurement, tables, and the (a) mc=0 byte-identity-to-wasb_hybrid confirmation are unchanged. Open follow-up unchanged: align the harness windowing with the served windowing before treating a harness lift as a served default.

Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

91. user (2026-06-14T11:02:08.453Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\commitmsg_reconcile.txt (file state is current in your context ? no need to Read it back)

92. user (2026-06-14T11:02:24.863Z)

[docs/served-gate-a-reconcile 869c870] docs(quality): reconcile served-side Gate-A entry with the #151 revert

```
1 file changed, 19 insertions(+), 14 deletions(-)
--- verify only my commit + only the doc ---
869c870 docs(quality): reconcile served-side Gate-A entry with the #151 revert
869c870 docs(quality): reconcile served-side Gate-A entry with the #151 revert
docs/QUALITY_ITERATIONS.md | 33 ++++++-----
1 file changed, 19 insertions(+), 14 deletions(-)
--- push + PR ---
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] docs/served-gate-a-reconcile -> docs/served-gate-a-reconcile
https://github.com/avidullu/badminton-highlight-indexer/pull/153
```

93. assistant (2026-06-14T11:02:30.074Z)

PR [#153](https://github.com/avidullu/badminton-highlight-indexer/pull/153) opened (docs-only, 1 file). Let me wait for CI and merge on green.

94. user (2026-06-14T11:04:33.371Z)

Command running in background with ID: bc7j0nc84. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\03c8ada-c18d-40e6-9ee1-144846fe3975\tasks\bc7j0nc84.output. You will be notified when it completes. To check interim output, use Read on that file path.

95. assistant (2026-06-14T11:04:50.089Z)

CI for the reconcile PR is running in the background; I'll merge it once it goes green. Quick status while that finishes:

****What happened:**** Your "commit the doc" landed as [PR #152](https://github.com/avidullu/badminton-highlight-indexer/pull/152) (merged, `53bc743`, CI green). But I then discovered a ****sibling session** had already acted on the same finding in parallel******:

- ****[#148](https://github.com/avidullu/badminton-highlight-indexer/pull/148)**** productionized a served-side litmus (`backend/eval/served\_gate\_a.py` + a regression test).
- ****[#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)**** ****reverted the #146 flip**** ? `config.json` `default\_segmenter` ? `gemini`, the `indexing.fusion` block removed, `FusionConfig.min\_crossings` default `2?0` (Gate A now OFF-by-default but still opt-in).

Their independent numbers ****match mine**** (served ?F1 ?0.008, ~12 rallies dropped, 2/6 losses) ? good corroboration that the eval?serve regression is real.

That made my freshly-merged #152 entry ****stale**** (it framed the rollback as "decision pending / mc=2 stays"). So [PR #153](https://github.com/avidullu/badminton-highlight-indexer/pull/153) reconciles it: reframed as the independent measurement corroborating #148/#151, "Outcome ? REVERTED" instead of "decision pending", pointing at the productionized guard. Memory checkpoint updated to match.

****Net live state:**** the Gate-A served-default flip is ****already reverted**** ? your original question (rollback to mc=0 / retune to mc=1 / leave mc=2) is effectively moot; production is back on the pre-#146 `gemini` default with Gate A opt-in. The one genuinely open item is the deeper lesson: ****align the experiment-harness windowing with the served windowing**** so a harness promotion actually predicts served behavior.

I'll confirm #153 merged once CI completes.

96. user (2026-06-14T11:04:50.173Z)

```
<task-notification>
<task-id>bc7j0nc84</task-id>
<tool-use-id>toolu_01XFNrchipiwuHWDCfpJ1cj7q</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\03c8ada-c18d-40e6-9ee1-144846fe3975\tasks\bc7j0nc84.output</output-file>
<status>completed</status>
<summary>Background command "Poll until all CI checks conclude on PR #153" completed (exit code 0)</summary>
</task-notification>
```

97. user (2026-06-14T11:05:00.747Z)

```
{"checks":[{"conclusion":"SUCCESS","name":"Ruff lint"}, {"conclusion":"SUCCESS","name":"Mypy (lenient baseline)"}, {"conclusion":"SUCCESS","name":"Syntax sweep + import smoke (no GPU stack)"}, {"conclusion":"SUCCESS","name":"Full test suite (offline, mocked)"}], "mergeStateStatus":"CLEAN", "mergeable":"MERGEABLE"}
```

98. assistant (2026-06-14T11:05:03.799Z)

All 4 lanes green, `CLEAN`. Merging the reconcile PR and returning to a clean master.

99. user (2026-06-14T11:05:12.675Z)

```
Fast-forward
 docs/QUALITY_ITERATIONS.md | 33 ++++++-----
 1 file changed, 19 insertions(+), 14 deletions(-)
Already on 'master'
Already up to date.
--- final state ---
a03f0f4 docs(quality): reconcile served-side Gate-A entry with the #151 revert (#153)
53bc743 docs(quality): served-side Gate-A check ? flip's offline win doesn't transfer to
production windowing (#152)
2cd469e revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve) (#151)
?? artifacts/
--- live config sanity (post-#151 revert) ---
default_segmenter = gemini | fusion block present: False
```

100. assistant (2026-06-14T11:05:35.565Z)

Done ? both PRs merged, master clean at `a03f0f4`, live config confirmed back to the pre-#146 state ('default_segmenter: gemini', no `fusion` block, Gate A off-by-default).

Final outcome

****Served-side verification (the task):****

- ****(a) No-regression invariant ? CONFIRMED ?**** At `min\_crossings=0`, the live `FusionSegmenter` seeding + handoff are byte-identical to `wasb\_hybrid` on all 6 golden videos ? verified on real served code, not just by review.
- ****(b) Regression found ?**** The offline `+0.074` win does ****not**** transfer to the production (coarse) windowing. At the shipped `mc=2`: mean F1 ?0.008, ****12 real-rally windows dropped**** (worst on doubles). Root cause: ****eval-harness windowing ? served windowing**** ? Gate-A's value was an artifact of the over-segmenting **proposal** windowing the harness uses but production doesn't serve.

****What landed:****

- [#152](https://github.com/avidullu/badminton-highlight-indexer/pull/152) ? the honest measurement recorded in `docs/QUALITY\_ITERATIONS.md` (your "commit the doc").
- [#153](https://github.com/avidullu/badminton-highlight-indexer/pull/153) ? reconciled that entry with reality, since a sibling session acted on the same finding in parallel.

****The parallel-session twist (worth knowing):**** while I was verifying, other slates independently reached the same conclusion and went further ?

****[#148](https://github.com/avidullu/badminton-highlight-indexer/pull/148)**** productionized a served-side litmus (`backend/eval/served\_gate\_a.py` + regression test) and

****[#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)**** reverted the flip. Their numbers match mine (?0.008, ~12 dropped, 2/6 losses) ? independent corroboration. So the rollback decision I queued for you is ****already executed****; nothing for you to decide there.

****One genuinely open item (the real lesson, not requested ? flagging only):**** the experiment harness promotes/refuses cues on a windowing production doesn't serve, so a harness lift doesn't reliably predict served behavior. Aligning the harness windowing with the served windowing (or re-running promotions on production windowing) would prevent the next eval?serve surprise. I can scope that as its own work item whenever you want ? it's the kind of thing worth a proper design pass.